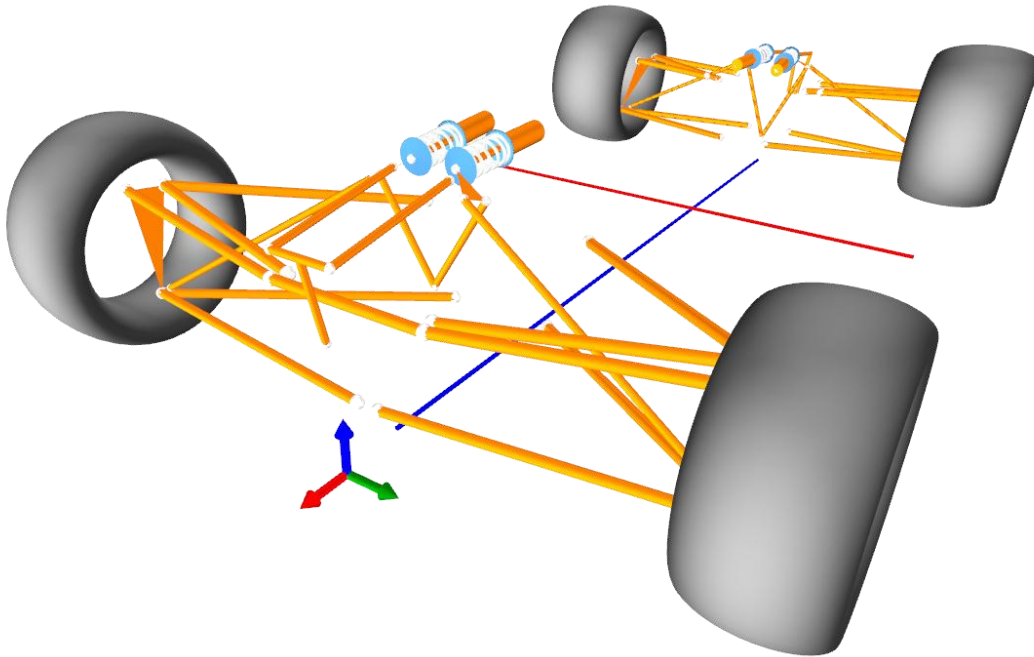


OptimumKinematics

Suspension Design and Analysis Software



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3 Welcome



Thank you for purchasing OptimumKinematics, the newest benchmark in kinematics analysis software. This help file contains information regarding the features and functions of OptimumKinematics.

3.1 Feedback

OptimumKinematics is a continually evolving program and we give high regard to any suggestions, comments, complaints and/or criticisms that OptimumKinematics users might have. Please contact us at softwaresupport@optimumg.com and we will endeavor to improve OptimumKinematics based on your feedback.

Within OptimumKinematics the user is able to report any issues, features or ideas. Users have the option to report feedback. Read the [Quick Start](#) section to see how to do this.

4 Introduction

OptimumKinematics has been developed with the goal of providing users with a powerful kinematics tool that is both easy to use, and easy to integrate into your existing workflow.

It can be used to rapidly simulate the kinematic properties of a suspension design, providing detailed output and visualization for a broad range of output channels.

OptimumKinematics builds upon the success of OptimumK, providing more scope for different suspensions, a greater selection of output channels and greater modularity between designs. The ability to animate and visualize motion, compare multiple suspension simulations back to back and import and output data in a standard format (for use in Microsoft® Excel or MATLAB®) all remain.

Major improvements for version two are:

- **New User Interface.** The philosophy and workflow has stayed the same in OptimumKinematics as the previous release. However, the new interface uses controls that are available with the new version of the .NET® Framework (Version 4 and higher). This is now the same standard control interface used by other Windows® products.
 - An interface layout similar to the Windows® products should reduce the learning period of OptimumKinematics.
 - The Ribbon Bar Control for quick access to all of the functionalities of the software. Buttons are filtered depending on the active control.
 - A multiple document interface (MDI) allows for flexibility and customization of the layout of the application. The ability to make edits to multiple points and instantly see the comparison is now possible.
 - Adjust the size and display and location of the controls and window panes to better organize the working space and maximize performance.
- **New Project Framework.** The project data is no longer saving in a single file. Each object (for instance; **Suspension**, **Motion** or **Analysis Results**) is saved in a unique file. Each file has a specific extension and icon and will be easily recognized on your hard drive. The **Project Tree** reflects the directory that is created in your hard drive. This has multiple advantages:
 - The risk of corruption and losing all of the data belonging to one project is decreased. An object can be corrupted or missing without affecting the rest of the objects belonging to the project.
 - The loading and saving performance has been improved by more than 30%
 - The sharing of files is faster and easier with the new proprietary file. There is no need to use Excel as a tool to share files.

- Better organization of your **Suspension**, **Motion** and **Analysis Results** objects with the introduction of a “Vehicle” level in the project tree.
- **New Design Tools.** The input forms have been redesigned for faster design and modifications of your suspensions and motions.
 - Suspension points can be edited directly in the 3D visualization.
 - The static outputs of your suspension are calculated automatically and display in real-time.
 - Add/Remove/Modify motion points directly in the chart.
 - The possibility to interpolate your motion points to have a smooth curve.
- **New Analysis Tools.** The tools that you know and use in OptimumK 1.1 have been redesigned and are now more powerful for faster and deeper analysis. The tools have been expanded to include:
 - Quick preview of the simulation result with a table or chart.
 - Display of the data is faster, thanks to the memory-mapping of the result file. This allows a faster reading of the simulation result data from the hard drive.
 - Table reports and chart data can be copied to the clipboard easily for use in other software such as Microsoft® Excel or MATLAB®.
 - Customizable graphs with more options for the title (font, color, etc.), axis (title, gridlines, primary and secondary, etc.), and legend.
 - More channels have been pre-calculated into the software to allow you to quickly analyze your simulation data from within the software. There is now no need for further post-software calculations.
 - Animated table, chart and 3D visualization for a complete review of your simulation. Animation is played on all tables, charts and 3D visualization in real-time.

4.1 Installation Requirements

4.1.1 Hardware Requirements

Processor

- Intel® Pentium 4™, Intel® Xeon™ and Intel® Core™
- AMD® Athlon™, AMD® Opteron™ and AMD® Turion™

Memory

- Minimum: 512MB RAM
- Recommended: 1GB RAM or more
- Virtual Memory recommended to be twice the amount of RAM

Storage

- At least 100MB of free disk space. Includes installation of OptimumKinematics and required software components (see Software Requirements)

Network

- Internet connectivity for License Activation and Deactivation.

Display Adapter

- Minimum: Microsoft® DirectX® 9.0c capable card with 32MB RAM
- Recommended: Microsoft® DirectX® 9.0c capable NVIDIA® GeForce® or ATI® Radeon® with 128MB RAM or higher.

Display Unit

- Minimum: 15" screen with resolution of 1024 x 768 pixels
- Recommended: 19" screen with resolution of 1280 x 1024 pixels
21" screen with resolution of 1280 x 1200 pixels

Other

- Mouse or other pointing device.

4.1.2 Software Requirements

Operating System

- Microsoft® Windows® XP (32 or 64-bit) or Microsoft® Windows® Vista (32 or 64-bit) or Microsoft® Windows® 7 (32 or 64-bit)

Components

- Microsoft® .NET® Framework 4.0 or higher
- Microsoft® Windows Installer 3.1 or higher

Other

- Microsoft® Excel™ version 10 or higher for import or export of data.
- SlimDX Redistributable for .NET 4.0 version 4.0.11.43 or higher for 3D visualization.
- Microsoft Office Database Engine 2007

4.2 Licensing

OptimumKinematics comes with a 14-day trial version. To further use the software after the initial period two options are available:

- Request a **Trial Key** to extend the trial period to be valid for a certain number of days or until a certain date.
- Purchase a **Full-Time Key** that has no expiration date. This type of key can be transferred from one computer to another (An internet connection required for activation and deactivation of the key).
 - **Activation.** In the **Backstage** (**Graphical User Interface**, Under File) select License and then **ApplyKey**. You will be asked for the Key, your Name and E-Mail address.
 - **Deactivation.** Keys must be deactivated on one computer before being activated on another computer. Deactivation must be done manually through the **Backstage** under 'License'.

In the case that an internet connection is not available for periodic activation, the user will have a small grace period to ensure that they do not lose access to OptimumKinematics.

5 Quick Start

The following section will help you to quickly learn how to utilize the main features of OptimumKinematics. For a more detailed overview of the software please refer to the [Detailed Overview](#).

5.1 Launching the Application

5.1.1 Graphical User Interface

When opening the software, you will arrive at the **Backstage**. This is where you will manage your projects and adjust **Licensing** and projects **Options Menu**.

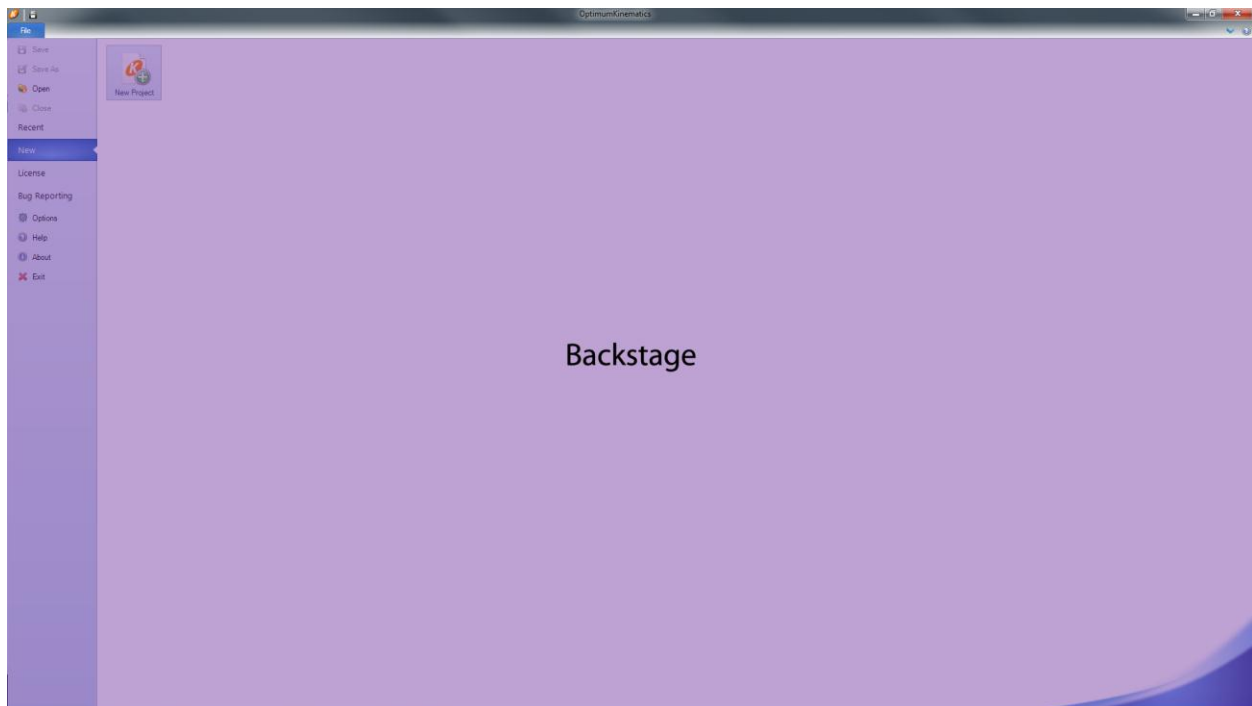


Figure 1 Backstage

Once a project has been created you will leave the **Backstage** and enter the Main GUI. The Main GUI is organized in the following way:

- The top is the **Ribbon Control Bar** with all shortcuts to functionalities of the software.
- The center is the **Document Manager** where you will be able to edit project files.
- The left contains the **Project Trees**, allowing you to organize project files.

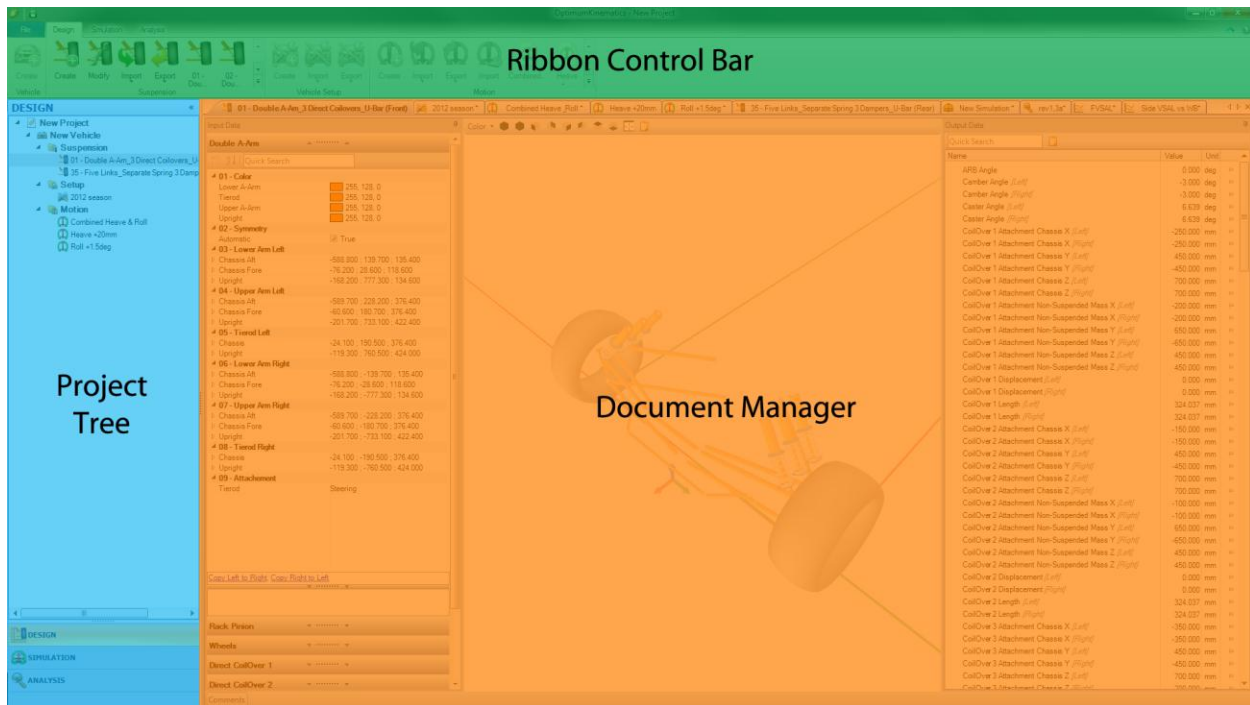


Figure 2 Graphical User Interface

The **Ribbon Control Bar** contains all the buttons that will help you interact with the application.

The **Document Manager** allows the user to have multiple tabbed documents open at the same time. The tabs can be dragged and dropped to be displayed side by side, or in any other configuration on the same screen. In the **Options Menu** the user has the ability to change the color of these tabs to better keep things organized.

The **Project Tree** allows you to manage all aspects of your project. Vehicles can be added and organized into differing subfolders, likewise with the suspensions, setups and motions in the project.

5.1.2 Creating a New Project

The first step is to create a New Project. This is found under the file tab. You will be asked to select a name and a directory for your project.

Upon creating a new project, a folder will be created on your hard drive in the selected directory. All the files related to your project will be saved in this location. Here are the different files that you could find:

Name	Extension	Icon
Project File	.O2Pro	
Suspension File	.O2Sus	
Vehicle Setup File	.O2Set	
Motion File	.O2Mot	
Simulation File	.O2Sim	
Result File	.O2Res	
Chart File	.O2Cha	
Table File	.O2Tab	
View File	.O2Vie	
User File	.O2Uso	

5.1.3 Opening an Existing Project

When it comes time to open an existing project, you will need to locate the directory in which the project was saved in. Within the folder you will find the OptimumKinematics Project File.

5.1.4 Saving a Project

The project can be saved using the Save button in the **Backstage** (**Graphical User Interface**, Under File) or using the shortcut (Ctrl + S). The project can be saved to another location using the Save As button in the **Backstage** or using the keyboard shortcut (Ctrl + Shift + S).

When closing the application, if the project has not been saved the application will ask if you want to save the project.

5.2 Options Menu

The following settings may be changed to adapt to your specific needs:

- Units
 - **Units of Length.** Meter, Kilometer, Centimeter, Millimeter, Inch, Thousandth of an Inch, Foot, Yard or Mile.
 - **Units of Angle.** Radian, Degree or Revolution.
- Referential
 - **Longitudinal Axis.** You may define this axis to either be the X, Y or Z and which direction you would prefer to be positive.
 - **Lateral Axis.** You may define this axis to either be the X, Y or Z and which direction you would prefer to be positive.
 - **Vertical Axis.** You may define this axis to either be the X, Y or Z and which direction you would prefer to be positive.
- Motion Steps
 - User has the option to override the Number of Steps for a given simulation. The default value is 20 steps.
- Roll/Pitch Axes
 - Moving Axis.
 - Fixed Axis on the Ground.
 - User Defined Fixed Axis.
- Display
 - Here the user can change the default colors used to associate the following:
 - Background Color
 - Roll Center
 - Roll Axis
 - Pitch Center
 - Pitch Axis
 - Contact Patch
 - Wheel Center
 - KPI Ground
 - Front View Instantaneous Center
 - Side View Instantaneous Center
 - Front View Instantaneous Axis
 - Side View Instantaneous Axis
 - Instantaneous Axis
- Names

- The user is asked to name the following Objects before they are created:
 - Vehicle
 - Motion
 - Setup
 - Simulation
 - Result
 - Chart
 - Table
 - View
- By unchecking the box next to the name you choose to let OptimumKinematics use the default name and not ask you for a different name.
- Numbers
 - Here the user may change the number of decimal places that are viewed. The default value is three decimal places.
- Documents Tab
 - The color of the Tab in the **Document Manager** and the option to Open After Creation may be changed for the following items:
 - Suspension
 - Setup
 - Motion
 - Simulation
 - Result
 - Chart
 - Table
 - View
- Result
 - Here the user can have OptimumKinematics Automatically Add Results to the **Document Manager**.

5.3 Design Overview

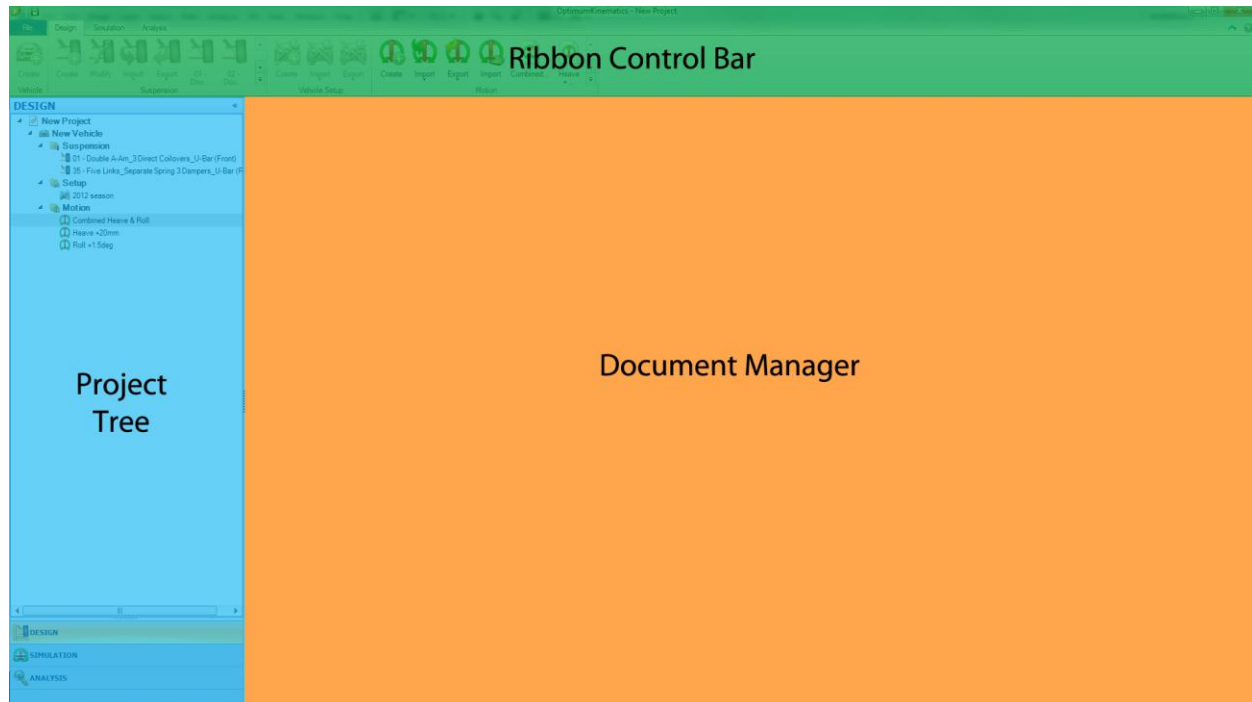


Figure 3 Design Window

The design section is where the **Suspension**, **Vehicle Setup** and **Motion** profiles are constructed and managed.

Selecting each of the following sections under the **Project Tree** will bring it up in the **Document Manager**.

- **Suspension** contains all the front and rear suspension systems that have been created, with certain static output channels available for immediate review in real time for each suspension.
- **Vehicle Setup** contains the vehicle setups and front and rear suspension configurations, with certain static output channels available for immediate review in real time.
- **Motion** contains the motion profiles for simulation.

5.3.1 Vehicle

The first step is to create a new vehicle by clicking on the Create Vehicle button on the **Ribbon Control Bar**. This new type of object in OptimumKinematics helps the user to have better organization. All elements created under the vehicle will belong to a library of objects available for the given vehicle. These objects can be copied, pasted or moved to another vehicle or sub folder for better organization.

5.3.2 Suspension

The next step is to build a suspension design. Three main options exist:

- **Creating a suspension.** You will be able to select the type of geometry, the type of actuation and anti-roll bars from a large library of elements.
- **Importing and Exporting Suspension.** You may choose to import from an Excel file, OptimumK 1.1 Excel file or from proprietary files.
- **Default Suspension Types.** Choose from one of the premade available files.

At this time you have the ability to adjust all of the connection points between the components to match that of your actual design.

5.3.3 Suspension Design

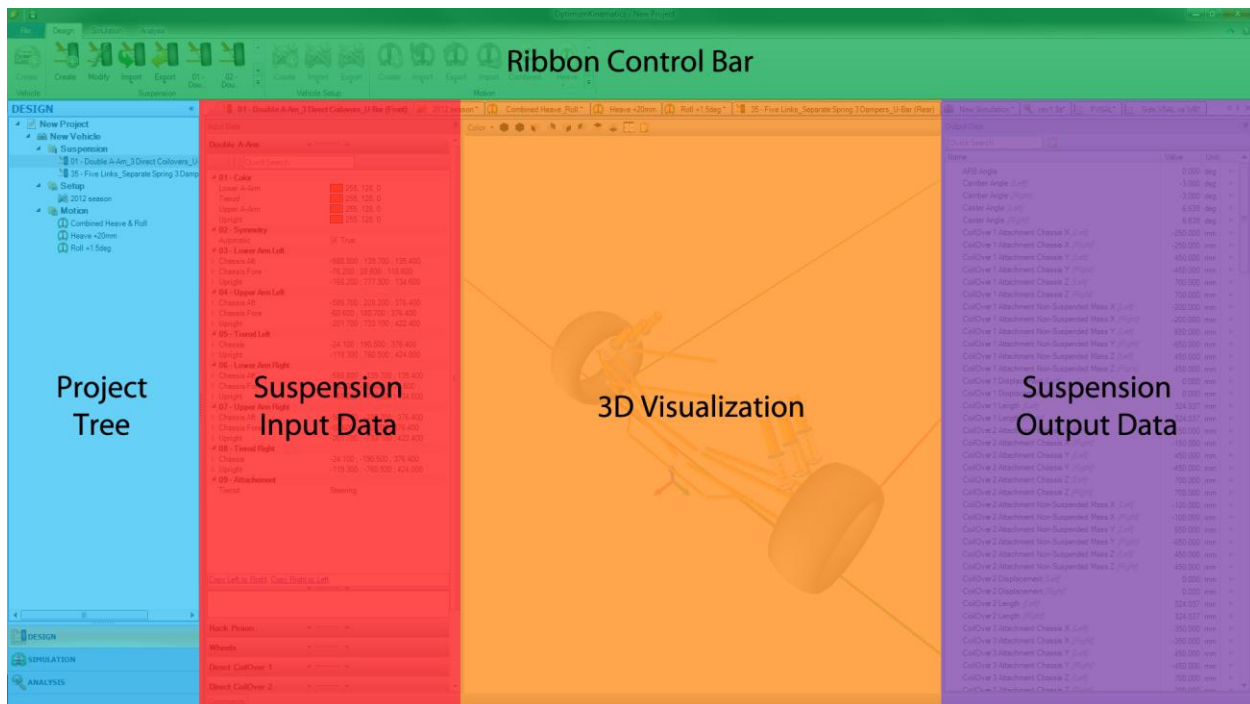


Figure 4 Suspension Design Window

The suspension design **Document Manager** includes the following sections:

- **Suspension Input Data.** This allows all suspension data points to be entered, as well as parameters for visualization colors, wheel and rim information as well as reference points.
- **3D Visualization.** Suspension data points can be edited by double-clicking on the suspension points in the **3D Visualization** window. The visualization can also be rotated, zoomed and panned by either left, middle and right clicking, and dragging the mouse.

Additional visualization of the roll axis, pitch axis and instant centers can be shown by enabling the display options in the **Options Menu**, located on the projects **Backstage** (**Graphical User Interface**, under File).

- **Suspension Output Data.** This pane displays the output data of all available data channels at static. Information such as roll center location, mechanical trail, scrub radius, caster and KPI angle, as well as all other available data channels are updated in real time as points are edited.
- **Comments.** Add comments to your design to help maintain organization.

The controls can be resized and minimized in order to maximize your working space. It is also important to remember that the suspension setups can be modified, renamed and exported.

5.3.4 Vehicle Setup

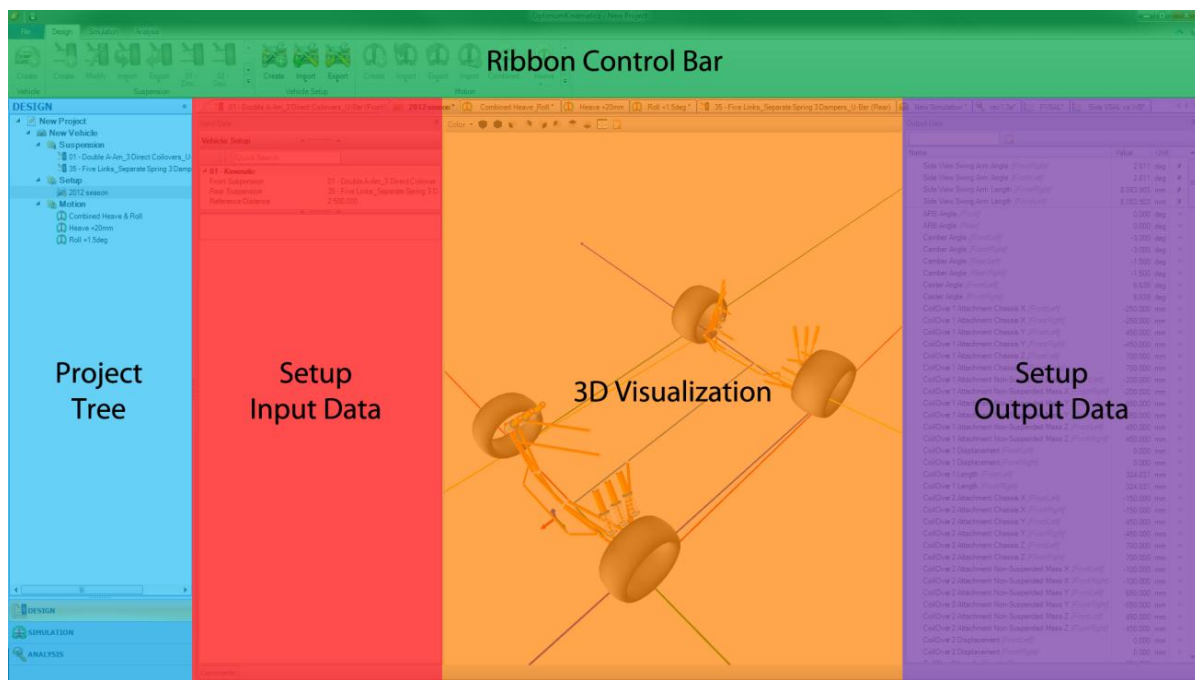


Figure 5 Vehicle Setup Window

The vehicle setup is a new feature of OptimumKinematics, and it is where front and rear suspensions are mated together to form a vehicle for simulation.

The vehicle setup design includes the following sections:

- **Setup Input Data.** This allows front and rear suspension setups to be assigned to a vehicle setup, and the reference distance. Note - this is the distance between the front

and rear reference planes and is not necessarily the same value as wheelbase.

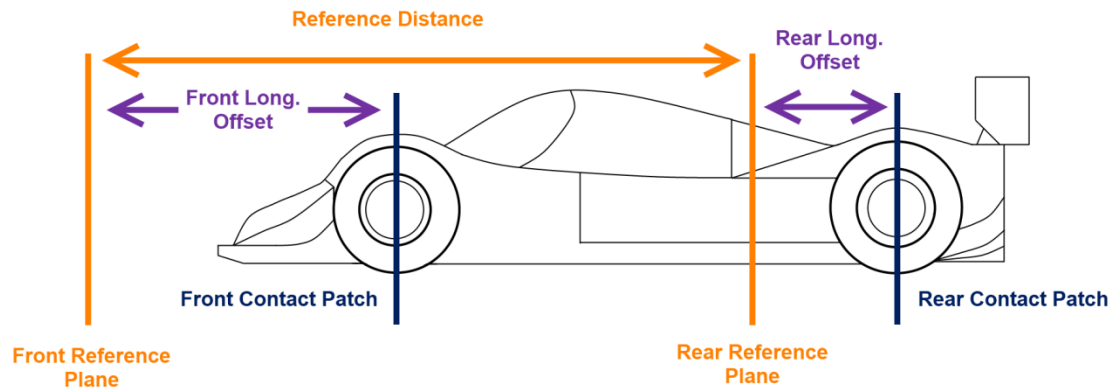


Figure 6 Vehicle Reference Distance

- **3D Visualization.** The visualization can be rotated, zoomed and panned by left, middle and right clicking, and dragging the mouse. Additional visualization of the roll axis, pitch axis and instant centers can be shown by enabling the display options in the [Options Menu](#), located on the projects [Backstage \(Graphical User Interface\)](#), Under File).
- **Setup Output Data.** Check the static output of your suspension such as roll center, mechanical trails, scrub radius, caster and KPI angle.
- **Comments.** Add comments to your design to help maintain organization.

The **Ribbon Control Bar** contains options to import setups from other OptimumKinematics Projects or from excel files. If importing from an OptimumKinematics Project, the .O2Set files will only contain references to the front and rear suspension files. The actual front and rear suspensions will not be imported. For this reason, we recommend exporting a setup from one project before import it to another, rather than directly locating the files on your hard drive.

5.3.5 Motion

Motion allows for the creation of motion paths for simulation. Motion profiles can be created by hand, imported from previous files or imported from logged track data.

Four options are available on the **Ribbon Control Bar** to create a motion:

- **Creating a Motion.** You have the ability to enter the points graphically or numerically.
- **Importing and Exporting Motion.** From an Excel file, OptimumK 1.1 Excel file or a proprietary file.
- **Import Track Data.** From a data acquisition system.

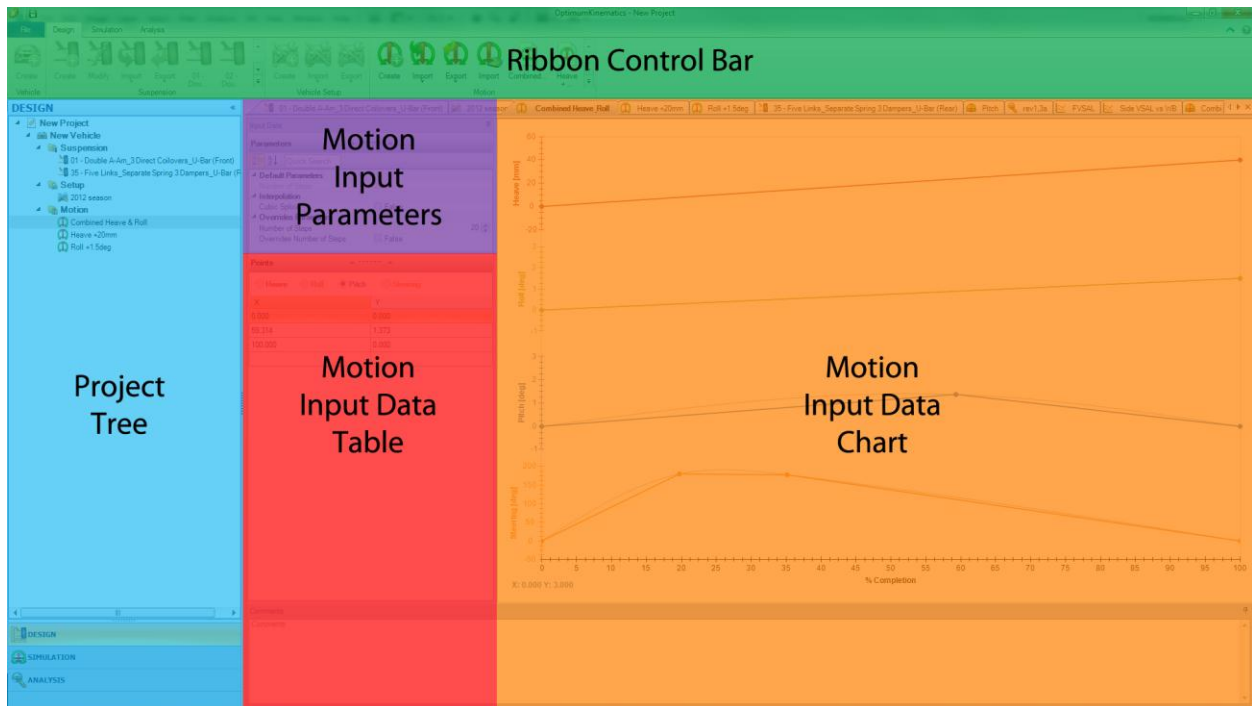


Figure 7 Motion Window

The motion **Document Manager** includes the following:

- **Motion Input Parameters** allows customization of the number of simulation steps and whether motion points should be interpolated.
- **Motion Input Data Table** allows motion data points to be entered, modified or deleted. Selecting between Heave, Roll, Pitch and Steering will switch between the different motion profiles to be edited. Any combination of the vehicle in different states of Heave, Roll, Pitch and Steering are possible.
- **Motion Input Data Chart** allows motion profile points to be added, removed or edited via a graphical interface. A Left Click will add a data point at the location of your mouse

and a Right Click will remove a data point at the location specified by your mouse. Ctrl + Left Click will move a data point. Clicking on the different types of motion will switch that form of motion to the active graph for editing, as indicated by the mouse cursor color.

- **Comments.** Add comments to your design to help maintain organization.

5.4 Simulation Overview

The simulation is a new type of object introduced in OptimumKinematics. The simulation **document manager** contains just one main section, the **Simulation Input Data** pane, which allows vehicle configurations and motion profile to be mated to each other, and simulations to be run.



Figure 8 Simulation Overview

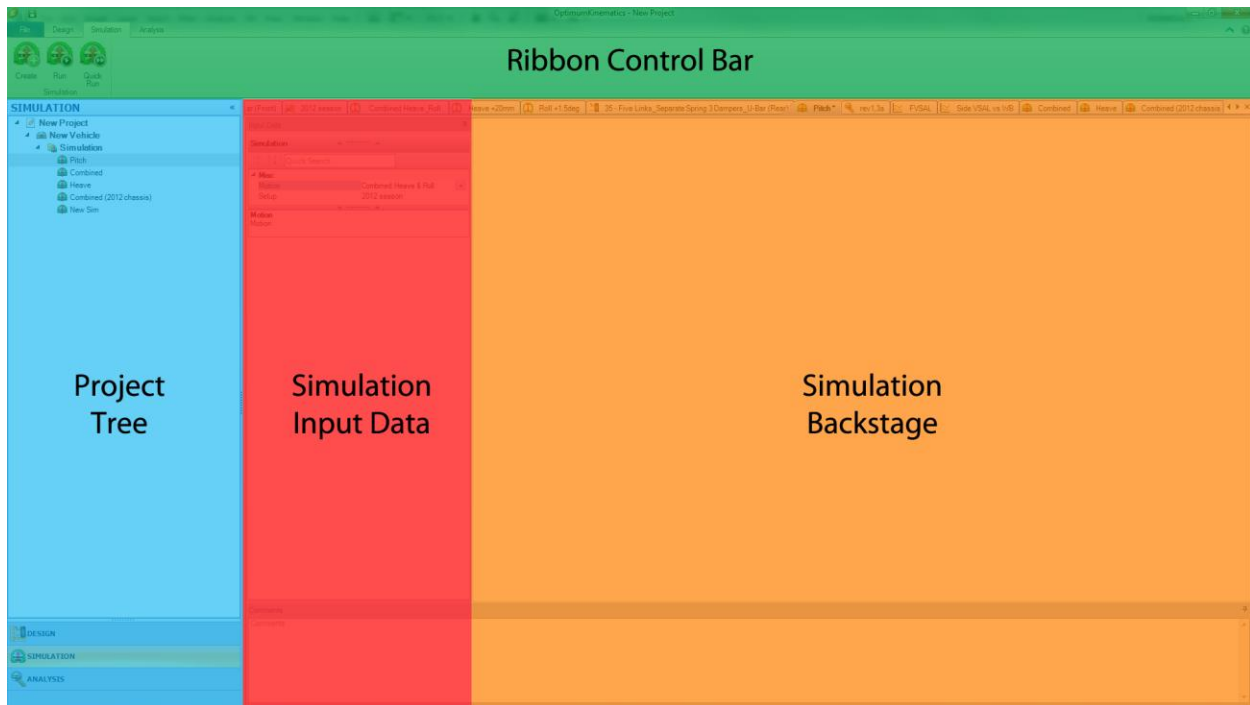


Figure 9 Simulation Window

Two options are available in the **Ribbon Control Bar** to create a simulation:

- **Create simulation.** This option will allow you to re-run the same simulation with the saved **Vehicle**, **Motion** and **Vehicle Setup**.
 - **Single Iteration.** Same properties as the quick run.
 - **Batch Run.** When utilizing the batch run feature you are able to simulate ONE vehicle setup and ONE motion over a VARIETY of user chosen pickup points and compare the results. See **Create/Run** in **Detailed Overview** for more information.
 - **Design of Experiment.** This feature allows you to select MULTIPLE vehicle setups and MULTIPLE motions and compare the results. See **Create/Run** in **Detailed Overview** for more information.
- **Quick Run simulation.** With this option you will need to select the **Vehicle**, **Motion** and **Vehicle Setup** each time.

Next step is to **Run** the simulation to acquire the Results. When running the simulation, a progress bar shows the completion of the simulation and 3D Visualization. A simulation can be cancelled or stopped at any time.

NOTE - If you stop the simulation you will be able to view the results up until the stop point, but if you cancel it the results will not be saved.

Once the simulation has finished OptimumKinematics will automatically show the Results Tab in the **Document Manager**.

When using the **Quick Run** feature, you will be asked to define which **Vehicle**, **Motion** and **Vehicle Setup**. This is perfect for getting to analysis in a hurry. If you plan on running the same simulation for multiple different types of motion, it would be beneficial to create the simulation so you can just click Run.

5.5 Analysis Overview

After the simulation is complete, a result file is created. A preview of the result file can be immediately seen in the **Document Manager** either in table or chart format. The selected channel or channels that are displayed in the preview chart (or table) is a global setting and will be common across all projects and results.

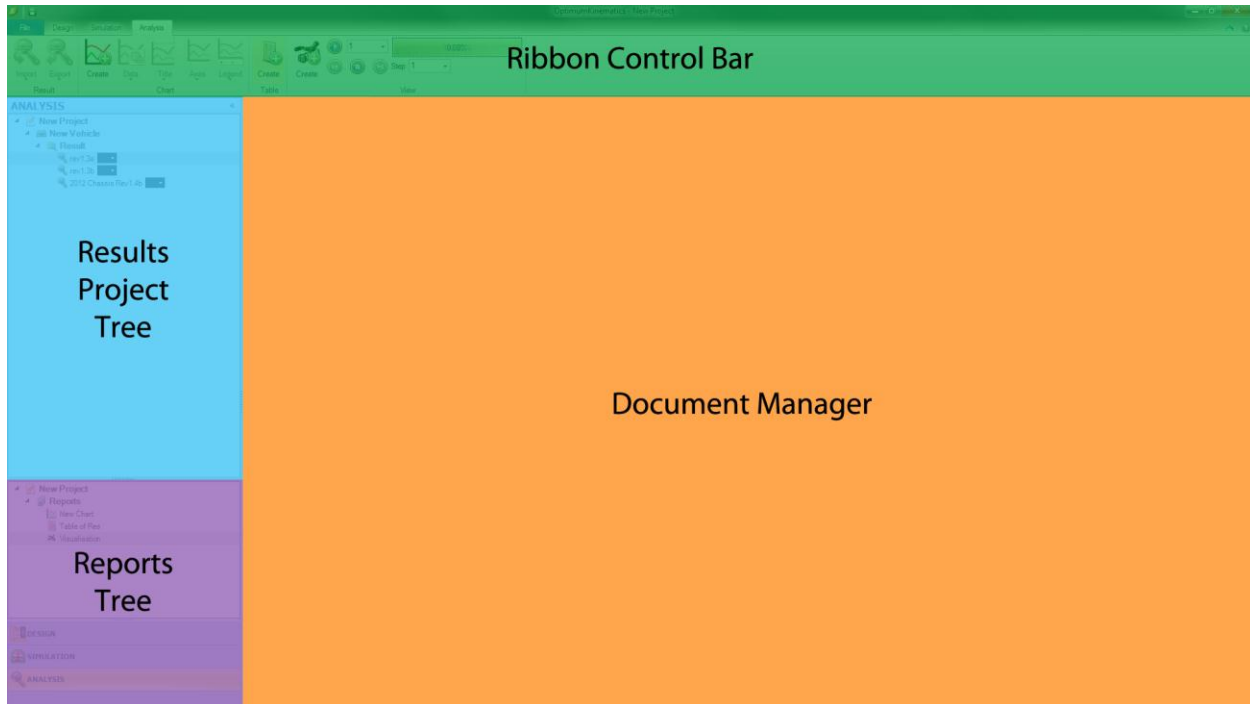


Figure 10 Analysis Window

Selecting results from the **Results Project Tree** will bring up a quick preview of the data in the **Document Manager**. After selecting the **Reports Tree**, A new report can be added. Three kinds of reports are available:

- **Chart.** Create a chart showing the selected data for the X and Y axis.
- **Table.** Create and compare tabular data between simulations.
- **View.** Create 3D Visualizations so you can see exactly what is happening in your simulation.

5.5.1 Chart

Report Charts allow the graphical plotting of two variables and their relationship to each other. Multiple results can be plotted on one chart, and a secondary axis can also be implemented. Charts are fully customizable using the buttons on the **Ribbon Control Bar** or by clicking inside the **Report Chart** Area. This is covered in greater detail in the **Detailed Overview** section.

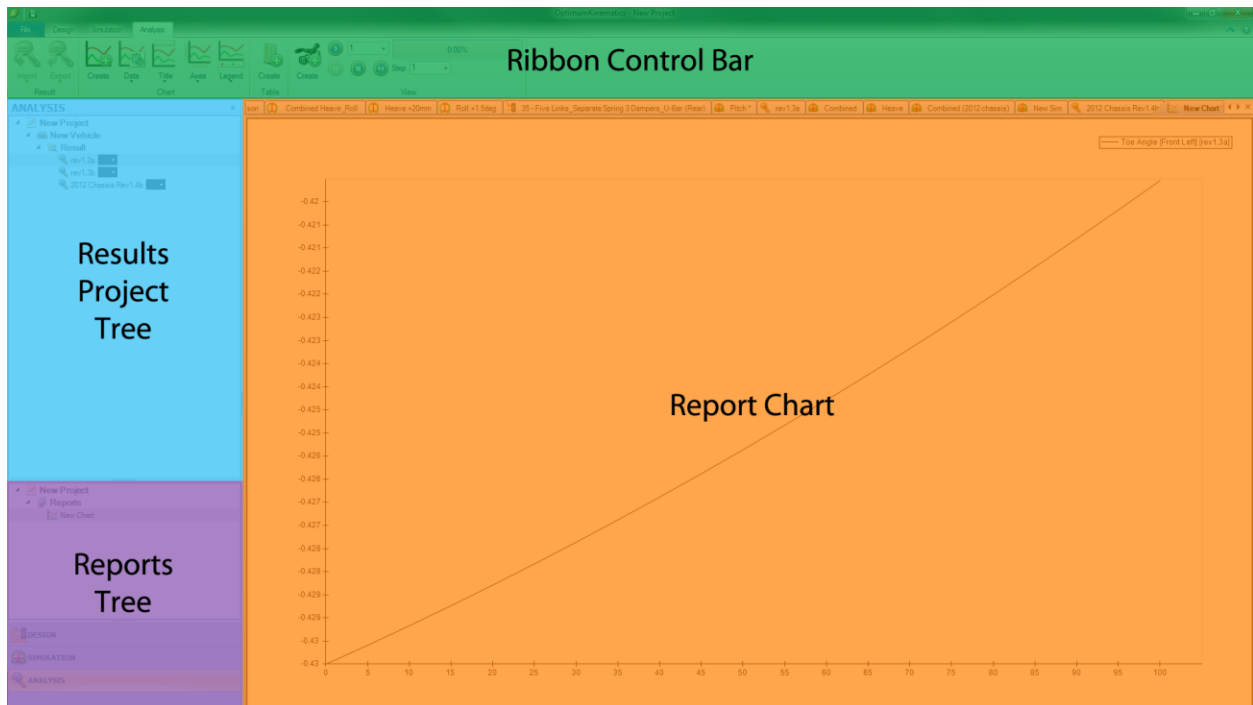


Figure 11 Analysis Report Chart

5.5.2 Table

Report Tables allow the tabular display of multiple channels across multiple runs next to each other. Channels for display are chosen through the **Reports Input Data** pane and results are chosen through the Results Input Data pane. Table data can be readily copied to the clipboard (Ctrl + C) for further analysis in external programs if required.

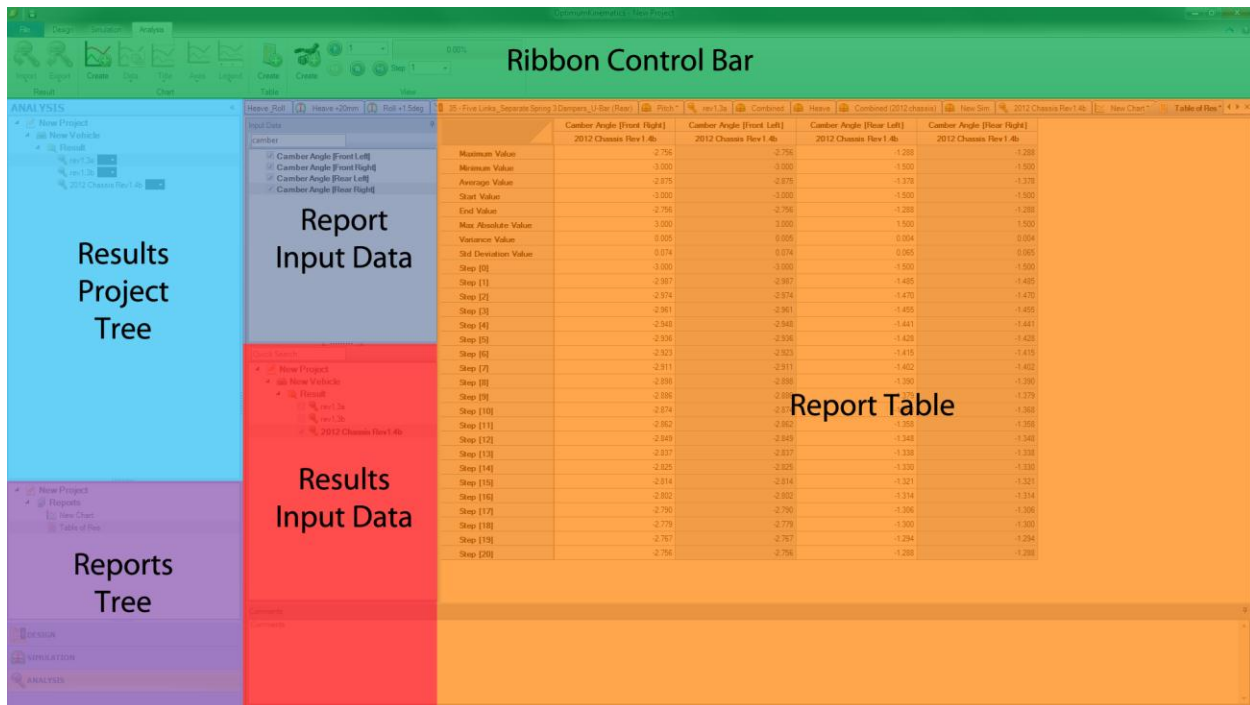


Figure 12 Analysis Report Table

5.5.3 View

Report Views allow 3D visualization and animated playback of a vehicle through a simulation run. New views can be added from the **Ribbon Control Bar**.

The view **Document Manager** contains the following:

- **Reports Input Data** contains a list of available simulation results for visualization.
- The **3D View** visualization can be rotated, zoomed and panned by either left, middle and right clicking, and dragging the mouse. Additional visualization of the roll axis, pitch axis and instant centers can be shown by enabling the display options in the **Options Menu**, located on the projects **Backstage** (**Graphical User Interface**, Under File).
- **Comments**. Add comments to your design to help maintain organization.

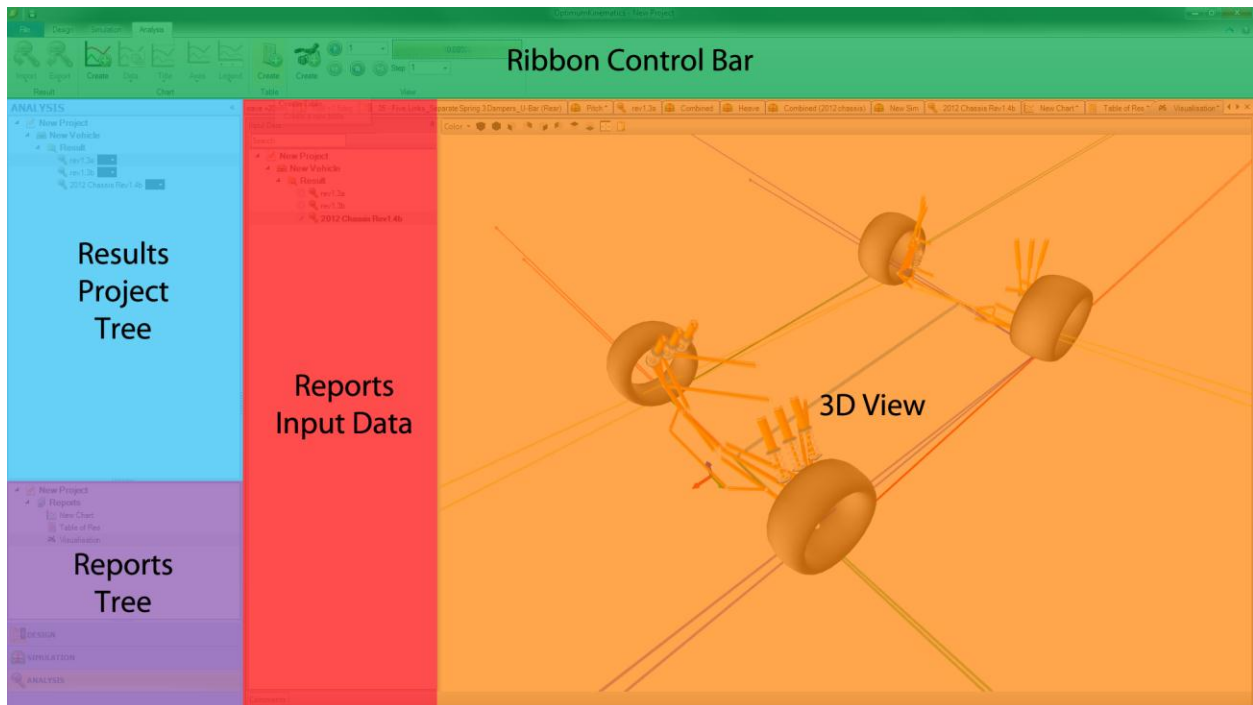


Figure 13 Analysis 3D View

6 Detailed Overview

The following section contains more detailed information than is given in the [Quick Start Guide](#). Menu options, simulation settings and software parameters are more fully explored. Definitions are provided for the available output data channels and the pre-existing suspension templates are outlined.

6.1 Reference System

By default, OptimumKinematics uses the following coordinate system.

Axis	Direction	Name	Color
Longitudinal	Points to Front or Rear of Car	X	Red
Lateral	Points to Left or Right of Car	Y	Green
Vertical	Points Vertically up or down	Z	Blue
Roll	Counter Clockwise Positive	Mx	Red
Pitch	Counter Clockwise Positive	My	Green
Yaw	Counter Clockwise Positive	Mz	Blue

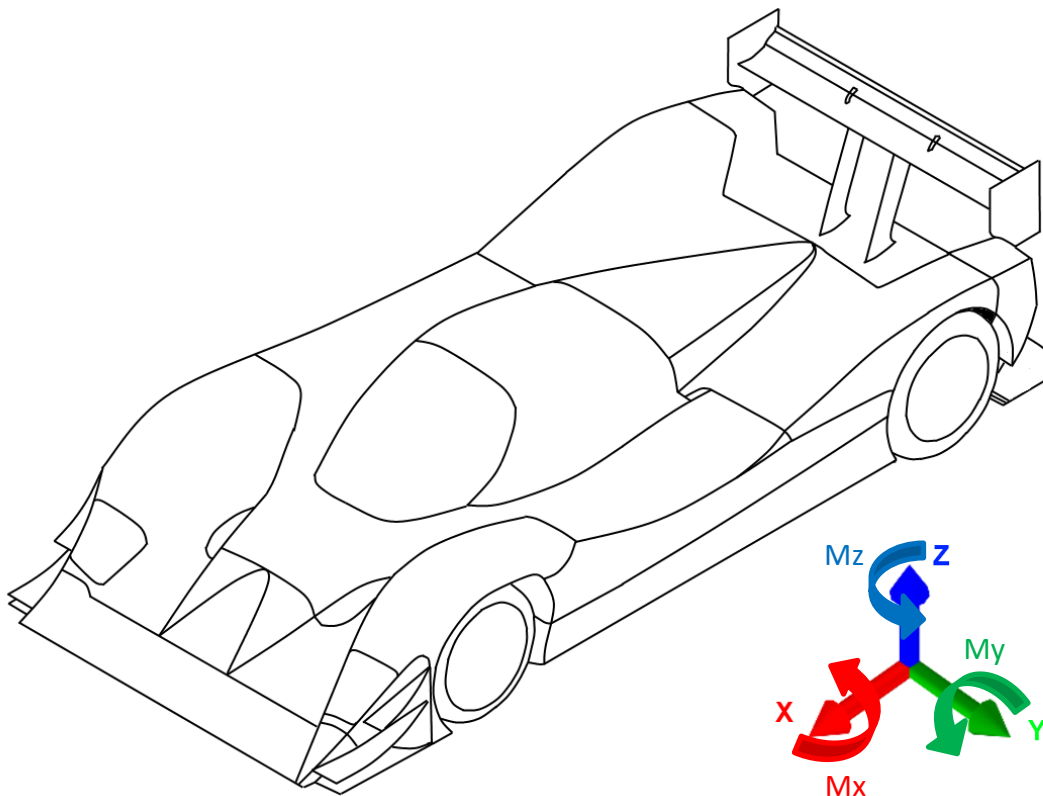


Figure 14 Default Vehicle Reference System

6.2 Design

The design section allows a user control over the front and rear suspension design, the vehicle setup and the simulated motion profiles.

The Design section is comprised of the **Project Tree**, the **Document Manager** and the **Ribbon Control Bar**.

The **Ribbon Control Bar** contains all relevant menu options for creating or modifying suspension, setup and motion profiles. It also contains the option to add a new **Vehicle** to the project.

A **Vehicle** must be added to the project before a suspension can be implemented.

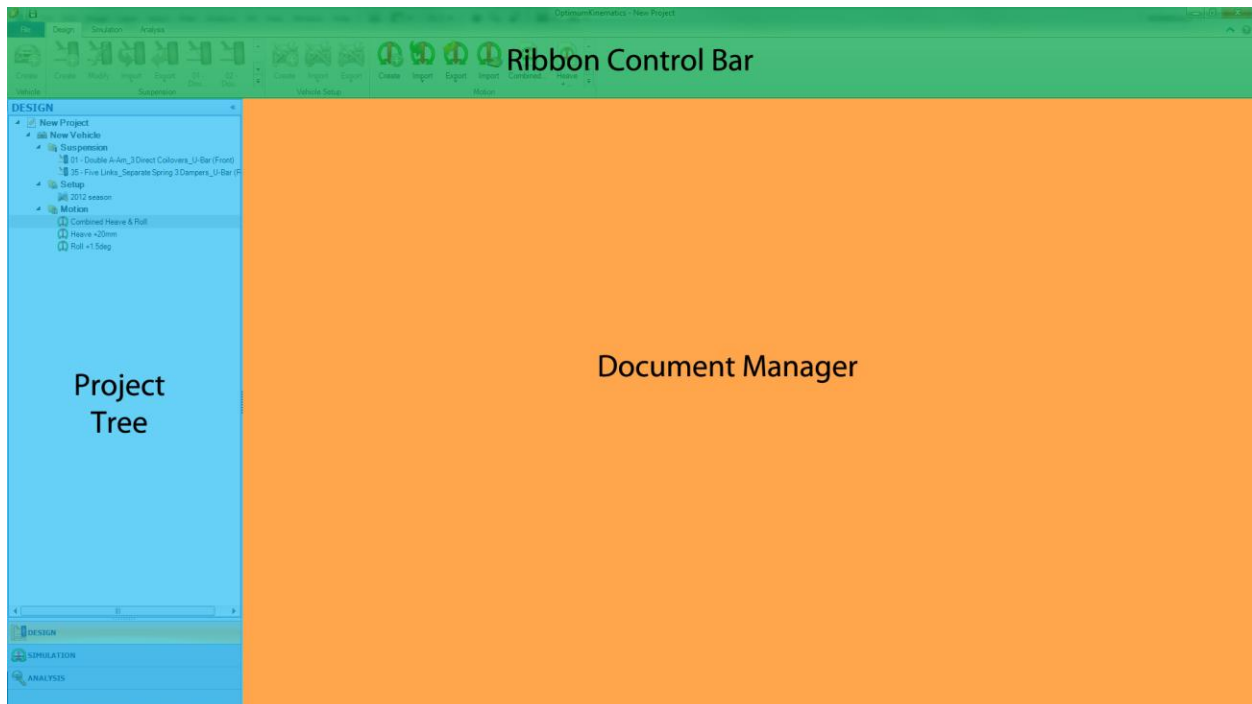


Figure 15 Design Window

6.2.1 Suspension Design

The suspension design **document manager** is comprised of three sections.

- The **Suspension Input Data** Pane
- The **3D Visualization Pane**
- The **Suspension Output Data** Pane

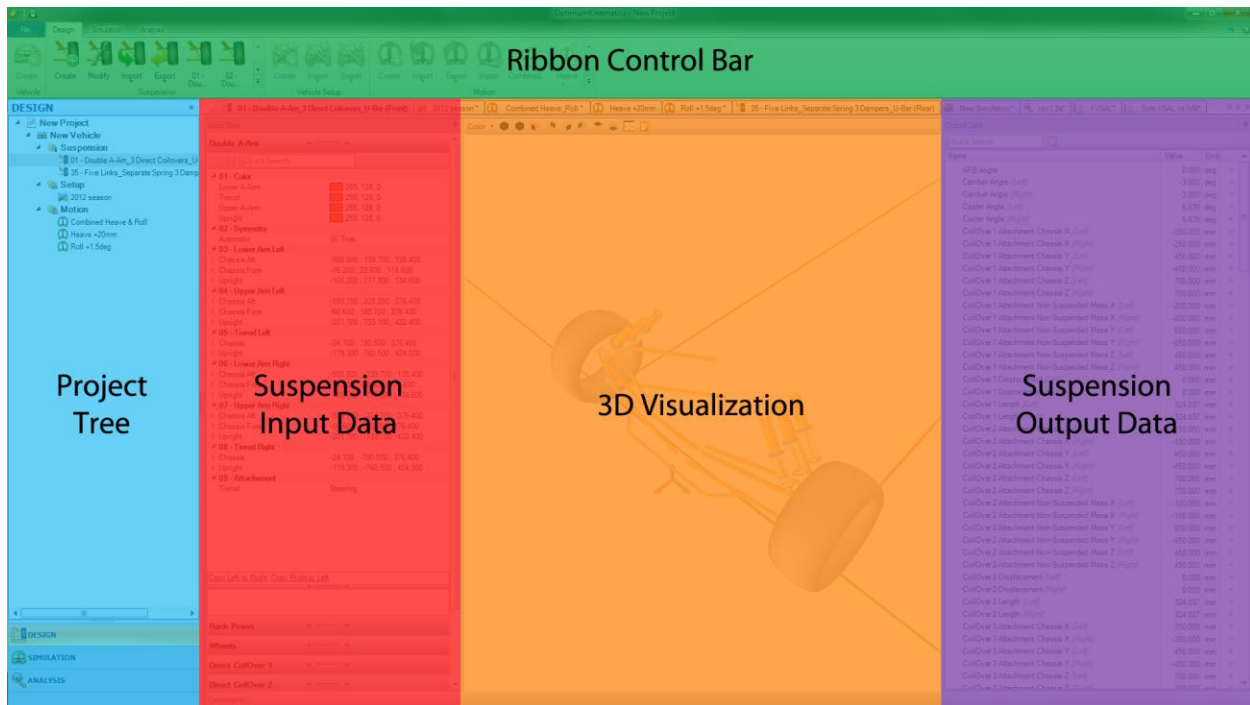


Figure 16 Suspension Design Window

6.2.1.1 Creating a suspension

After a **Vehicle** has been added to the selected project, Selecting **Create Suspension** from the **Ribbon Control Bar** will allow you to define a suspension setup – This can be seen in the following figure.

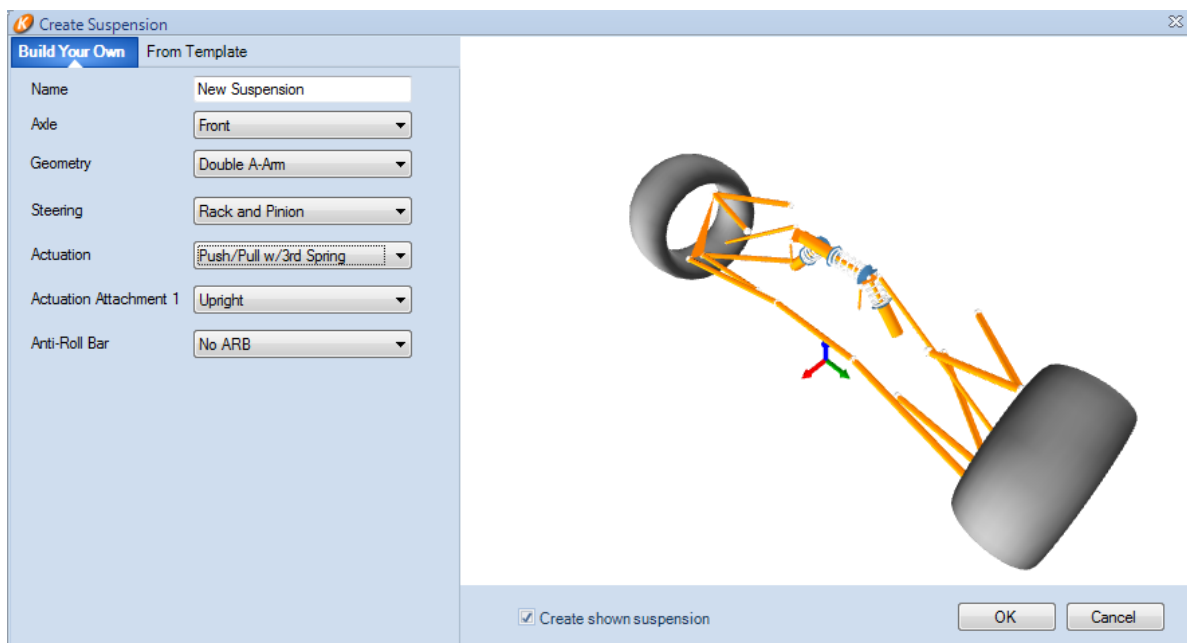


Figure 17 Creating a Suspension

The options that are available are as follows:

- Axle.
 - Front.
 - Rear.
- Geometry.
 - Double A-Arm.
 - Mac Pherson.
 - Mac Pherson Pivot Arm (**Front Only**).
 - Five Links (**Rear Only**).
 - Live Axle, 2 A-Arms (**Rear Only**).
 - Live Axle, 2 Trailing Arms w/ Panhard Bar (**Rear Only**).
 - Live Axle, 4 Trailing Arms w/ Watts Linkage (**Rear Only**).
- Steering (**Front only**).
 - Rack and Pinion.
 - Recirculating Ball.
- Actuation.
 - Direct CoilOver.
 - Separate Springs/Dampers.
 - Push/Pull.
 - Torsion Bar.
 - MonoShock Rotational.
 - MonoShock Slider.
 - Push/Pull w/ 3rd Spring.
- Number CoilOvers. You are able to have up to 3 CoilOvers.
- Actuation Attachment. Depending on the number of CoilOvers, or Springs/Dampers you will be able to choose the attachment
 - Upright.
 - Lower A-Arm.
 - Upper A-Arm.
 - Chassis (**Rear Only**).
 - Axle (**Rear Only**).
- Anti-Roll Bar.
 - U-Bar.
 - U-Bar Rocker.
 - T-Bar.
 - T-Bar w/ 3rd Spring.

6.2.1.2 Input Data

After a suspension has been created, additional suspension parameters can be entered in the Suspension Input Data pane. This pane defines all of the input parameters for a given suspension, including the location of the end points for all suspension members, steering geometry properties, wheel and rim information and any non-suspension reference points of your choosing, such as center of gravity or lowest bodywork points.

The location of each point can either be given as a list of semicolon (;) separated x, y, z points (IE - x;y;z) or the input item may be expanded and each x, y, z point entered individually. The values for all points should reflect their location when the car is at static.

Alternatively, a suspension point may be double clicked upon in the 3d visualization window – allowing the x, y, z coordinates to be adjusted directly from the visualization pane. (Figure 18)

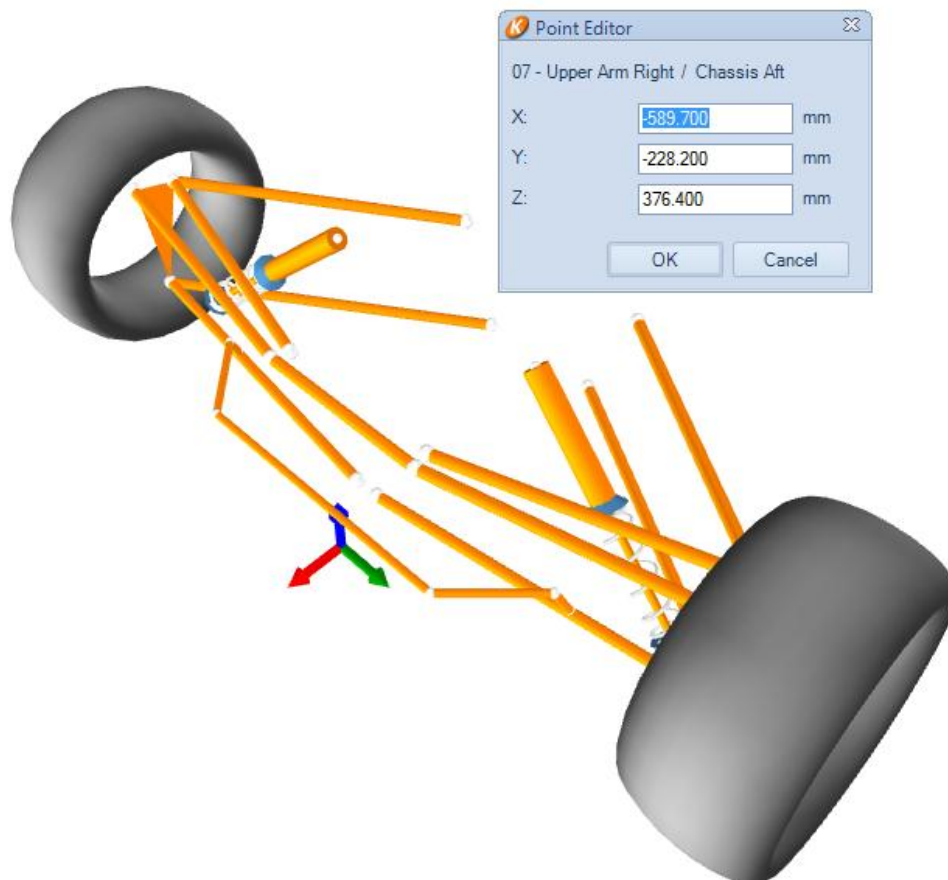


Figure 18 Point Editor

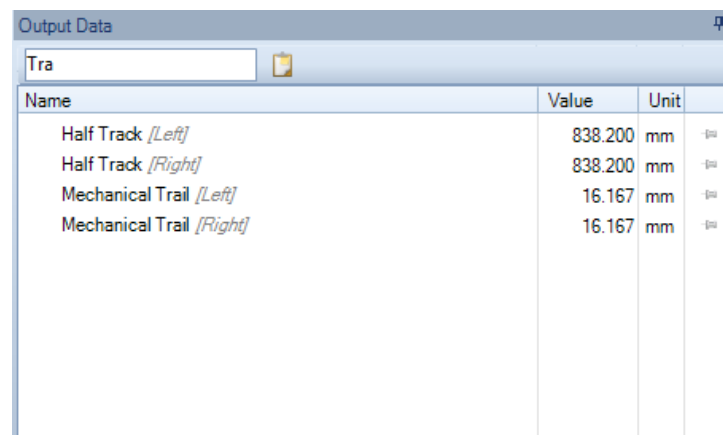
A brief description of what a selected suspension point represent is also shown beneath each section.

OptimumKinematics has many premade front and rear suspension setups to choose from. Within the suspension setups the user maintains the ability to modify any of the existing setups or create a suspension setup from scratch.

6.2.1.3 Output Data

After the information on the input tab has been completed, the corresponding information regarding the newly create suspension is available under the output tab.

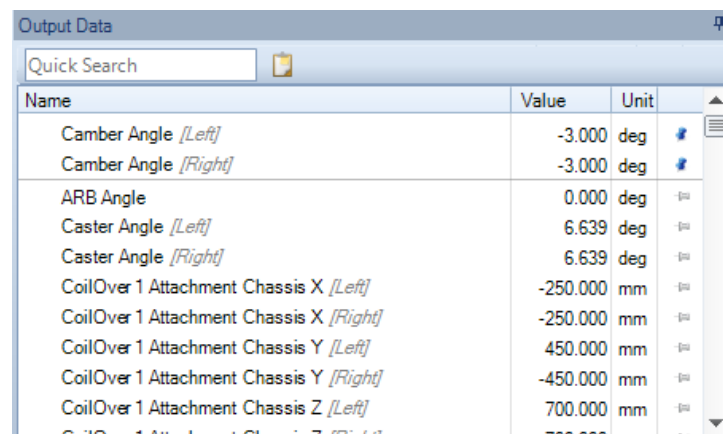
Output channels can be quickly sorted through, via the quick search box. Search results will be displayed if a channel contains the search string anywhere inside the channel name.



Name	Value	Unit	
Half Track <i>[Left]</i>	838.200	mm	
Half Track <i>[Right]</i>	838.200	mm	
Mechanical Trail <i>[Left]</i>	16.167	mm	
Mechanical Trail <i>[Right]</i>	16.167	mm	

Figure 19 Output Data (Quick Search)

Output items of interest may be 'pinned' to the top of the list, ensuring that they are easier to find at all times.



Name	Value	Unit	
Camber Angle <i>[Left]</i>	-3.000	deg	
Camber Angle <i>[Right]</i>	-3.000	deg	
ARB Angle	0.000	deg	
Caster Angle <i>[Left]</i>	6.639	deg	
Caster Angle <i>[Right]</i>	6.639	deg	
CoilOver 1 Attachment Chassis X <i>[Left]</i>	-250.000	mm	
CoilOver 1 Attachment Chassis X <i>[Right]</i>	-250.000	mm	
CoilOver 1 Attachment Chassis Y <i>[Left]</i>	450.000	mm	
CoilOver 1 Attachment Chassis Y <i>[Right]</i>	-450.000	mm	
CoilOver 1 Attachment Chassis Z <i>[Left]</i>	700.000	mm	
CoilOver 1 Attachment Chassis Z <i>[Right]</i>	700.000	mm	

Figure 20 Output Data (Pinned)

6.2.2 Modify Suspension

Modifying suspension geometry allows you to ensure that the geometry created matches that of your car.

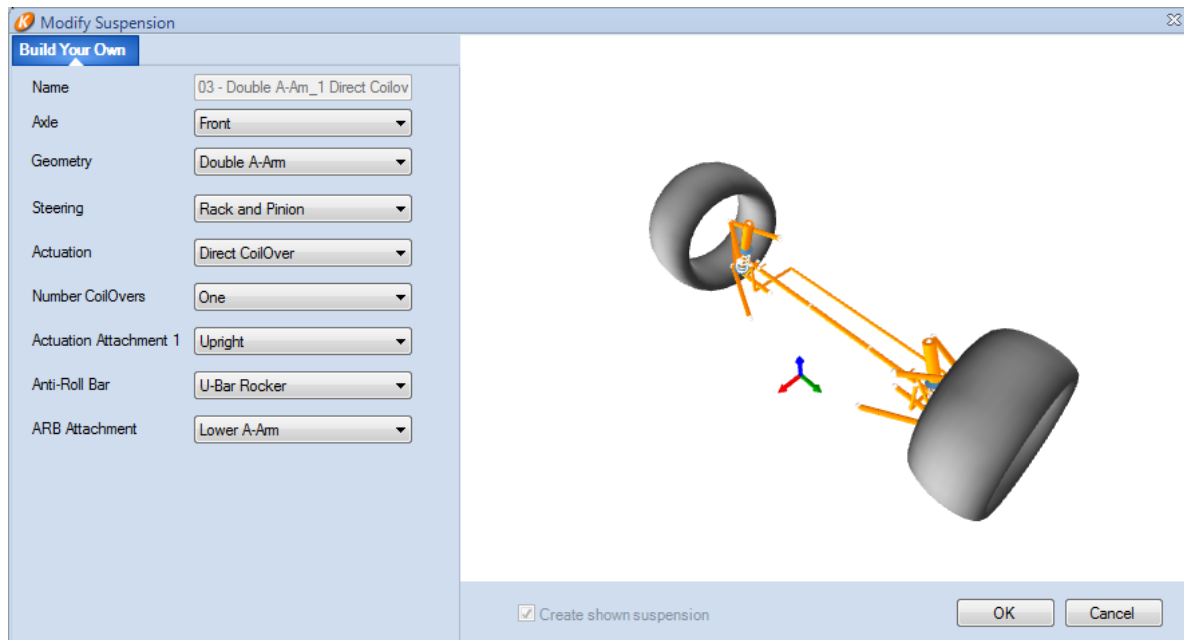


Figure 21 Modifying a Suspension

6.2.3 3D Visualization

The **document manager** will show a 3d visualization of the modeled suspension system. When designing a front or rear suspension, it will only display that axle. When entering a vehicle setup, it will display the whole vehicle suspension.

- Left click + Drag: **Rotate**
- Middle click + Drag: **Zoom**
- Right click + Drag: **Pan**
- Right click: **Adjust visualization options**

The following visualization options are available:

- Adjust Background Color
- Projection
 - Parallel
 - Perspective
- Switch to a predefined view:
 - Isometric

- Front
- Back
- Left
- Right
- Top
- Bottom
- Fill mode:
 - Point
 - Wireframe
 - Solid (Default)
- Copy the visualization area to the clipboard
- Fit the suspension visualization to the window.

Predefined views, fitting to window, copying to clipboard and adjusting the background color are also accessible from the toolbar at the top.

6.2.4 Importing and Exporting Suspension

Suspension configurations can be imported and exported from the **Ribbon Control Bar**. Both OptimumKinematics projects and Excel files can be imported – allowing OptimumK v1 files to be imported into OptimumKinematics through excel file exports from v1.

6.2.5 Creating a Vehicle Setup

Once you have the suspension setup of your choice the next step is to create a vehicle setup. With **Setup** highlighted within the **Design Tree** you are able to **Create** a new vehicle setup.

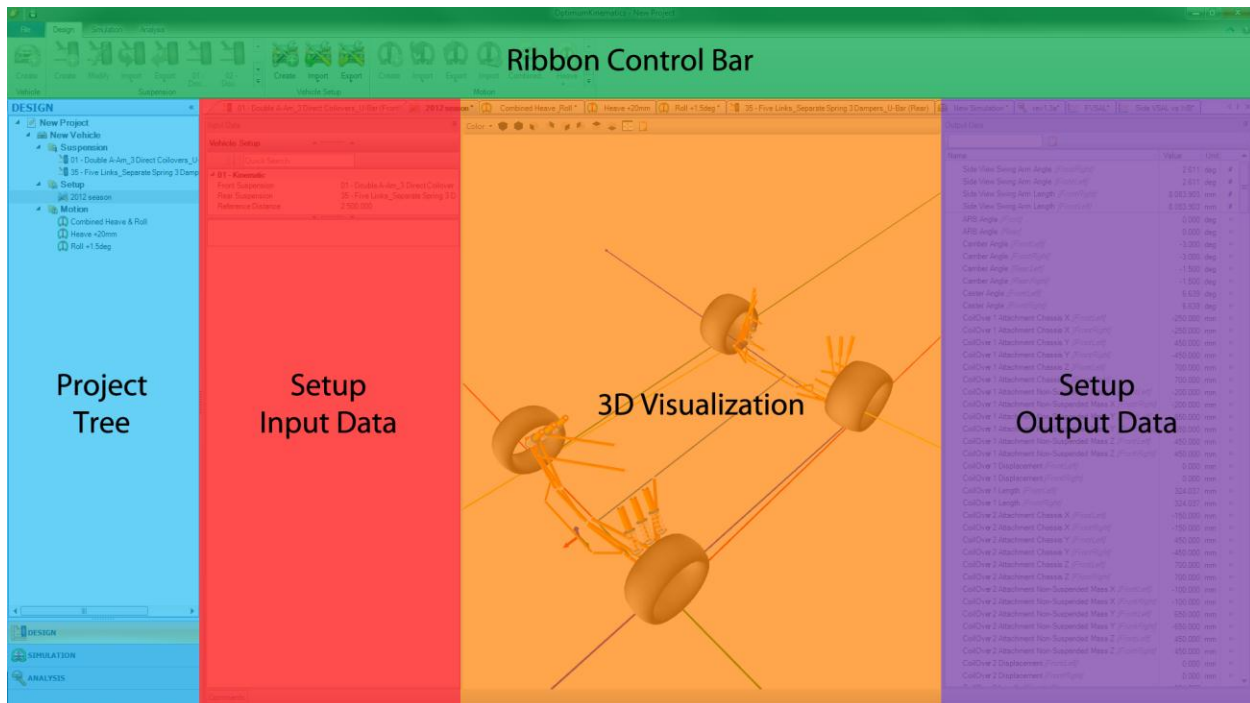


Figure 22 Creating a Vehicle Setup

With the “New Setup” created you can now assign suspension setups to the front and rear of a vehicle and apply a reference distance. It should be noted that this is the distance between the front and rear reference planes and is not necessarily the same value as wheelbase.

6.2.6 Importing and Exporting a Vehicle Setup

Use the Import and Export features to save and reuse your vehicle setups. You will notice that when it is time to Export the vehicle setup you have two options for the file type:

- OptimumK Vehicle Setup File.
- Excel File.

The default .O2Veh setup files that can be found inside your project directory, (located on your hard drive) only contain information on which suspension files are in use, and the corresponding reference distance.

To import a vehicle setup you simply highlight **Setup** from the **Design Tree**, click on the **Import** button located on the **Ribbon Control Bar**, select the file that you would like to import and click Open when you are finished.

6.2.7 Creating a Motion

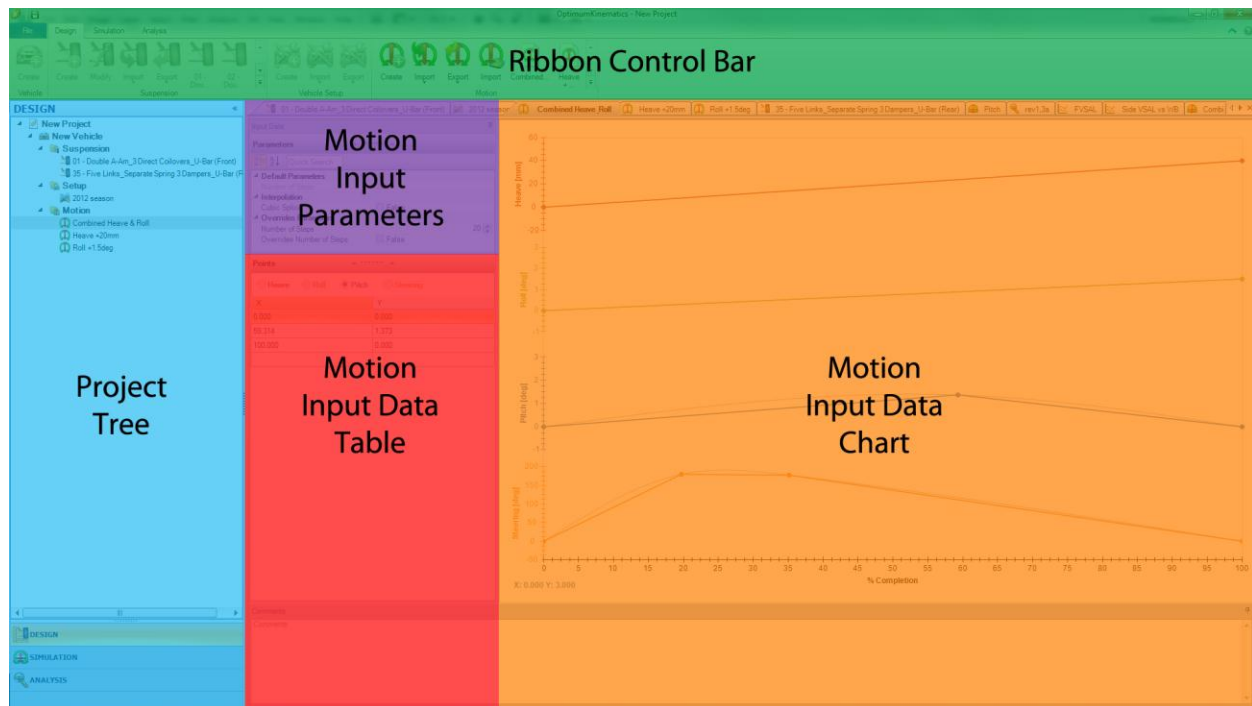


Figure 23 Creating a Motion

After selecting 'Create Motion' you will have the option to add points to the Heave, Roll, Pitch and Steering motion profiles.

Within the **Motion Input Data Chart**:

- Clicking on a Heave, Roll, Pitch or Steering graph will set that as the active motion profile to edit.
- Left click at a point will add a point.
- Right click on a point will delete the selected point.
- Ctrl+Left Click on a point will drag the selected point.

Within the **Motion Input Data Table**:

- Selecting from the Heave, Roll, Pitch, or Steering radio buttons will set that as the active motion profile to edit.
- Right clicking on items in the X,Y table of points will allow a point to be deleted.
- Points may also be copied to and from excel

The user also has the option to interpolate the motion, resulting in smoother motion paths. The user may also override the number of steps in the simulation.

6.2.8 Importing and Exporting Motion Profiles

The **Ribbon Control Bar** contains options to either Import motion files from other OptimumKinematics projects or previously exported data in either an OptimumKinematics or Excel format.

Important: Make sure that the units that are in the File that you are importing match the units you have selected in the **Options Menu**.

Here is an example of what an exported OptimumKinematics Motion File would look like. This is the format that OptimumKinematics is looking for when you are importing a Motion File of this type.

	A	B	C	D	E	F	G	H	I
1		Name:	New Motion						
2		Type:	Motion	Created by:		Created on:	1/18/2012 14:23		
3		Version:	2.0.1	Modified by:		Modified on:	1/18/2012 14:48		
4	Comments:								
5									
6									
7									
8	User Options	Length	millimetre	mm					
9		Angle	degree	deg					
10		Coordinate System	[[1,0,0] [0,1,0] [0,0,1]]						
11									
12	Motion	Heave		Pitch		Roll		Steering	
13		X	Y	X	Y	X	Y	X	Y
14		0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
15		14.408	-1.663	23.870	0.427	49.151	1.195	8.610	0.779
16		43.564	-0.412	51.492	-0.422	85.059	-1.204	31.969	-0.624
17		56.777	0.759	65.303	0.683	100.000	0.000	42.285	0.658
18		73.146	-1.118	81.586	-0.634			76.471	-0.418
19		100.000	0.000	100.000	0.000			100.000	0.000

Figure 24 OptimumKinematics Motion File

Once you have selected the proper file, OptimumKinematics will automatically show the Motion in the **Document Manager**. At this time you are able to modify the values or run the Simulation.

6.2.9 Import Track Data

When importing track data you need not worry about the formatting of the Excel or CSV file. You will manually select the data from the file to ensure that the proper data is assigned to the proper channels.

NOTE - The track import function is only an approximation of the chassis position as it assumes a constant motion ratio, track, wheelbase, etc.

The following options are available for importing track data:

- Utilizing three ride height sensors in known locations.

- Utilizing damper position sensors on a vehicle with known motion ratios.

The following example shows importing track data using ride height sensors; with 2 front sensors and 1 rear sensors. Located on the **Ribbon Control Bar** you will find the Import button for track data.

Once the data file has been opened OptimumKinematics will show in a window very similar to that of Excel.

The next step is to define the layout of your the sensors - this is found when you click on the **Sensor Configuration** button at the top left corner of the data to import.

The **Sensor Configuration** window will now open and let you choose the Sensor Type, Sensor Combination and the ability to insert the distance values. Again, you need to remember to keep matching units between these distances and the chosen units in OptimumKinematics. The units can be changed by accessing the **Options Menu**.

By clicking **OK** in the **Sensor Configuration** window you will be brought back to the **Import Motion** window and you can click on the **Select Data** button to bring you to the **Select Ranges** window shown below.

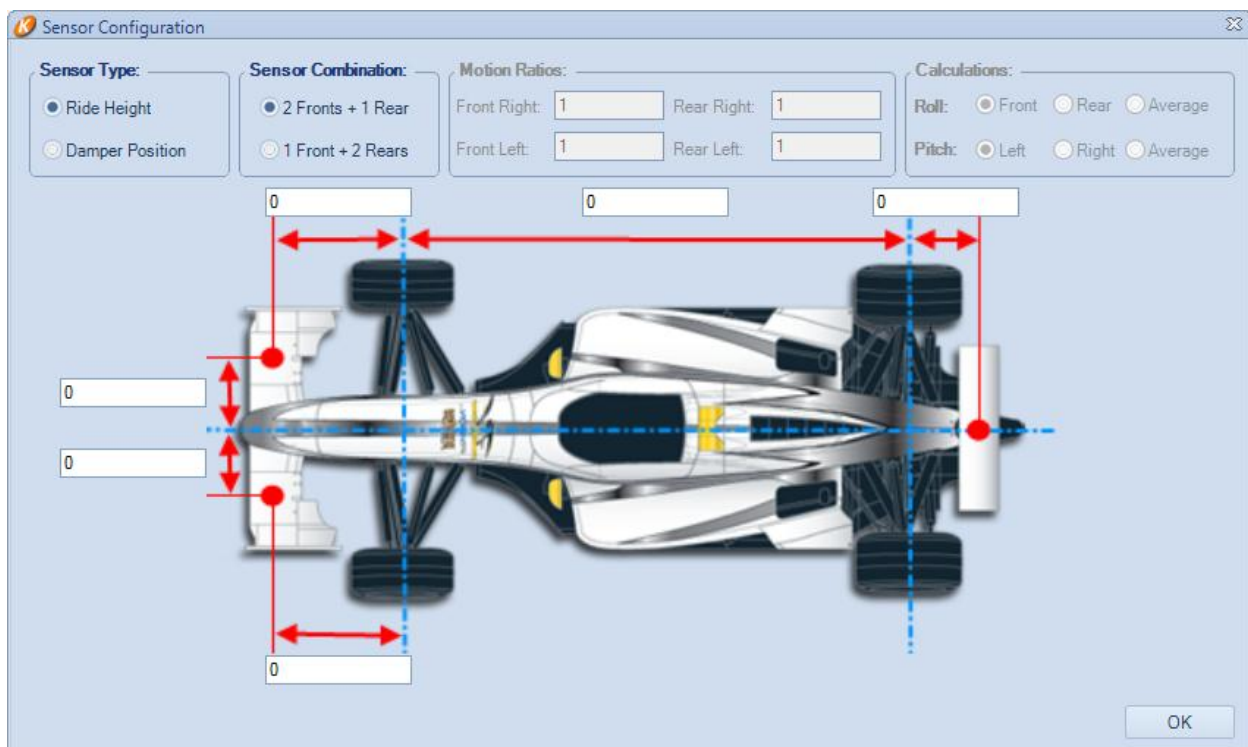


Figure 25 Example Sensor Configuration

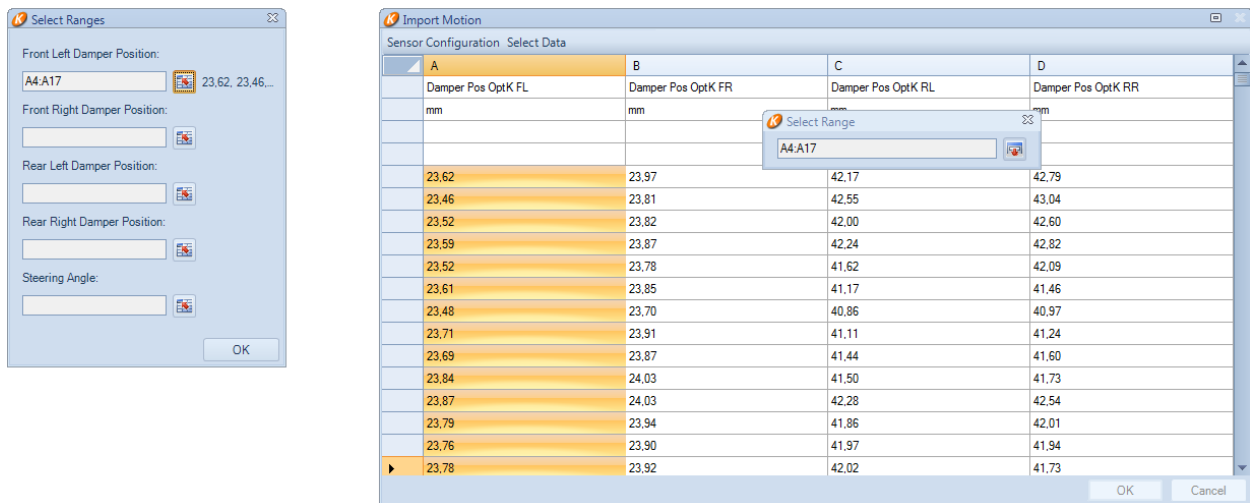


Figure 26 Example Import Motion File

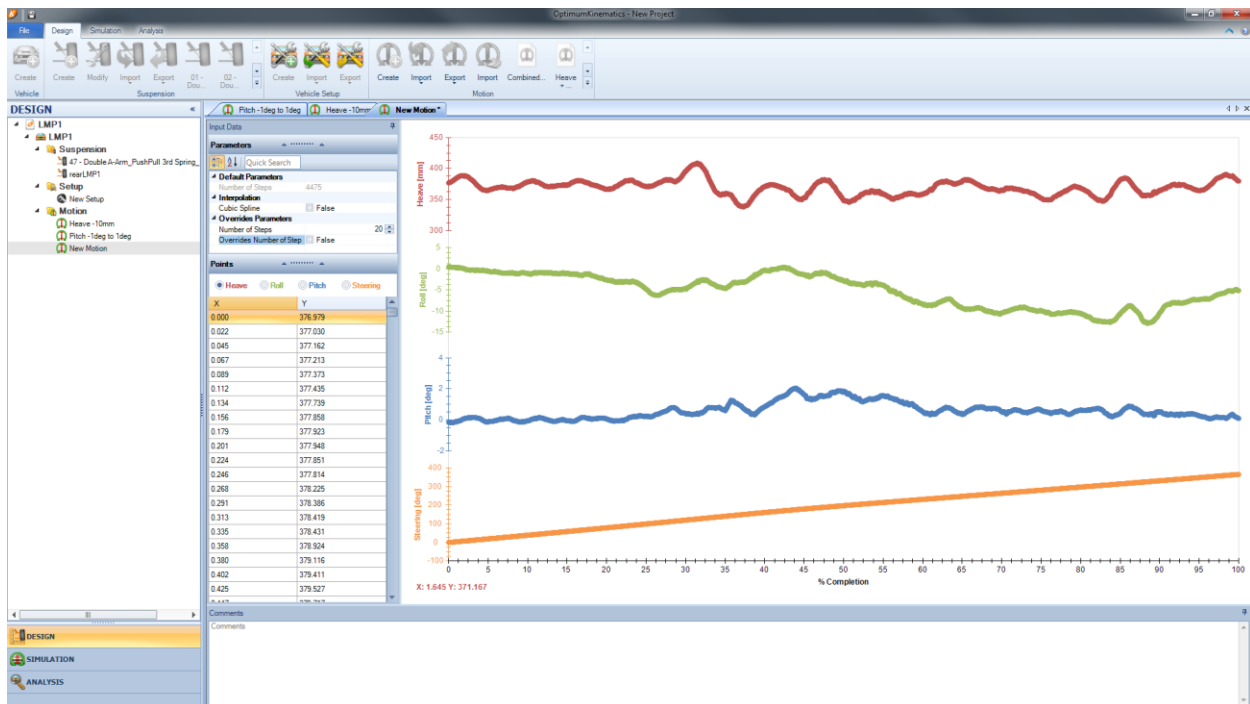


Figure 27 Example Imported Motion

6.3 Simulation

The simulation management settings are a new feature introduced in OptimumKinematics. The simulation **Document Manager** contains just one main section, the **Simulation Input Data** pane, which allows vehicle configurations and motion profile to be mated to each other, and simulations to be run.

Two options are available to create a simulation:

- **Quick Run.** With this option you will need to select the **Vehicle**, **Motion** and **Vehicle Setup** each time.
- **Create/Run.** This option will allow you to re-run the same simulation with the saved **Vehicle**, **Motion** and **Vehicle Setup**.

6.3.1 Quick Run

When using the **Quick Run** feature, you will be asked to define which **Vehicle**, **Motion** and **Vehicle Setup**. This is perfect for getting to analysis in a hurry. If you plan on running the same simulation for multiple different types of motion, it would be beneficial to create the simulation so you can just click Run.

6.3.2 Create/Run

Next step is to **Run** the simulation to acquire the Results. When running the simulation, a progress bar shows the completion of the simulation and 3D Visualization. A simulation can be cancelled or stopped at any time.

NOTE - If you stop the simulation you will be able to view the results up until the stop point, but if you cancel it the results will not be saved.

Figure 28 shows the Batch Run (Multiple Iterations) feature in OptimumKinematics.

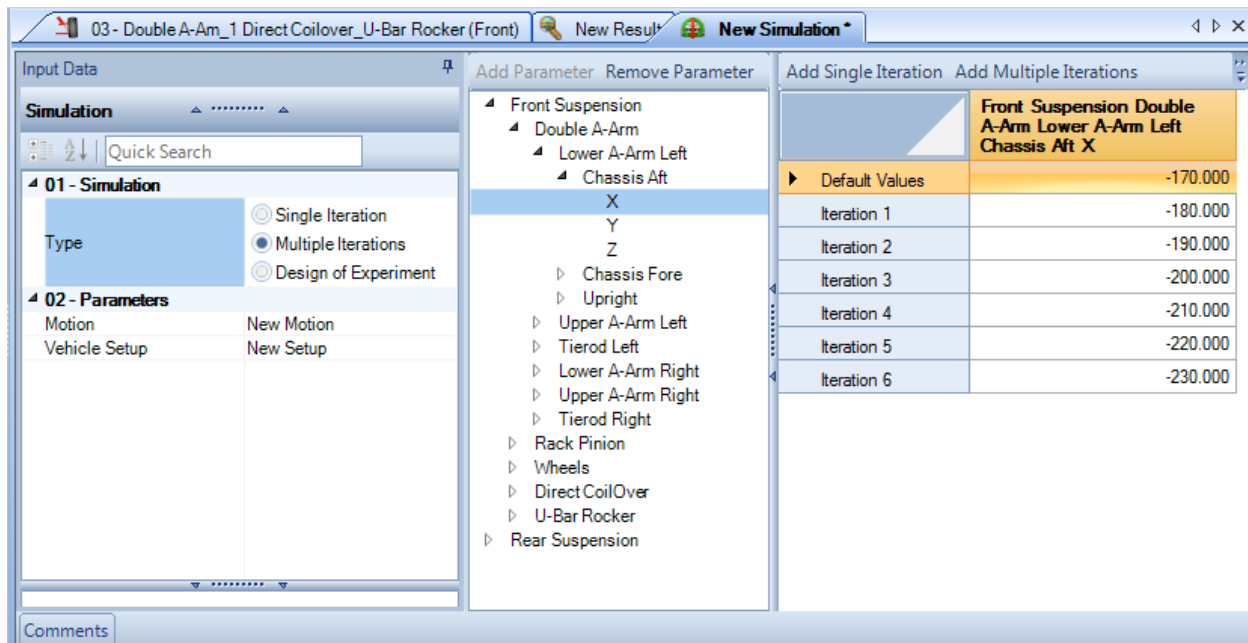


Figure 28 Batch Run (Multiple Iterations)

When running multiple iterations you are able to see how changing the location of one or multiple points can change the location of key features such as roll centers, pitch center and more! You are able to view these results side by side to compare your different design options. Utilizing this feature you are able to have only one **Motion** and one **Vehicle Setup**.

Figure 29 shows the Design of Experiment feature in OptimumKinematics.

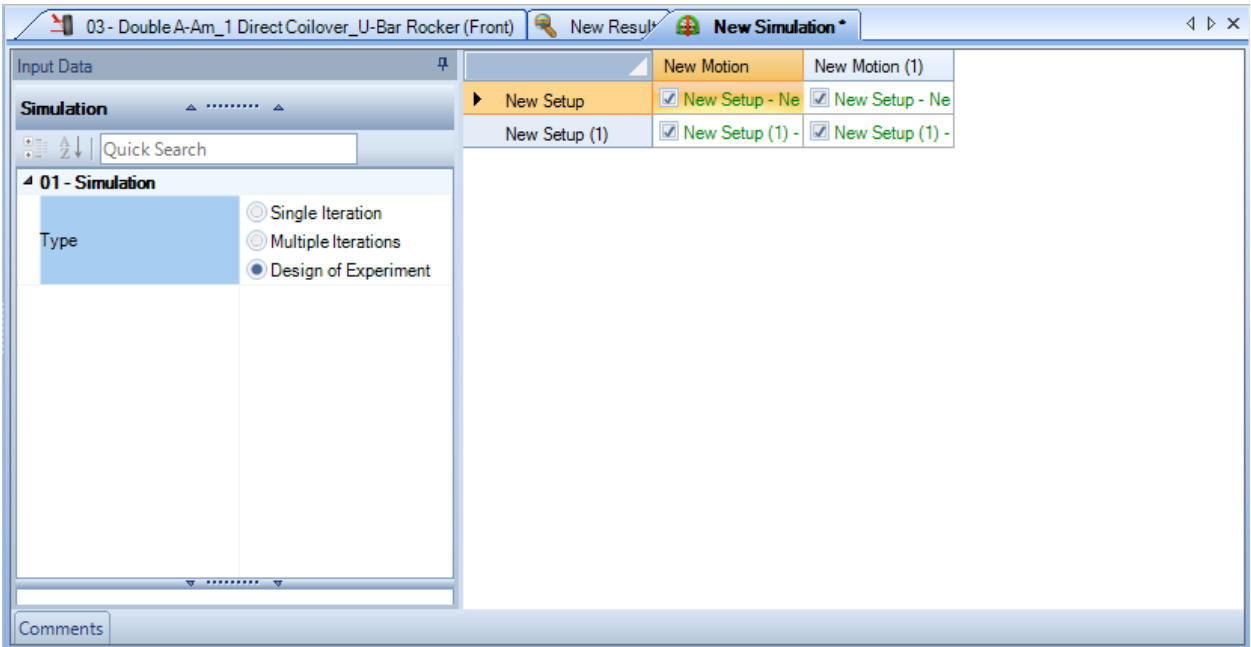


Figure 29 Design of Experiment (DOE)

When running a simulation using the Design of Experiment feature you have the ability to select multiple **Motions** and multiple **Vehicle Setups** and view the results side by side.

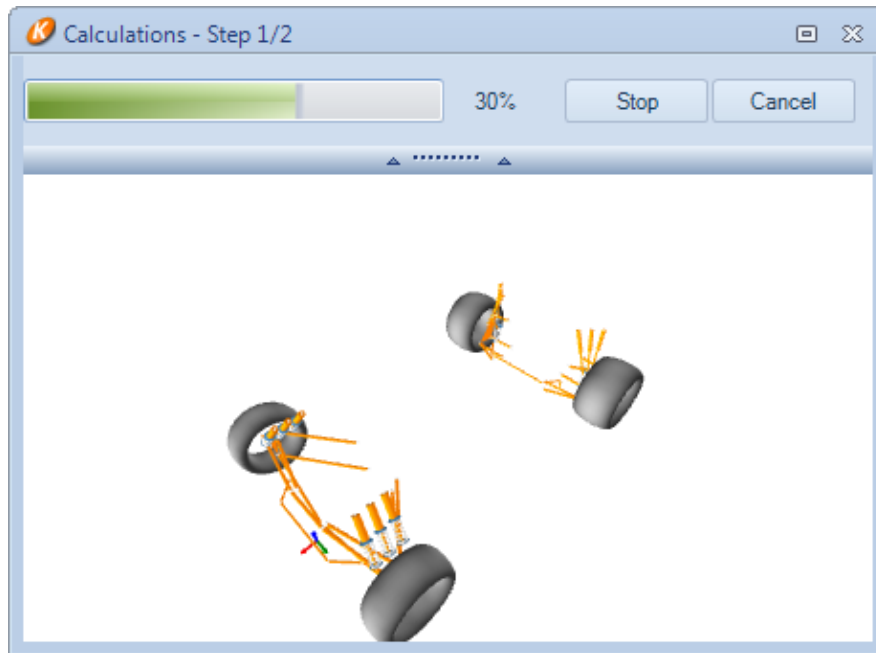


Figure 30 Calculating a Simulation

Once the simulation has finished OptimumKinematics will automatically show the Results Tab in the Results **Document Manager**.

6.4 Analysis

The analysis environment contains two **Project Trees** in the left hand pane; the **Results tree** and the **Reports tree**. Both are detailed below.

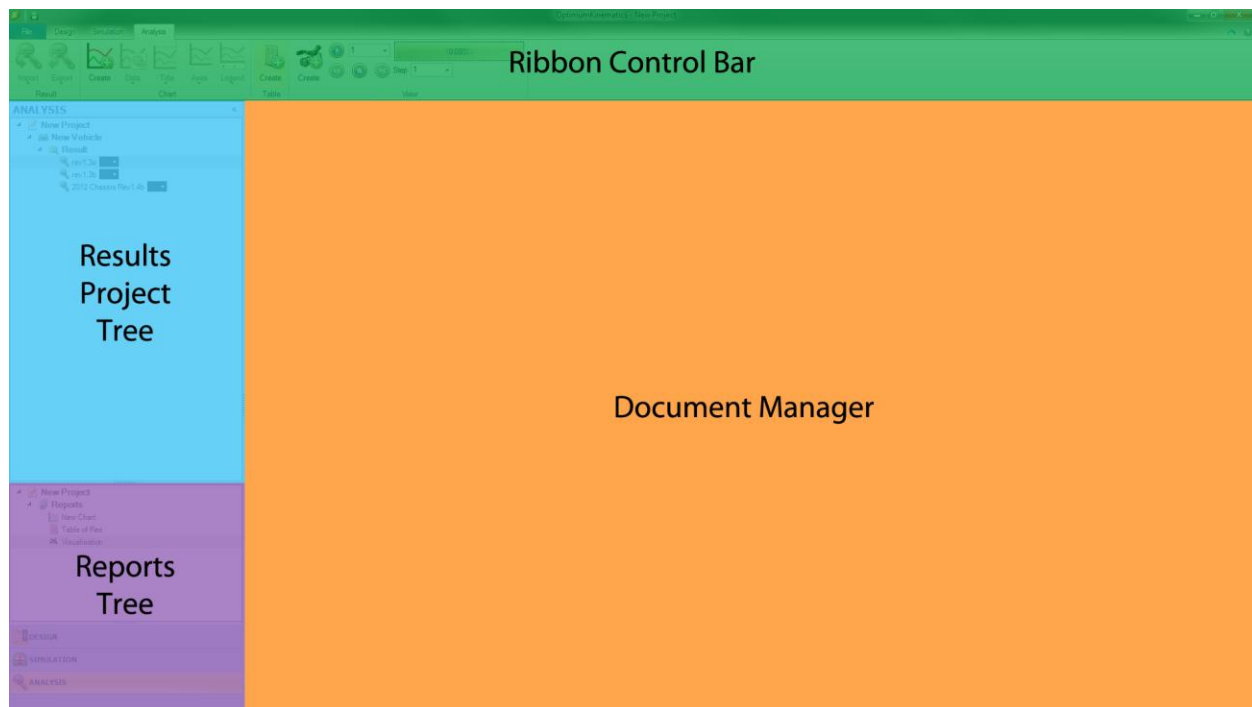


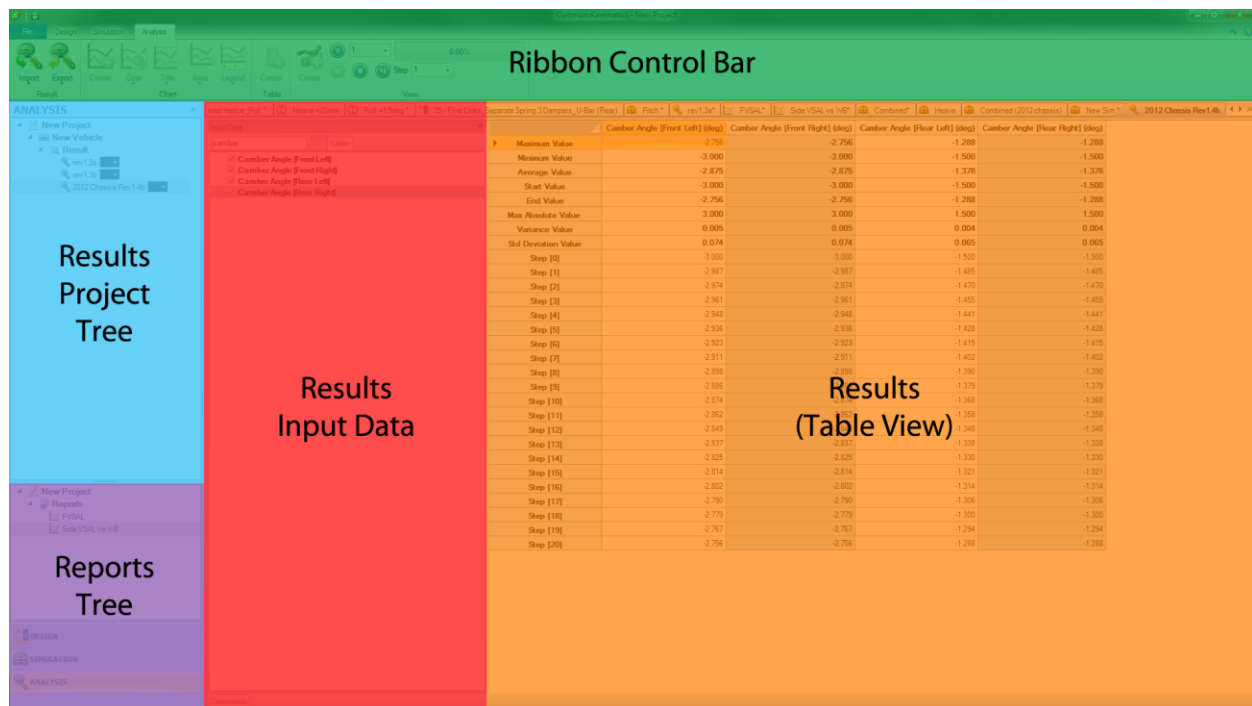
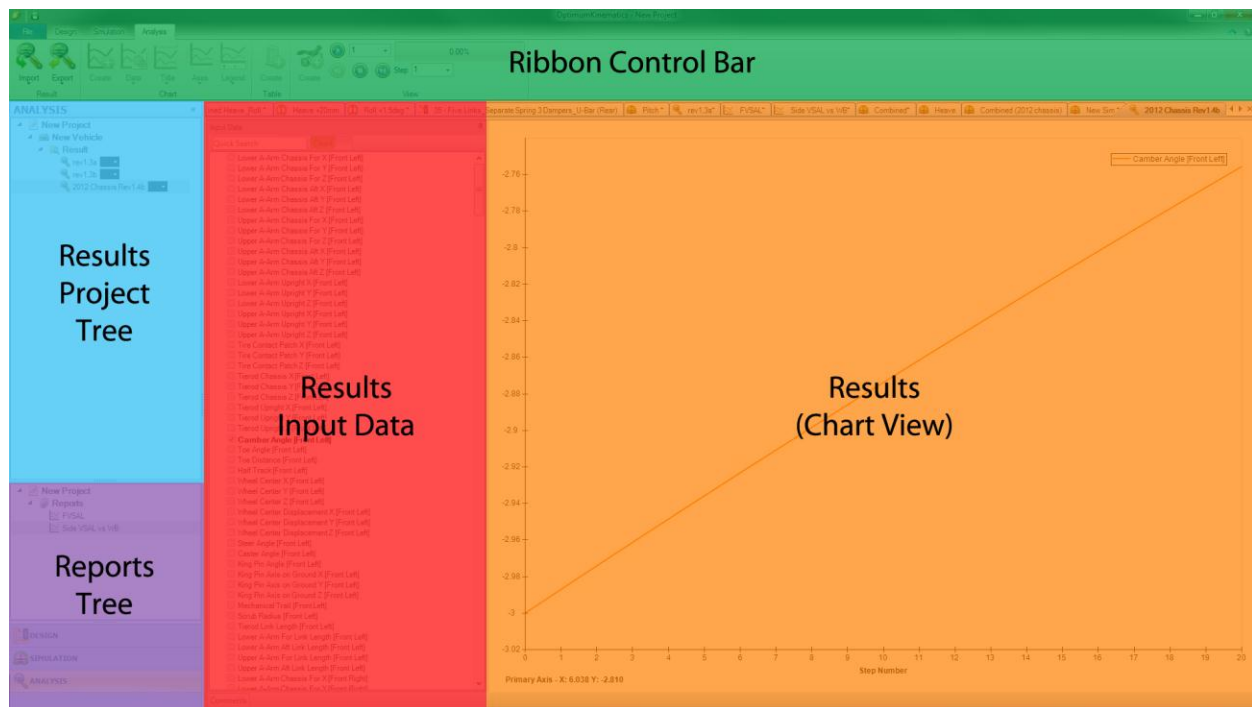
Figure 31 Analysis

6.4.1 Analysis Results

Immediately after a simulation has been run, the results are available to view in the **Document Manager**. For every simulation run, a new set of results is generated.

After selecting a simulations result, the reported channels can be viewed in either chart form or table form. The channels that are reported in this section are global settings - the same channels will be shown here between different simulation results, and different OptimumKinematics projects.

For a full list of result channels, see [Available Results Channels](#).



6.4.2 Import/Export Results

Results file can be exported to the following files.

- OptimumKinematics [.O2res]
- Excel [.xlsx and .xls]
- CSV [.csv]

Results from previous simulations may be imported from any .O2Res files.

6.4.3 Analysis reports

The analysis **Reports Tree** contains the reports that have been generated for a project.

The following reports are available:

- 2D Charts
- Tables
- Visualisations

To add a report to a project, select the project in the **Reports Tree**, and add the desired report by selecting create chart, create table or create view.

6.4.3.1 Creating a Chart

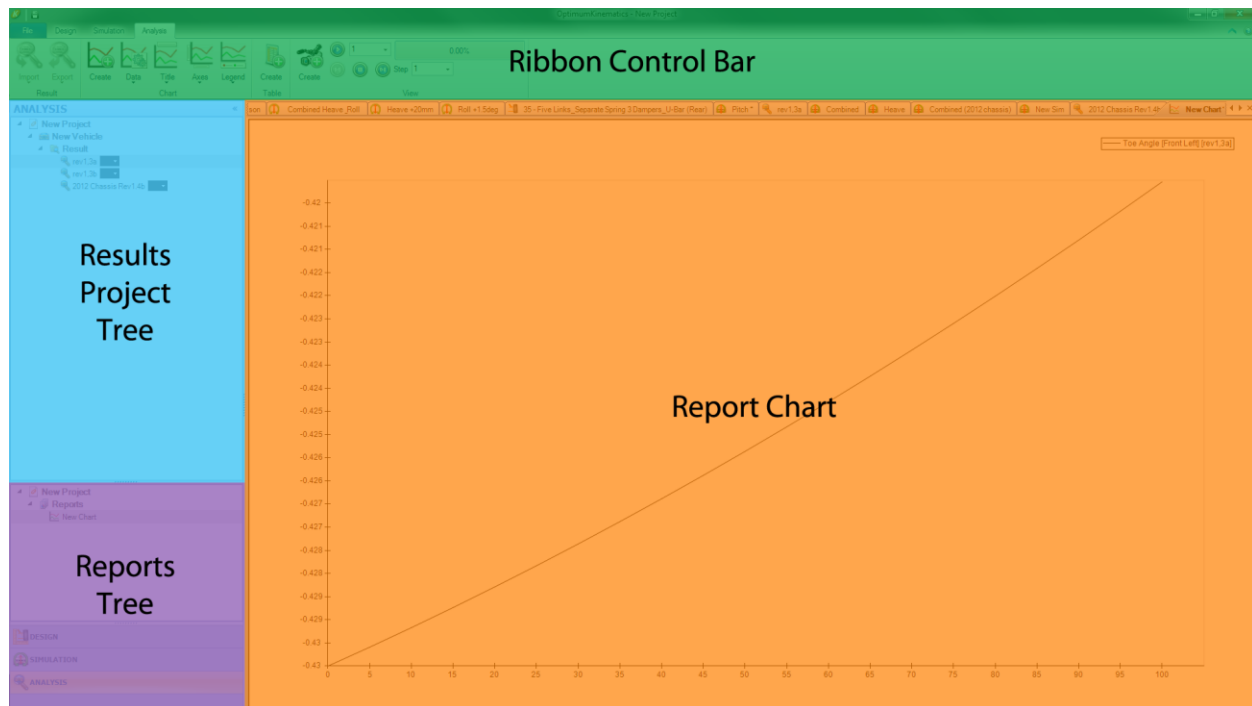


Figure 34 Creating a Chart

Clicking add chart will bring up the Chart Series Editor. A chart may contain multiple series of data.

Each series consists of the following:

- Series Name
- X Channel
- Y Channel
- Selected Results
- Line options
- Marker options

If a chart contains multiple series, it is possible to plot certain channels against a secondary axis.

When graphing a single selected result – the graphed color will be the color selected under ‘line options’, when multiple results are graphed on the same chart, the graphed color will be that of the color nominated under the results **Project Tree**.

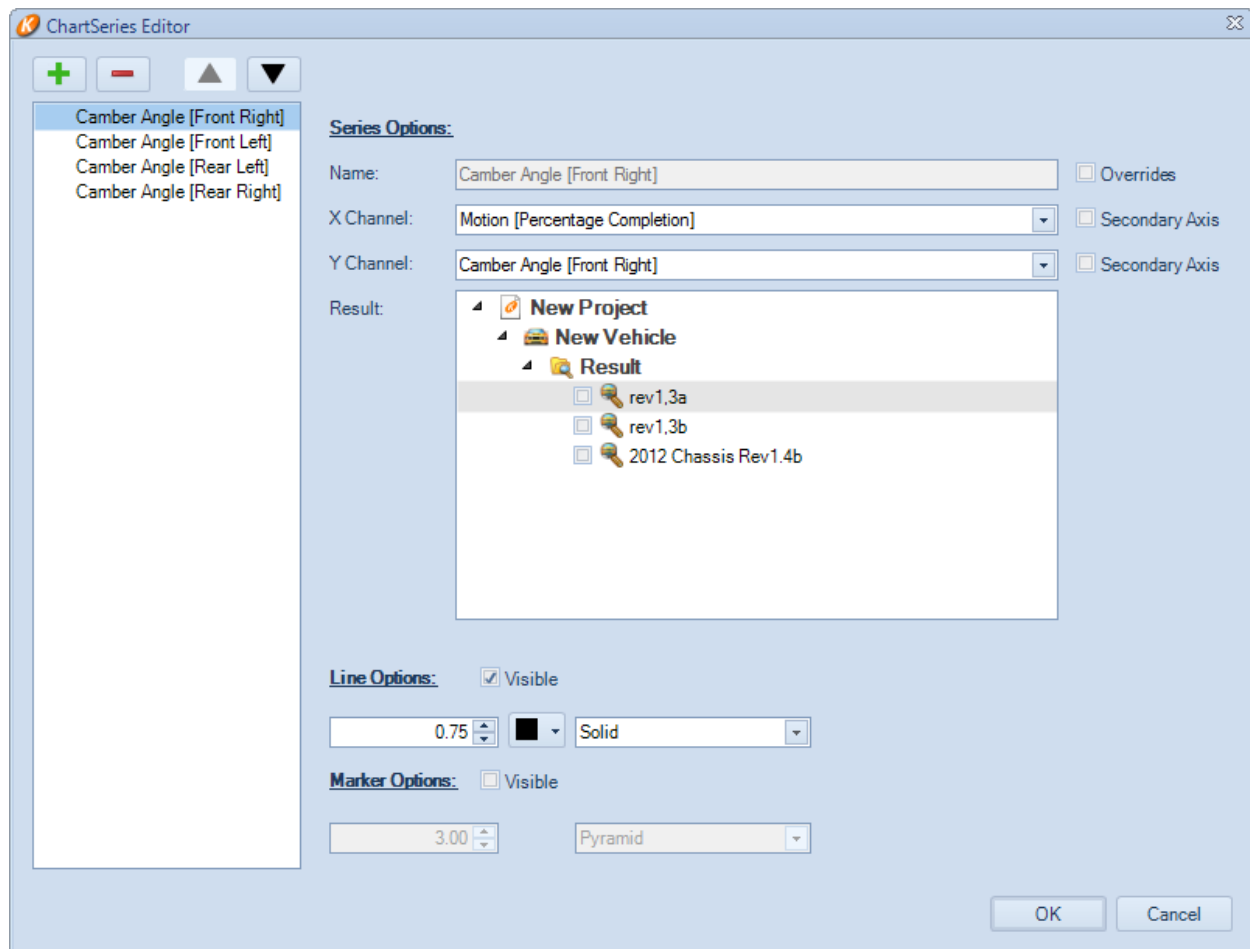


Figure 35 ChartSeries Editor

Each chart can displays results from multiple simulation runs. The selected results can be chosen via the Chart Series editor, via the Data menu option (on the **Ribbon Control Bar**) or by right clicking on the chart.

Axis, Title and Legend options are accessible from the **Ribbon Control Bar**.

The following options are accessible by right clicking on the chart.

- Title
- Legend
- Horizontal Axis
- Vertical Axis
- Tools
 - Zoom/Pan
 - Cursor
- Data

- Add/Remove Series
- Add Result
- Remove Result
- Clear Result
- Copy to Clipboard
- Copy Data to Clipboard
- Save as Picture.

6.4.3.2 Creating a Table

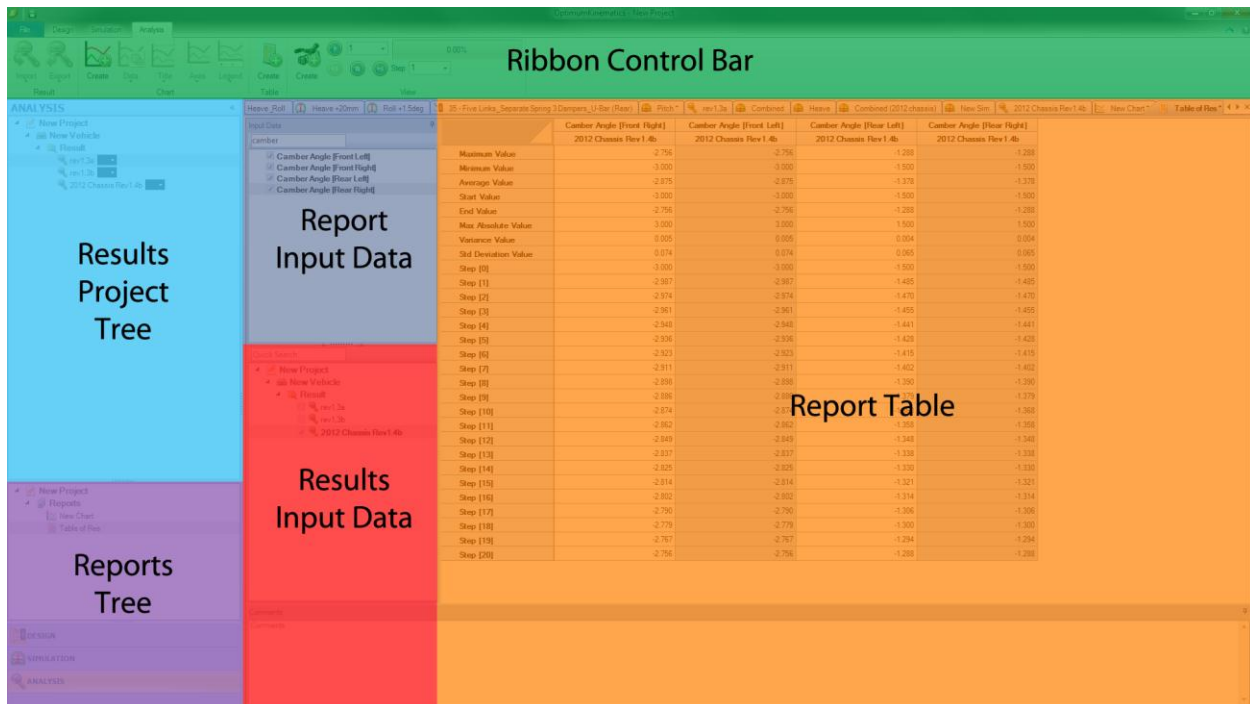


Figure 36 Creating a Table

Adding a table will allow a user to see the numerical values at each step in the simulation.

Tables also include the following calculated values:

- Maximum Value
- Minimum Value
- Average Value
- Start Value
- End Value
- Maximum Absolute Value
- Variance Value
- Standard Deviation Value.

Selecting a channel and a simulation result will display the information in the **Document Manager**. Values for each result will be displayed for each selected channel.

The values shown in a table report can be readily copied to the clipboard, for further processing in Excel or MATLAB®.

6.4.3.3 Creating a View

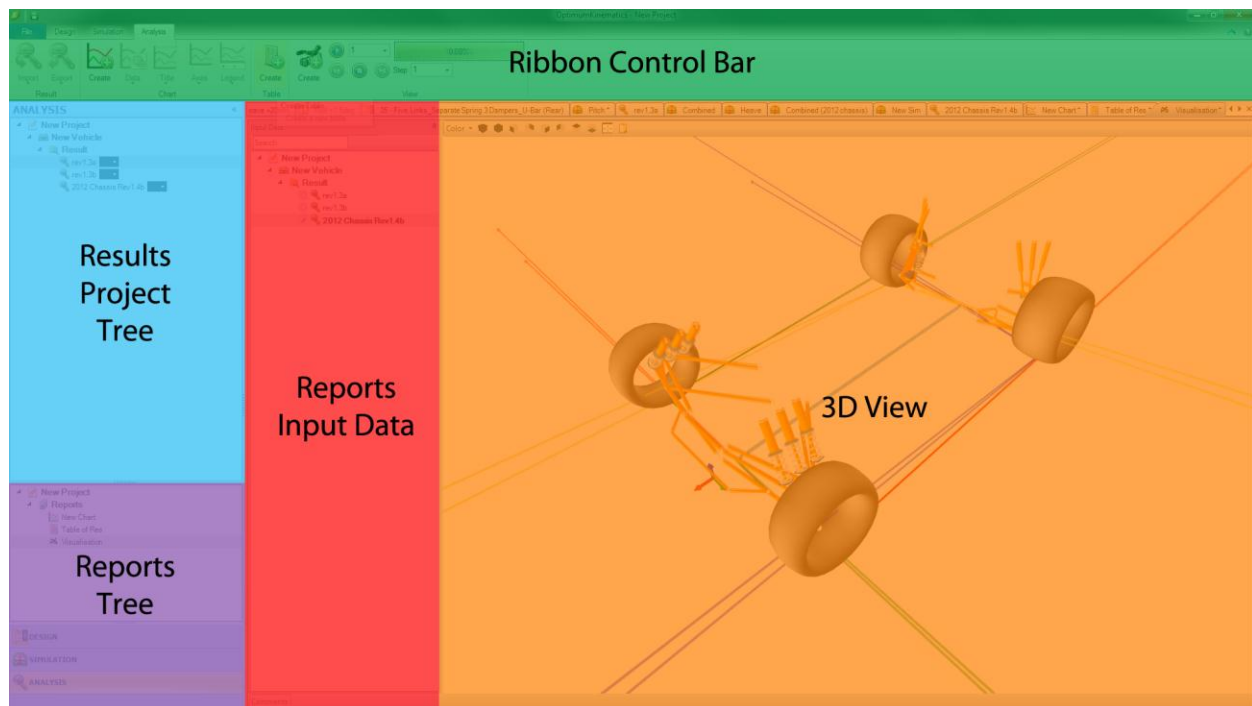


Figure 37 Creating a View

A view is used to visualize the motion of a suspension throughout a given simulation. After selecting 'Create a View', individual result files can be selected and played back. The Reports Input Data pane lists the available simulation results for visualization.

6.4.3.4 Animating Charts, Tables and 3D Views



Figure 38 View Playback Controls

The playback controls can be found on the **Ribbon Control Bar**. They allow a user to replay the simulation steps, watching animations of the key values or the whole suspension system through its range of motion. The playback speed and the step size can be adjusted, and individual frames can be navigated through. Selecting the progress bar directly will allow the user to either skip or pan to a specific stage of the motion completion.

7 Frequently Asked Questions

7.1 I lost my License Key

License keys can be retrieved from your account on our client services center, <http://www.login.optimumg.com>. Account details for the client services center can be found in the confirmation email sent when OptimumKinematics was purchased. If you require further assistance please contact softwaresupport@optimumg.com

7.2 How many computers can I install the software on?

A license key allows you to use the software on a single computer, but it is possible to have the software installed on multiple computers. If the software is installed on multiple computers, it cannot be used by more than one computer at any one time. This process is managed via activating and deactivating a license key manually, which does require an internet connection.

7.3 I have no internet access to activate my license key.

If a license key is attempting to be activated (or reactivated), and a license key cannot be found, a user will have a small grace period to connect the computer to the internet. During this grace period, the program will be fully functional, but will only be able to start for a further five starts.

7.4 I can't install OptimumKinematics correctly.

OptimumKinematics must be installed to a location that you have permission to install to. If you are unable to install OptimumKinematics due to permission issues, please contact your network administrator.

7.5 Can I import suspensions from previous OptimumK Versions?

Yes, OptimumK v1 suspensions may be imported into OptimumKinematics. They must first be exported from OptimumK v1 as an excel file, which OptimumKinematics can then import.

7.6 Why are my instant centers, roll centers or pitch centers not calculated?

This occurs due to the instant centers being located at infinity – as is the case for truly parallel suspension setups. A method of solving this is to slightly offset the inboard or outboard pickup points by a fractional amount of one unit, allowing the instant center location to be solved.

7.7 When running a simulation, I receive an error message stating that solver couldn't meet the required accuracy.

This occurs when the motion paths used in the simulation exceed that of what the suspension system is physically capable of. For instance, if excessively large values for heave, roll, pitch or steering are sent for simulation, the suspension is not capable of that range of motion.

7.8 Can I Import Track Data into OptimumKinematics?

Yes, track data can be imported into OptimumK as a motion path. Data can either come from three ride height sensors located around the car, or from suspension travel data gather from damper position sensors. The imported file format can be either Excel or .CSV files.

7.9 Can I add to the default suspension templates on my Hard Drive?

The location of the default suspension templates is under 'C:\Program Files\OptimumG\OptimumKinematics\Templates'. Adding exported suspension files (*.O2sus) will make these templates readily available to import into a project.

7.10 What is the difference between Geometric and Kinematic Roll Centers?

Geometric roll centers are calculated by taking the intersection between various planes of suspension members to locate the instant centers, then taking an axis from the instant centers to the contact patch. At the point that the instant axis cross, lies the roll center. This differs for the kinematic roll center calculation, as the instant centers are not found from the intersection of the various planes, but are found by iteratively applying a slight disturbance to the wheel center location, allowing movement in all six degrees. The kinematic method accounts for the effects of bump steer and steering, and can be readily applied to any independent suspension type.

7.11 Nothing shows on the 3D Visualization.

If you are using NVIDIA then go under Control panel, Appearance and Personalization. You should be able to open the NVIDIA Control Panel. The window should be similar to Figure 37.

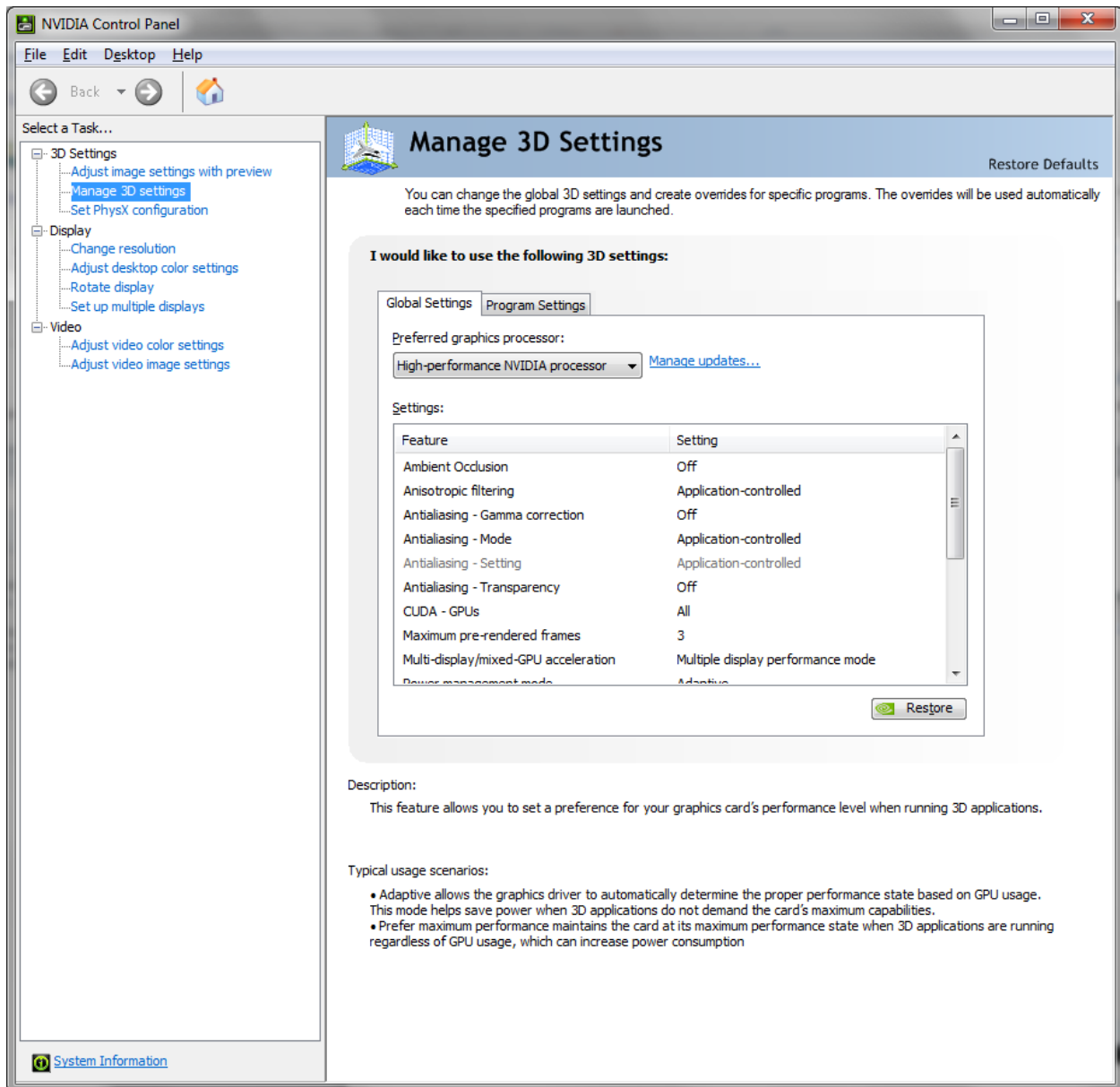


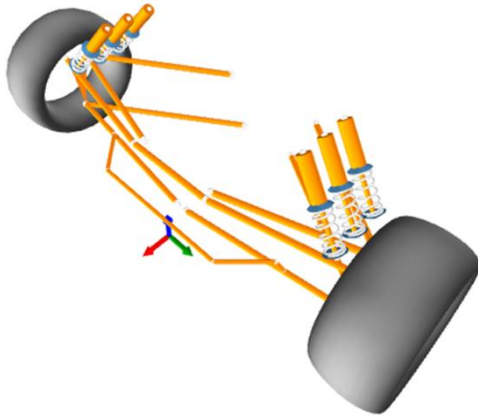
Figure 39 NVIDIA Control Panel

Select 'Manage 3D Settings' on the left tree and select 'High-Performance NVIDIA Processor' for the preferred graphics processor.

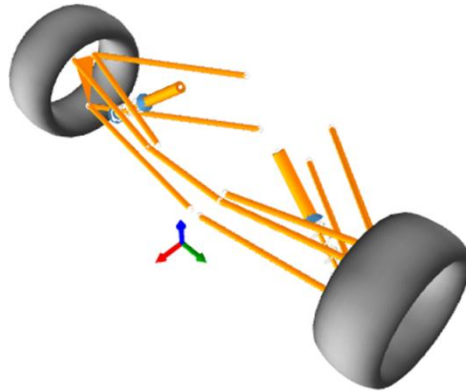
8 Reference & Appendix

8.1 Default Suspension Types

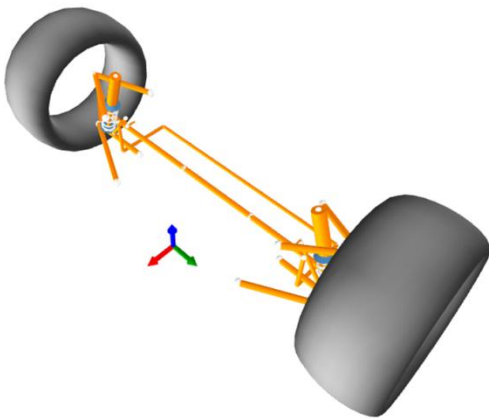
8.1.1 Front



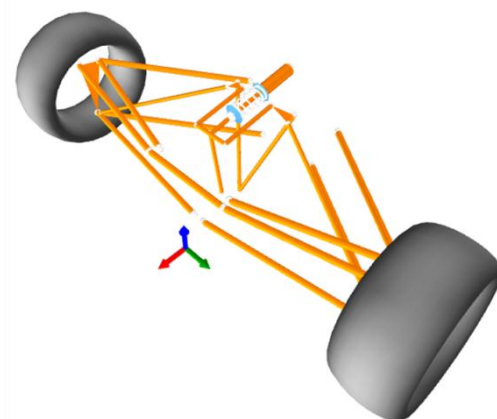
01-Double A-Arm_3 Direct CoilOvers_U-Bar (Front)



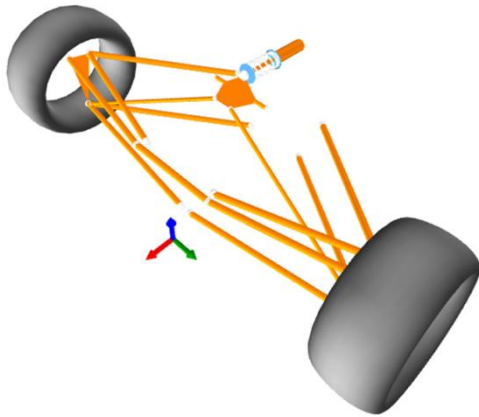
02-Double A-Arm_1 Direct CoilOver_U-Bar (Front)



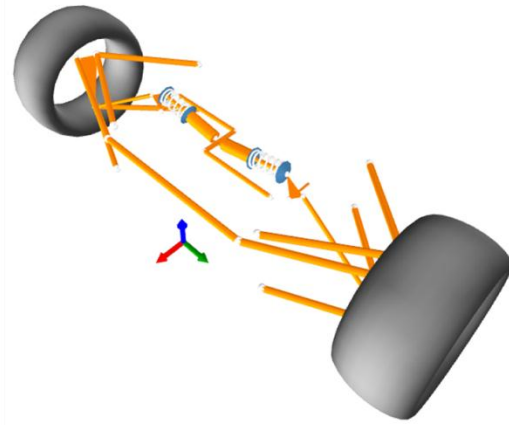
03-Double A-Arm_1 Direct CoilOver_U-Bar Rocker (Front)



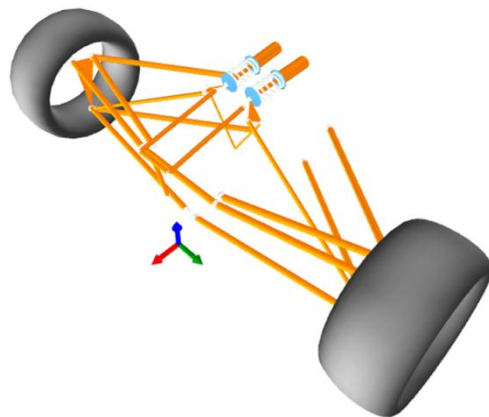
04-Double A-Arm_MonoShock Rotate_T-Bar (Front)



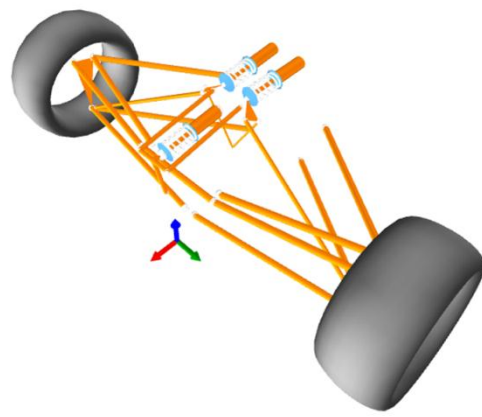
05-Double A-Arm_MonoShock Slide (Front)



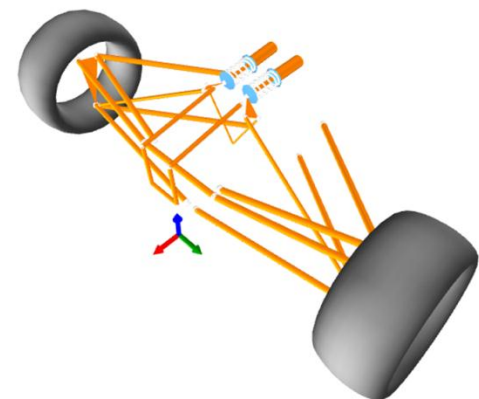
06-Double A-Arm_PushPull_T-Bar Longitudinal (Front)



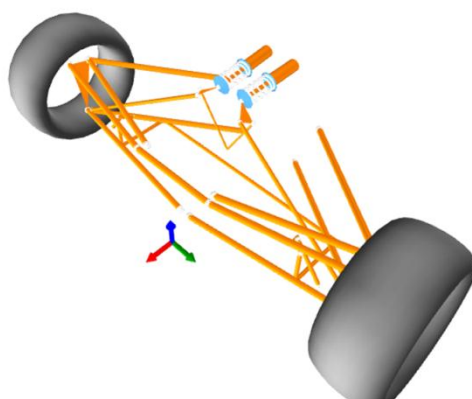
07-Double A-Arm_PushPull_T-Bar (Front)



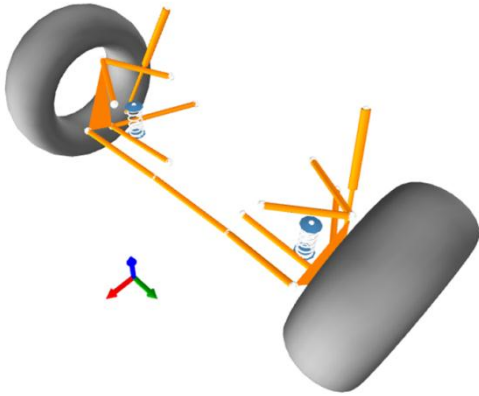
08-Double A-Arm_PushPull_T-Bar 3rd Spring (Front)



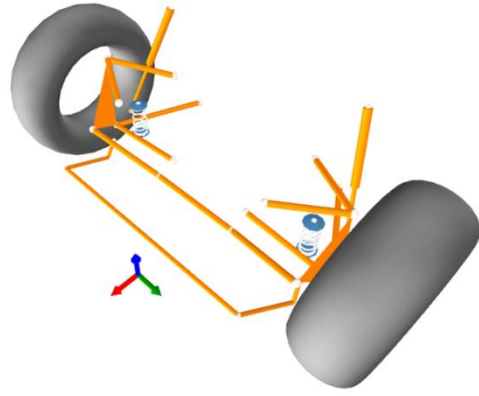
09-Double A-Arm_PushPull_U-Bar (Front)



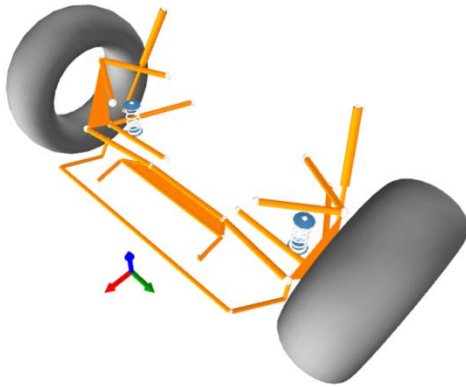
10-Double A-Arm_PushPull_U-Bar Rocker (Front)



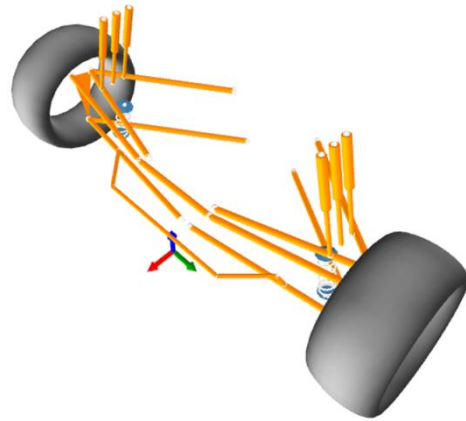
11-Double A-Arm_Seperate Spring 1 Damper (Front)



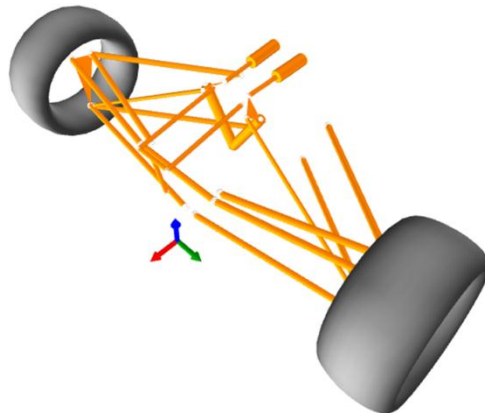
12-Double A-Arm_Seperate Spring 1 Damper_U-Bar (Front)



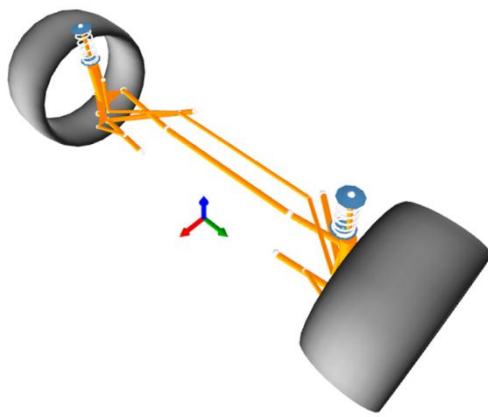
13-Double A-Arm_Seperate Spring 1 Damper_U-Bar_R-Ball (Front)



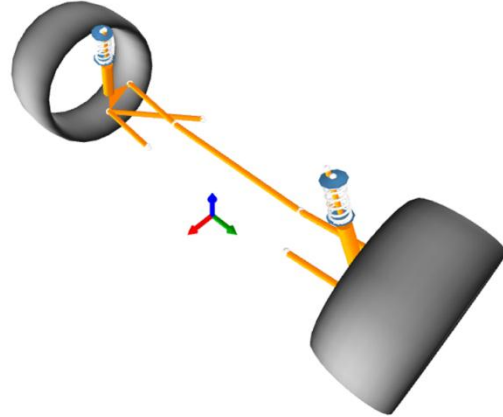
14-Double A-Arm_Seperate Spring 3 Damper_U-Bar (Front)



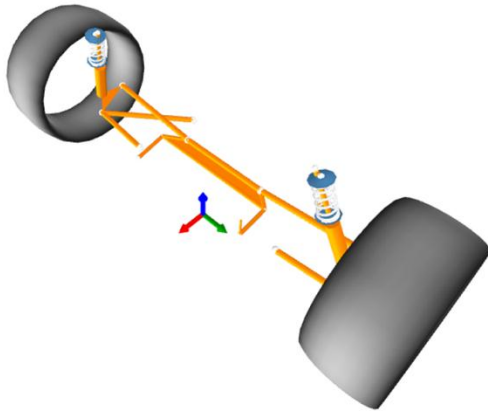
15-Double A-Arm_Torsion Bar_T-Bar (Front)



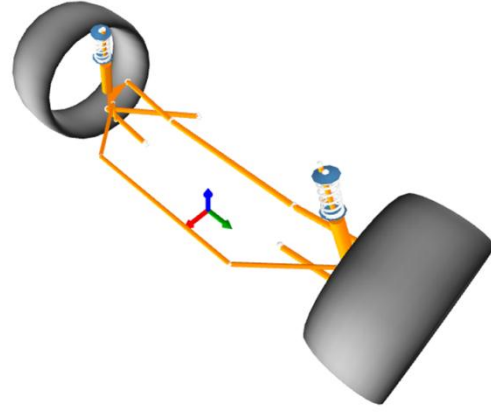
36-Mac Pherson Pivot Arm_U-Bar (Front)



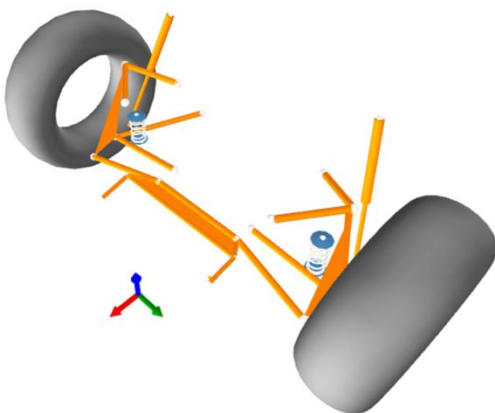
37-Mac Pherson (Front)



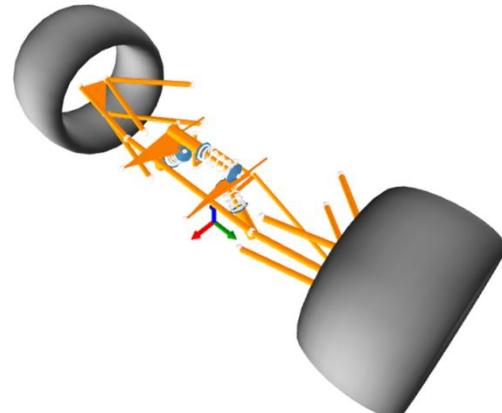
38-Mac Pherson_R-Ball (Front)



39-Mac Pherson_U-Ball (Front)

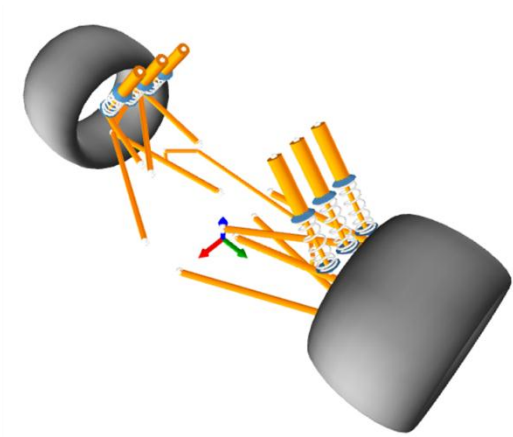


42-Double A-Arm_Seperate Spring 1 Damper_R-Ball (Front)

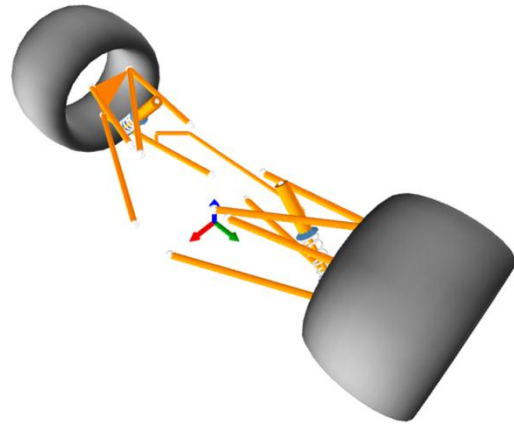


47-Double A-Arm_PushPull 3rd Spring_U-Bar (Front)

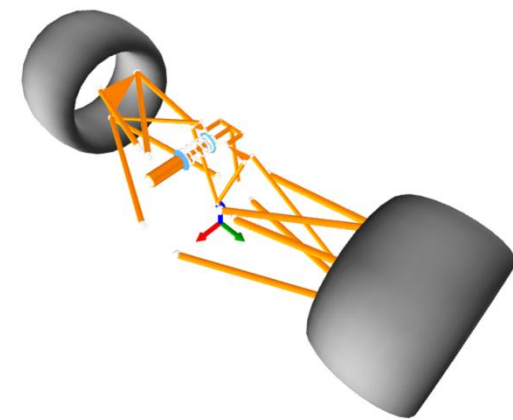
8.1.2 Rear



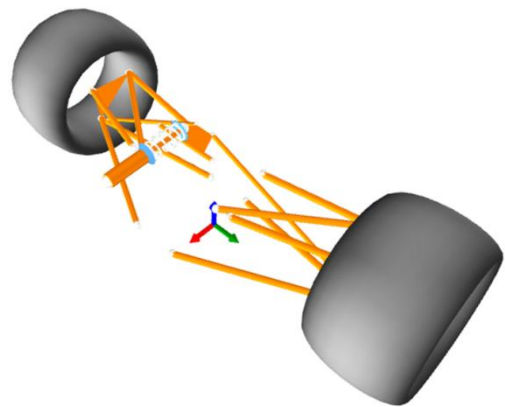
16-Double A-Arm (Chassis)_3 Direct Coilovers_U-Bar (Rear)



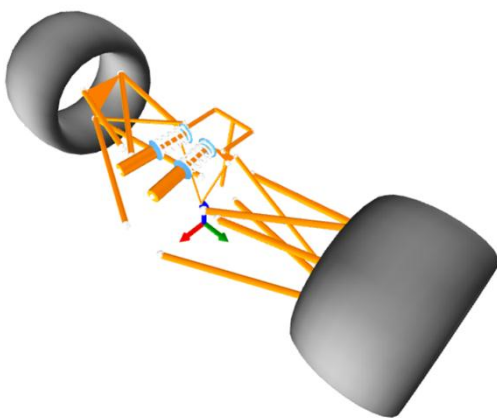
17-Double A-Arm (Chassis)_1 Direct Coilover_U-Bar (Rear)



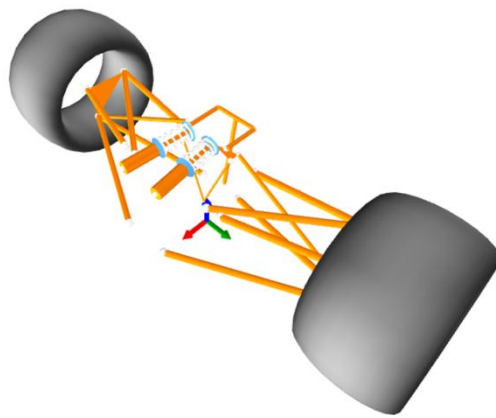
18-Double A-Arm (Chassis)_MonoShock Rotate_T-Bar (Rear)



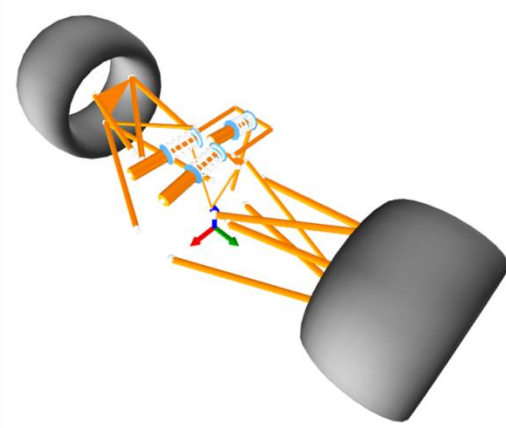
19-Double A-Arm (Chassis)_MonoShock Slide (Rear)



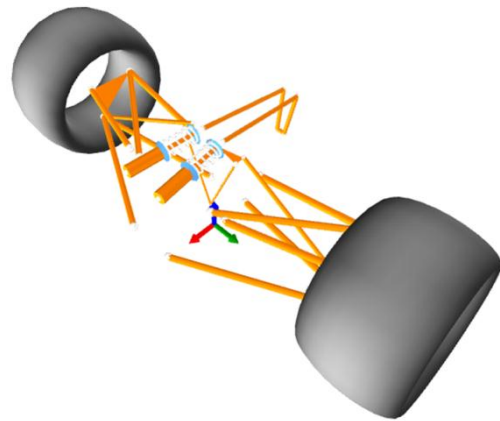
20-Double A-Arm (Chassis)_PushPull_T-Bar (Rear)



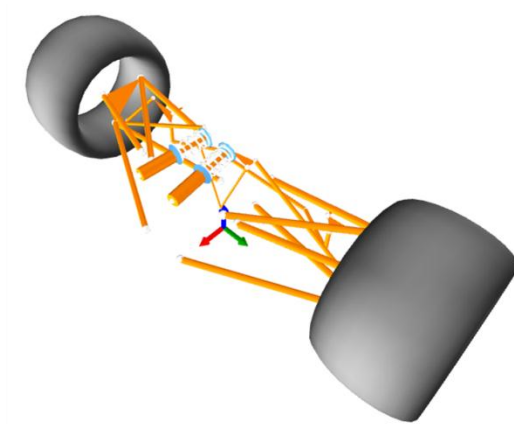
20 Bis-Double A-Arm (Upper A-Arm)_PushPull_T-Bar (Rear)



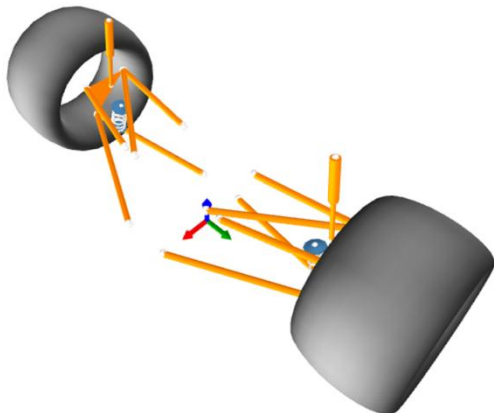
21-Double A-Arm (Chassis)_PushPull_T-Bar 3rd Spring (Rear)



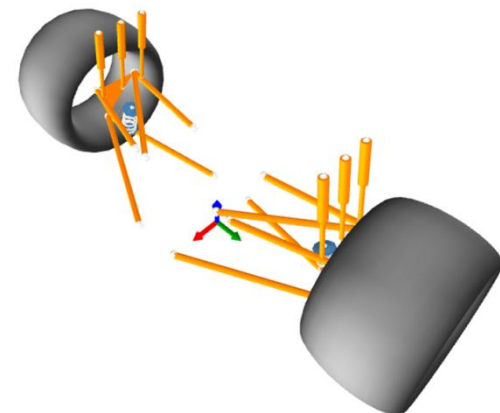
22-Double A-Arm (Chassis)_PushPull_U-Bar (Rear)



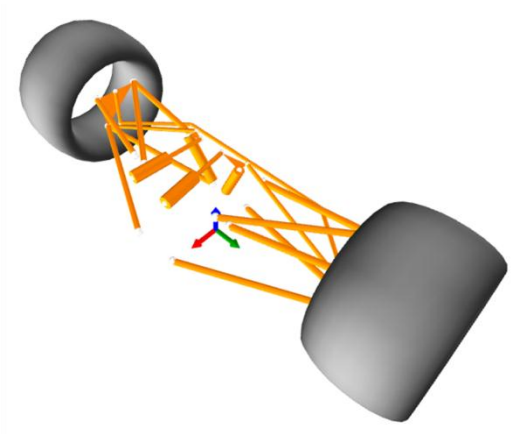
23-Double A-Arm (Chassis)_PushPull_U-Bar Rocker (Rear)



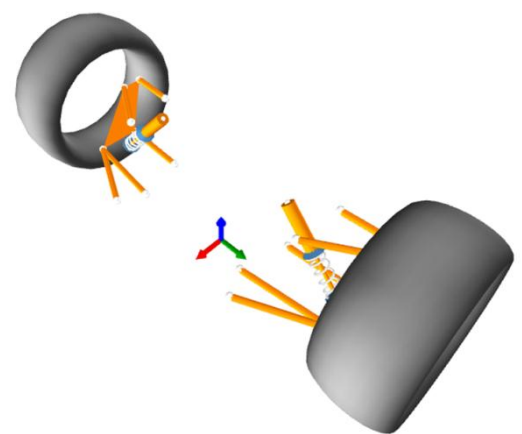
24-Double A-Arm (Chassis)_Seperate Spring 1 Damper (Rear)



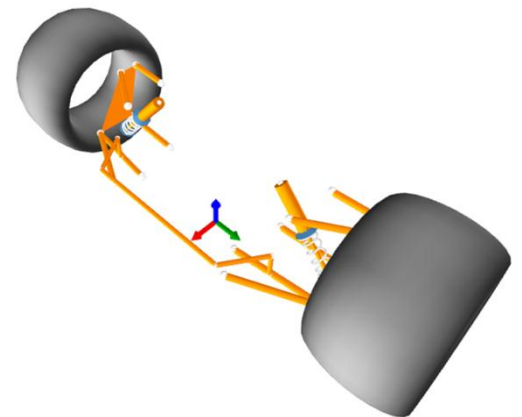
25-Double A-Arm (Chassis)_Seperate Spring 3 Dampers (Rear)



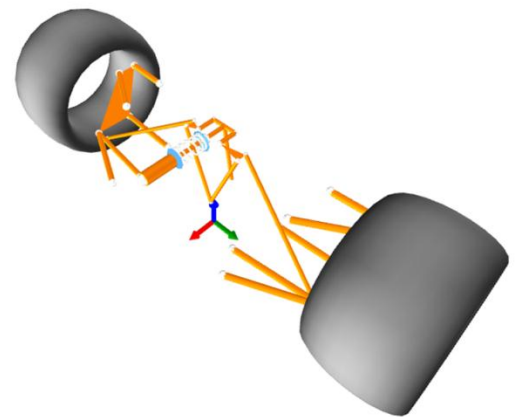
26-Double A-Arm (Chassis)_Torsion Bar_T-Bar (Rear)



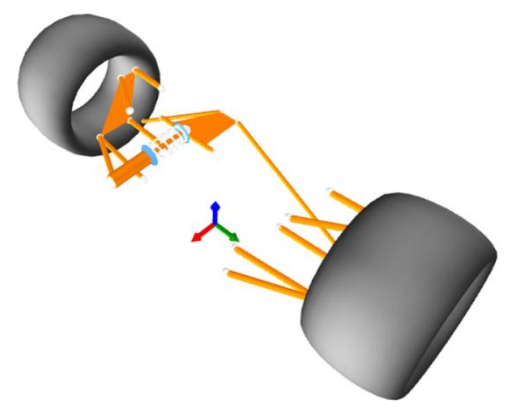
27-Five Links_1 Direct CoilOver (Rear)



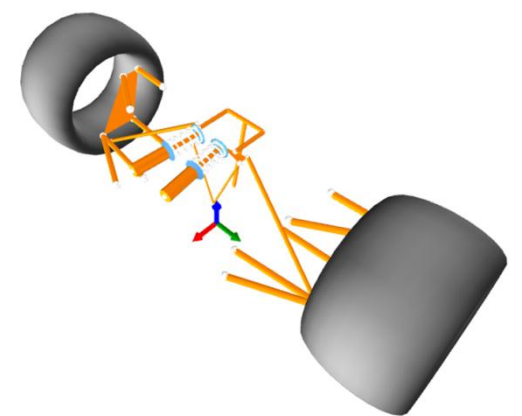
28-Five Links_1 Direct CoilOver_U-Bar (Rear)



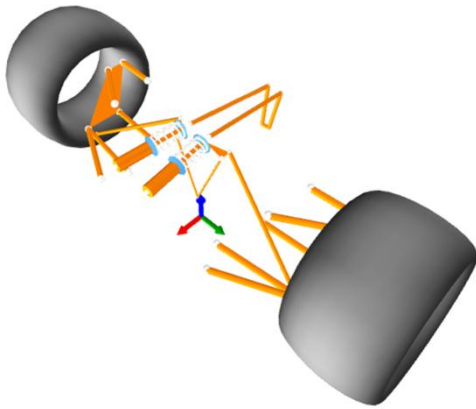
29-Five Links_MonoShock Rotate_T-Bar (Rear)



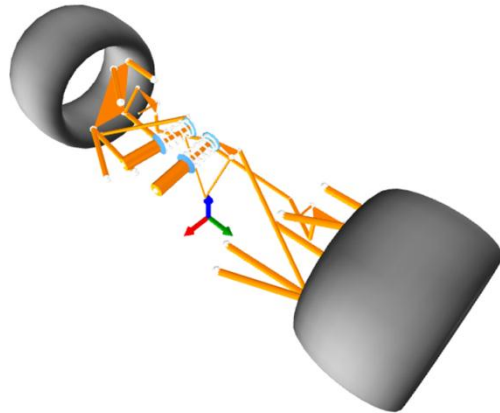
30-Five Links_MonoShock Slide (Rear)



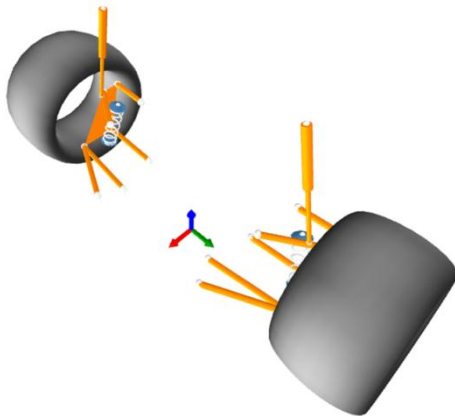
31-Five Links_PushPull_T-Bar (Rear)



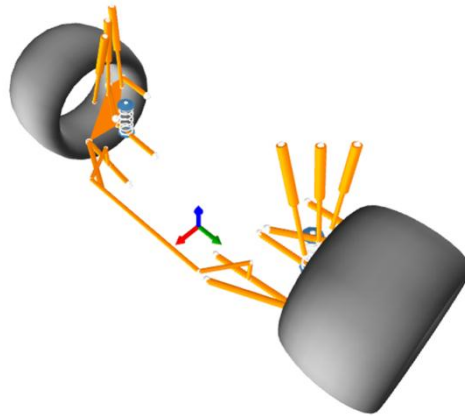
32-Five Links_PushPull_U-Bar (Rear)



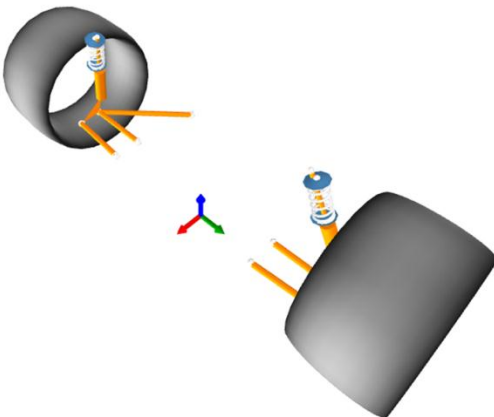
33-Five Links_PushPull_U-Bar Rocker (Rear)



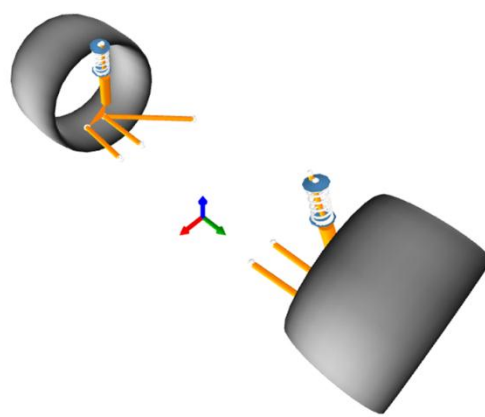
34-Five Links_Seperate Spring 1 Damper (Rear)



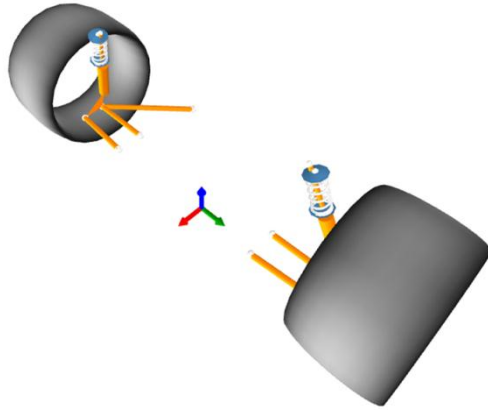
35-Five Links_Seperate Spring 3 Dampers_U-Bar (Rear)



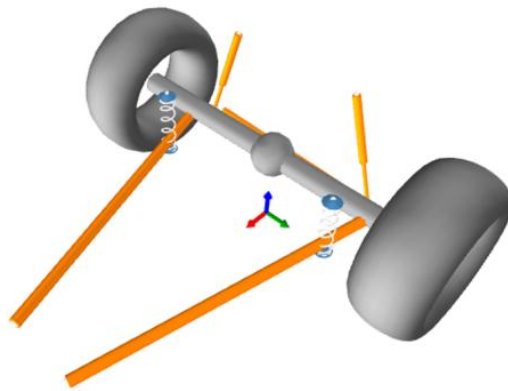
40-Mac Pherson (Chassis)(Rear)



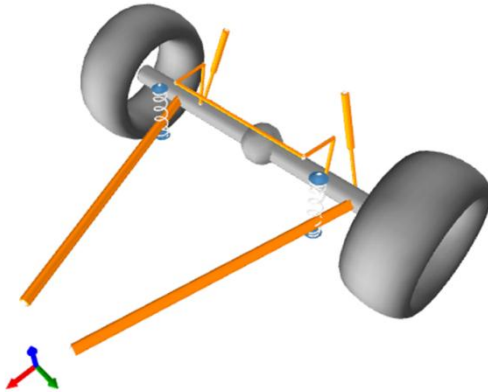
40 Bis-Mac Pherson (Lower A-Arm)(Rear)



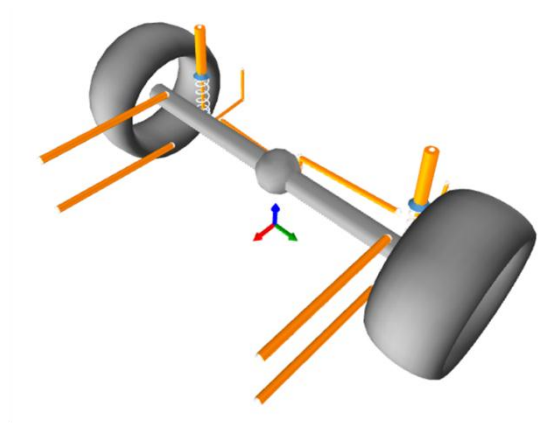
41-Mac Pherson_U-Bar (Rear)



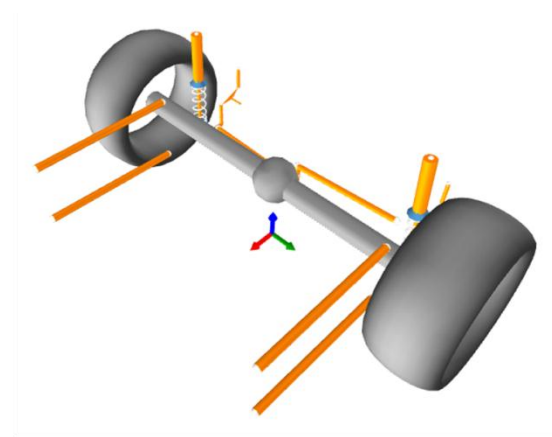
43-Live Axle 2 Trailing Arms Panhard Bar_Seperate Spring Damper (Rear)



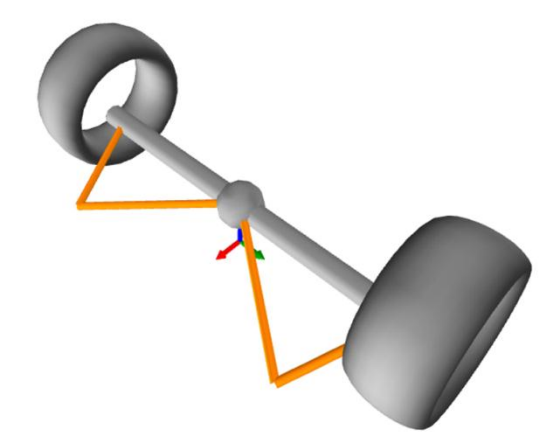
44-Live Axle 2 Trailing Arms Panhard Bar_Seperate Spring Damper_U-Bar (Rear)



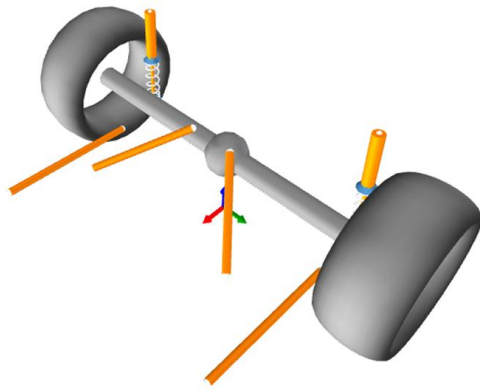
45-Live Axle 4 Trailing Arms Watts Linkage_1 Direct CoilOver_U-Bar (Rear)



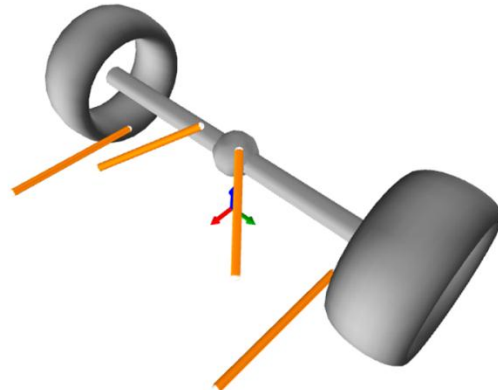
46-Live Axle 4 Trailing Arms Watts Linkage_1 Direct CoilOver_U-Bar Rocker (Rear)



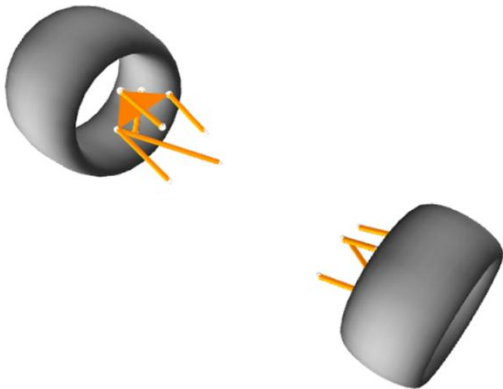
48-LiveAxle 2 A-Arm (Rear)



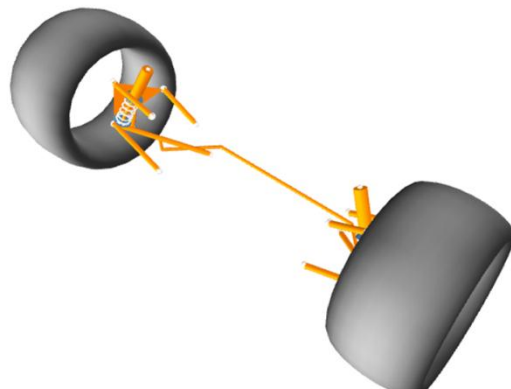
49-LiveAxle 4 Trailing Arms_1 Direct CoilOver (Rear)



50-LiveAxle 4 Trailing Arms (Rear)



51-Test Multilink F458 (Rear)



52-Test Multilink F458_1 Direct CoilOver_U-Bar (Rear)

8.2 Default Motion Types

Default motion types include:

- **Combined Heave & Roll.** 20mm of positive heave and 1.5° of positive roll.
- **Heave +20mm.** 20mm of positive heave. (see [Positive Heave](#))
- **Heave +40mm.** 40mm of positive heave. (see [Positive Heave](#))
- **Pitch +1.5deg.** 1.5° of positive pitch. (see [Positive Pitch](#))
- **Roll +1.5deg.** 1.5° of positive roll. (see [Positive Roll](#))
- **Steering +270deg.** 270° of inboard steering tie rods towards the default positive lateral axis. (see [Positive Steering](#))

8.3 Glossary and Definitions

8.3.1 Motion Definitions

8.3.1.1 Positive Heave

Positive heave is defined as the chassis body moving along the positive vertical axis. (Default axis : Z)

8.3.1.2 Positive Roll

Positive roll is defined as an anticlockwise rotation about the longitudinal axis (default: X), when the longitudinal axis is viewed head on.

8.3.1.3 Positive Pitch

Positive pitch is defined as an anticlockwise rotation about the lateral axis (default: Y), when the lateral axis is viewed head on.

8.3.1.4 Positive Steering

Positive Steering is defined as migration of the inboard steering tie rods towards the default positive lateral axis.

8.3.2 Input Definitions

8.3.2.1 Tire Diameter

The tire diameter is the outside diameter of the tire. This value can be given as the loaded, unloaded or rolling diameter of the tire.

8.3.2.2 Tire Width

The tire width is the overall width of the tire. It is used for visualization purposes only.

8.3.2.3 Rim Diameter

The Rim diameter is the diameter of the rim edge, not the tire bead seat. This is used as a reference point to measure toe distance.

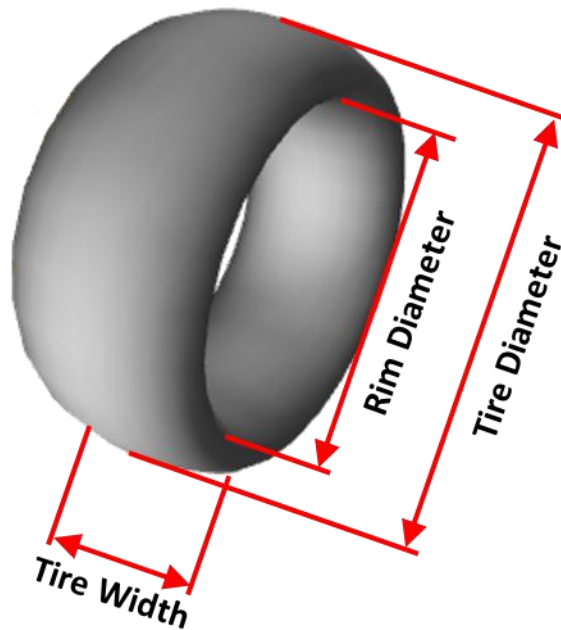


Figure 40 Rim Diameter

8.3.2.4 Halftrack

The halftrack is the horizontal difference from the center of the tire contact patch to the longitudinal axis.

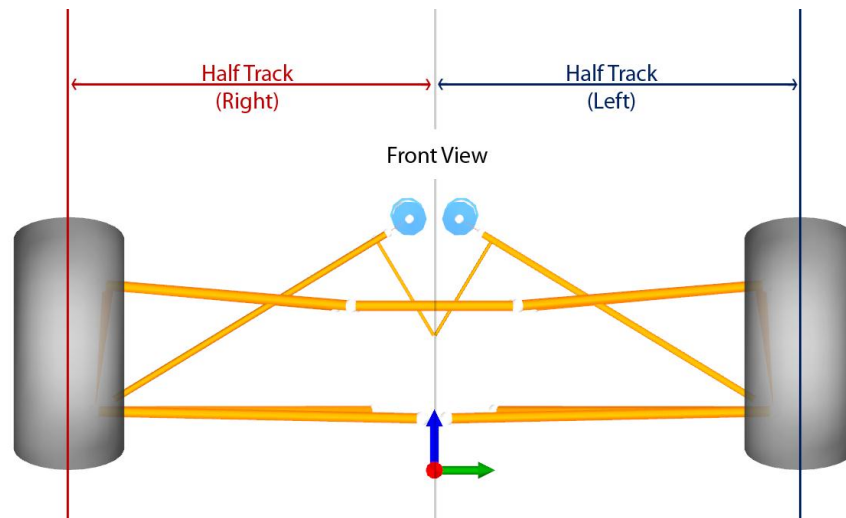


Figure 41 Halftrack

8.3.2.5 Reference Distance

The reference distance refers to the distance between the front and rear reference planes.

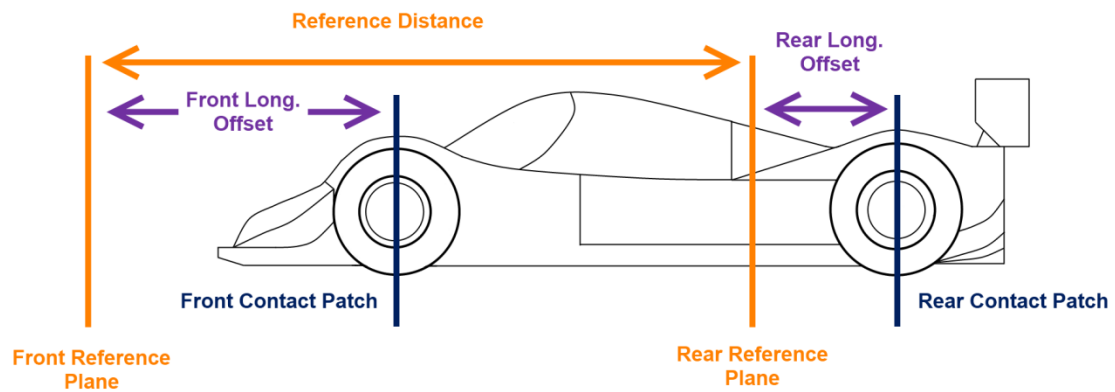


Figure 42 Reference Distance

8.3.3 Output definitions

8.3.3.1 Camber Angle

Camber is defined as the inclination angle between the side plane (vertical-longitudinal plane) and the rim plane lying on the centerline of the rim. Positive camber is defined as the tops of the wheels tipping away from the vehicle.

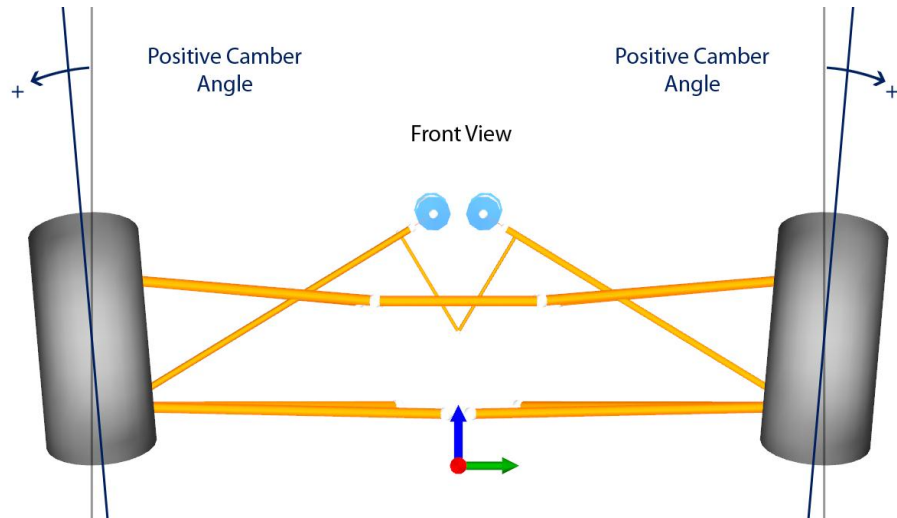


Figure 43 Camber Angle

8.3.3.2 Toe Angle

Toe is defined as the angular deflection from the vehicles centerline and the centerline of the rim. Positive toe (toe out) is defined as a wheel splaying out from the direction of travel. Toe Angle carries the same sign as Toe Distance.

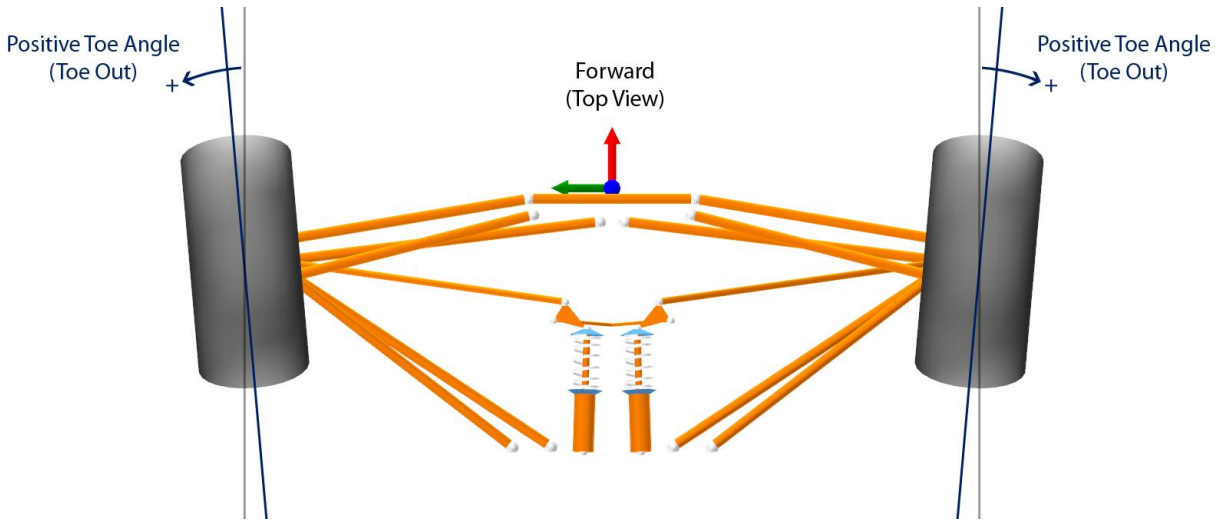


Figure 44 Toe Angle

8.3.3.3 Motion Ratio

The motion ratio is calculated in the following way:

$$\frac{\text{Wheel Displacement}}{\text{Spring Displacement}} \text{ and } \frac{\text{Roll Angle}}{\text{ARB Angular Displacement}}$$

8.3.3.4 Toe distance

Toe distance is defined as the total lateral distance between the leading and trailing edge of one side of a vehicles rim. Toe distance is taken to have the same sign as the angular toe measurement. Positive toe angle equates to positive toe distance.

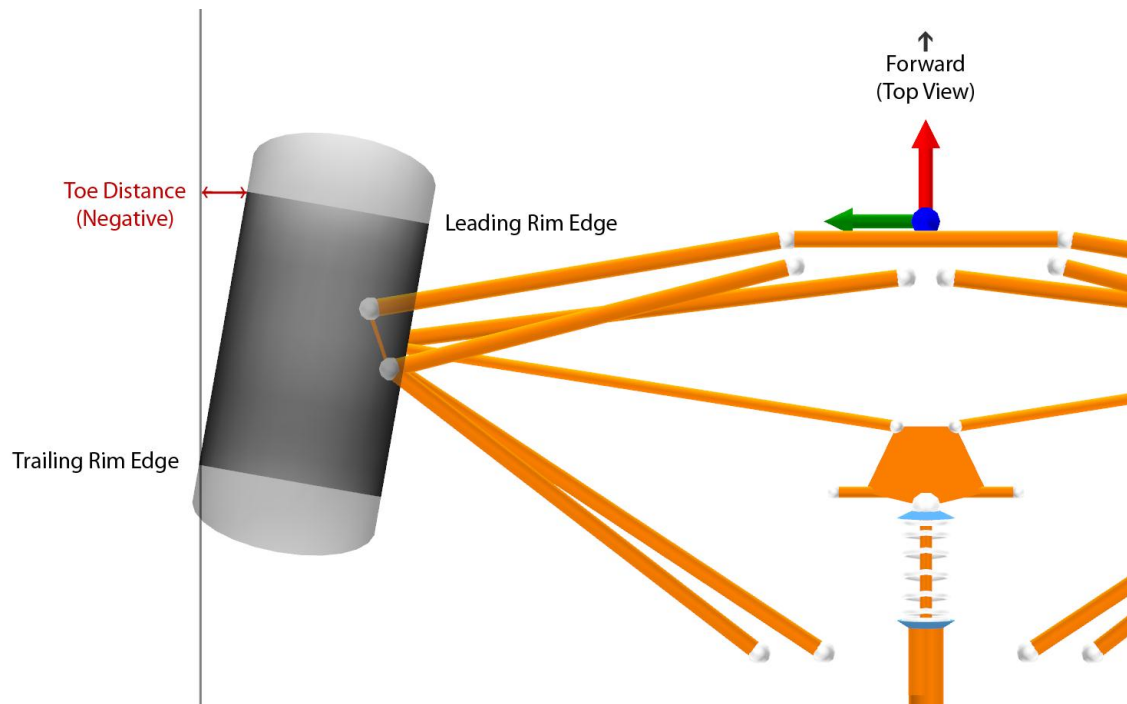


Figure 45 Toe Distance

8.3.3.5 Caster Angle

Caster is defined as the angle between the steering axis and the wheel centerline extending perpendicular from the contact patch, viewed perpendicular to the side view (vertical-longitudinal plane).

Positive caster is defined as the steering axis tilting back from the wheel centerline in side view (perpendicular to the longitudinal-vertical axis).

8.3.3.6 Mechanical Trail

Mechanical Trail is defined as the distance between the intersection of the steering axis and the ground measured to the center of the contact patch, viewed perpendicular to the vertical-longitudinal plane.

Positive mechanical trail is defined as the steering axis intersecting the ground plane before the contact patch.

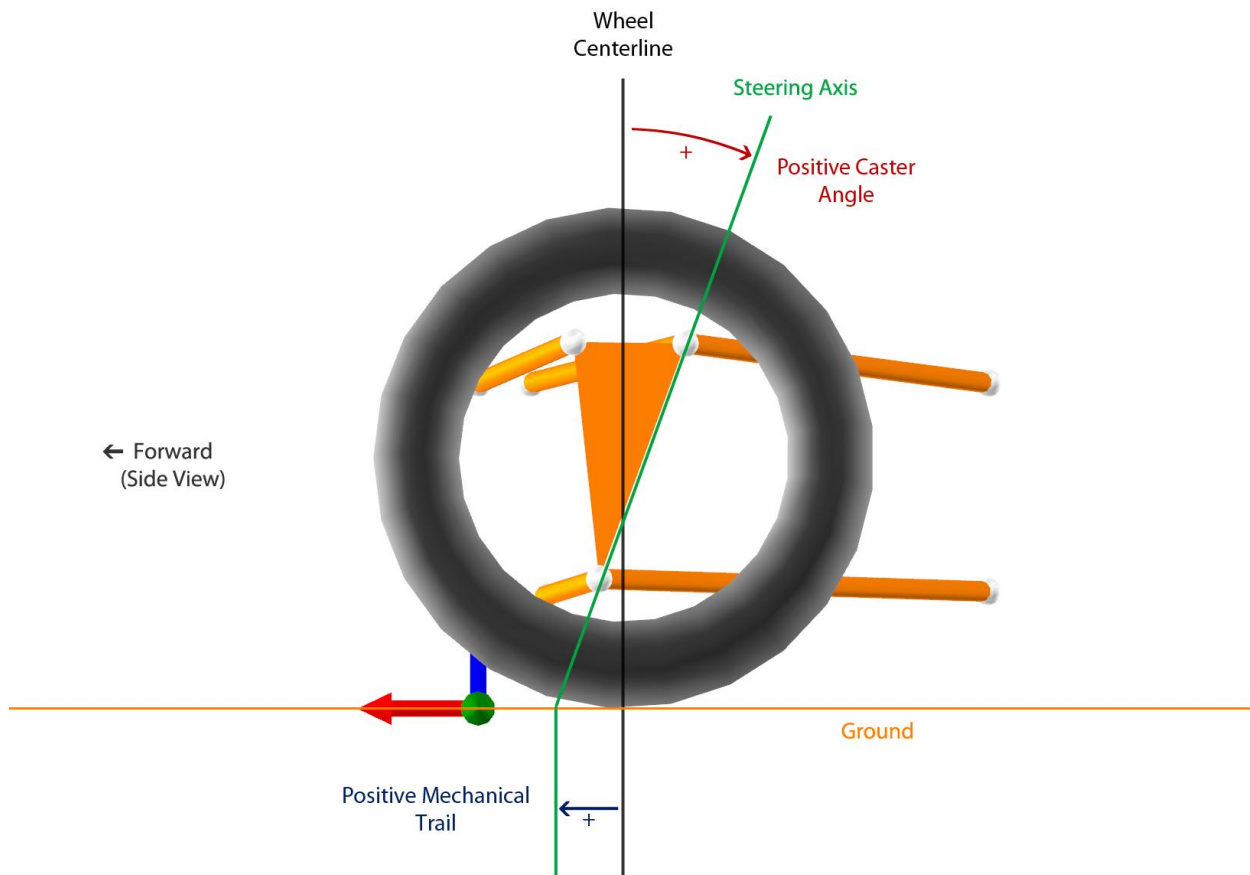


Figure 46 Mechanical Trail

8.3.3.7 King Pin Angle

King pin angle is defined as the angle between the steering axis and an axis extending perpendicular from the contact patch, viewed front on (perpendicular to the vertical-lateral plane.)

Positive king pin angle is defined as the top of the steering axis being closer to the vehicle centerline. The

8.3.3.8 Scrub Radius

Scrub Radius is defined as the distance between the intersection of the steering axis and the ground measured to the center of the contact patch, viewed perpendicular to the vertical-lateral plane.

Positive scrub radius is defined as the steering axis intersecting the ground plane between the vehicle centerline and the contact patch.

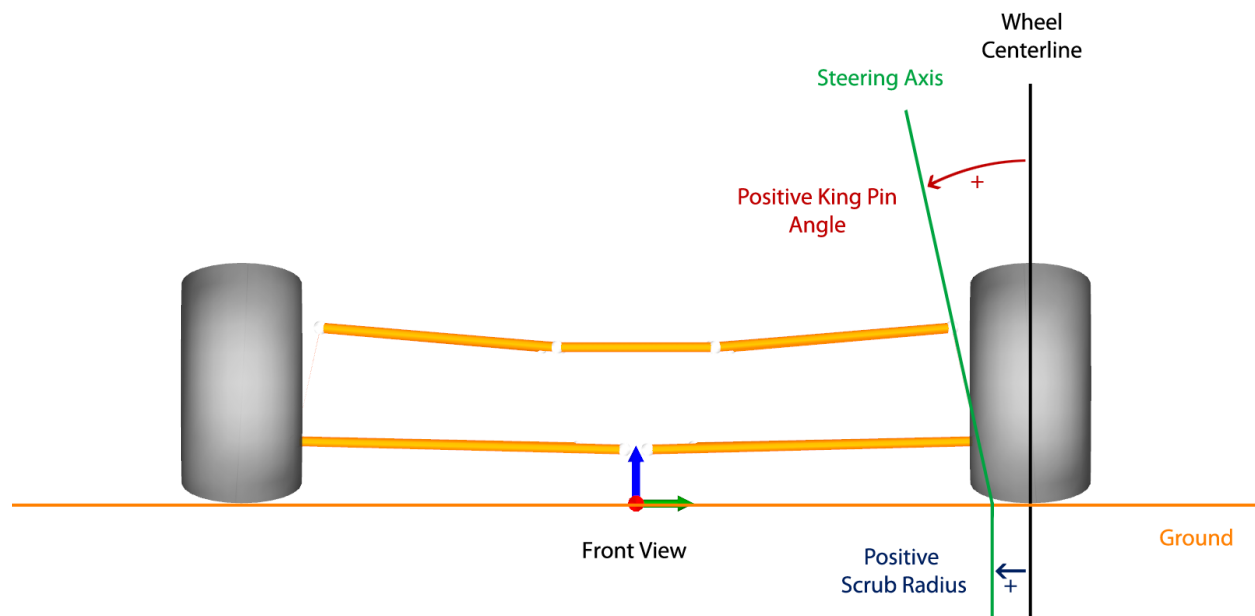


Figure 47 Scrub Radius

8.3.3.9 Front View Instantaneous center

The front view instant center point is where the instant axis intersects a vertical plane between the two front or rear tire contact patches.

8.3.3.10 Front View Virtual Swing Arm Length

The front view virtual swing arm length is defined as the distance between the contact patch and the front view instant center.

8.3.3.11 Front View Virtual Swing Arm Angle

The front view virtual swing arm angle is defined as the angle formed between the ground plane and the front view virtual swing arm, viewed from the front. Positive angles occur when the instant center is located above the ground.

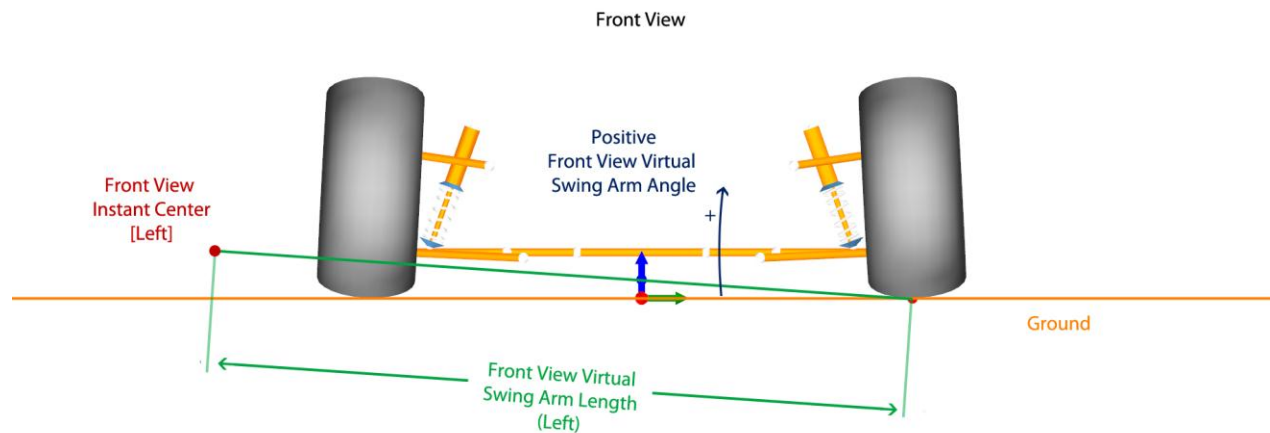


Figure 48 Front View Virtual Swing Arm Angle

8.3.3.12 Side view instantaneous center

The Side view instant center point is where the instant axis intersects a vertical plane between the two left or right contact patches.

8.3.3.13 Side View Virtual Swing Arm Length

The side view virtual swing arm length is defined as the distance between the contact patch and the side view instant center.

8.3.3.14 Side View Virtual Swing Arm Angle

The side view virtual swing arm angle is defined as the angle that the side view virtual swing arm forms with the ground plane, viewed from the side (longitudinal-vertical) plane.

8.3.3.15 Side Virtual Swing Arm Length vs. Wheelbase

The side view virtual swing arm length vs. wheelbase is defined as the side view virtual swing arm length divided by wheelbase.

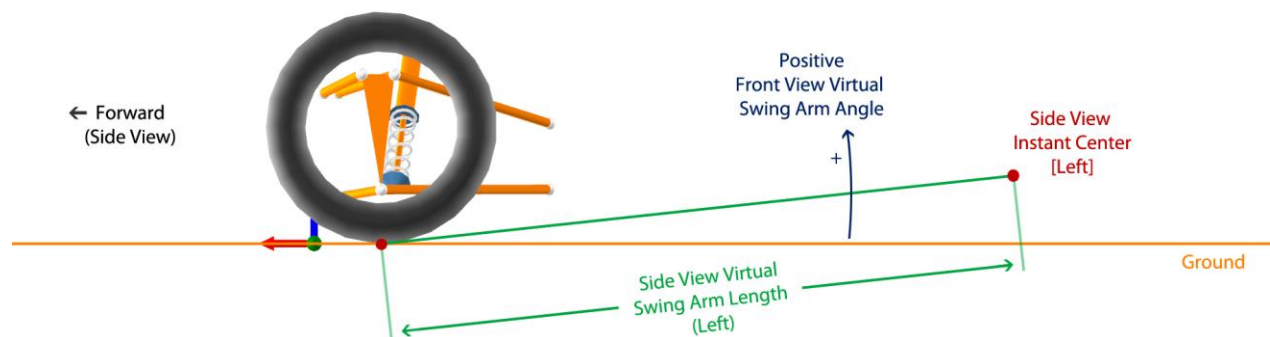


Figure 49 Side Virtual Swing Arm Length vs. Wheelbase

8.3.3.16 Instantaneous Axis

The instantaneous axis is the axis which the non-suspended mass rotates around as the suspension articulates. Each corner of the car has its own instantaneous axis. The instantaneous axis is located by moving the wheel up and down by a small increment and determining the point about which the wheel rotates, taking into account all suspension links –including tie rods and suspension.

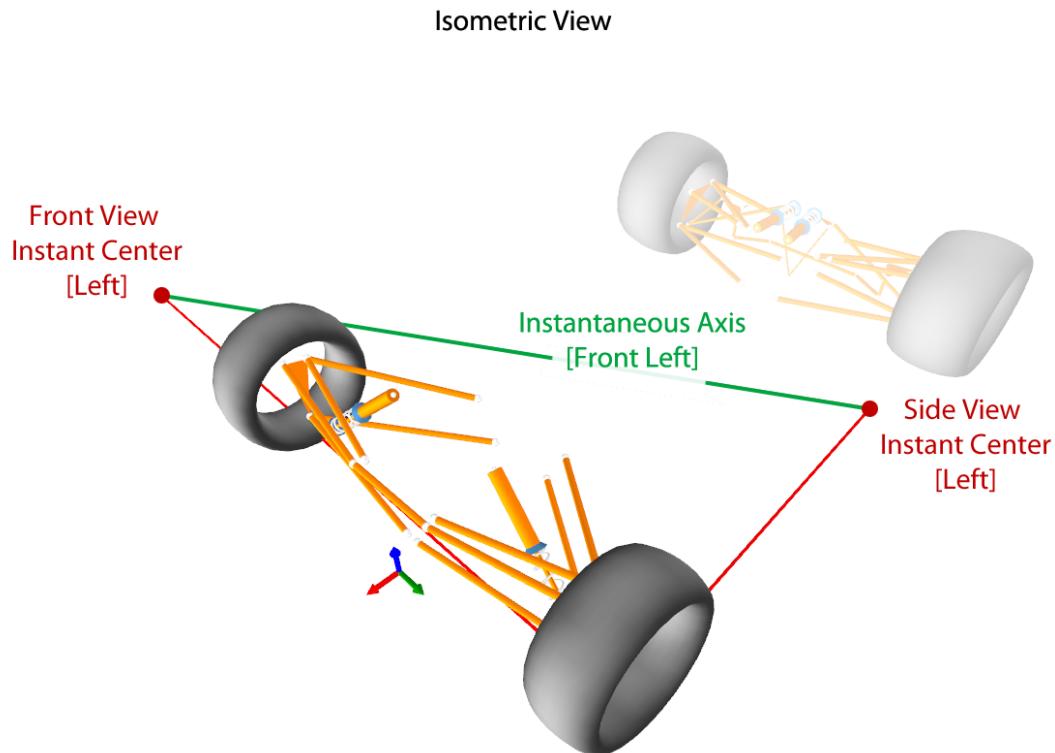


Figure 50 Instantaneous Axis

8.3.3.17 Kinematic Pitch Center

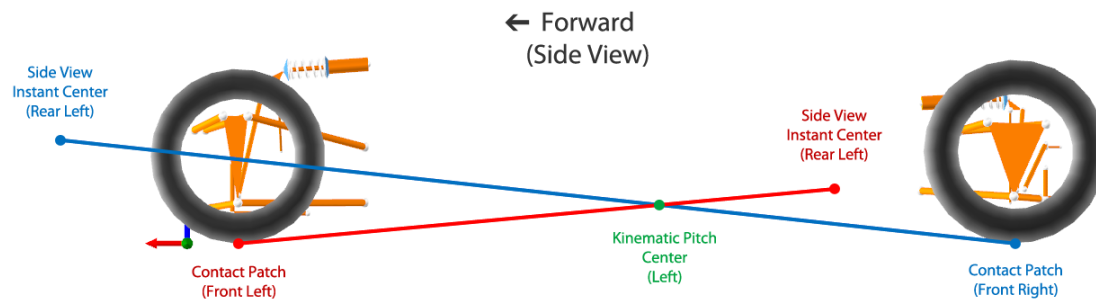


Figure 51 Kinematic Pitch Center

8.3.3.18 Kinematic Pitch Axis

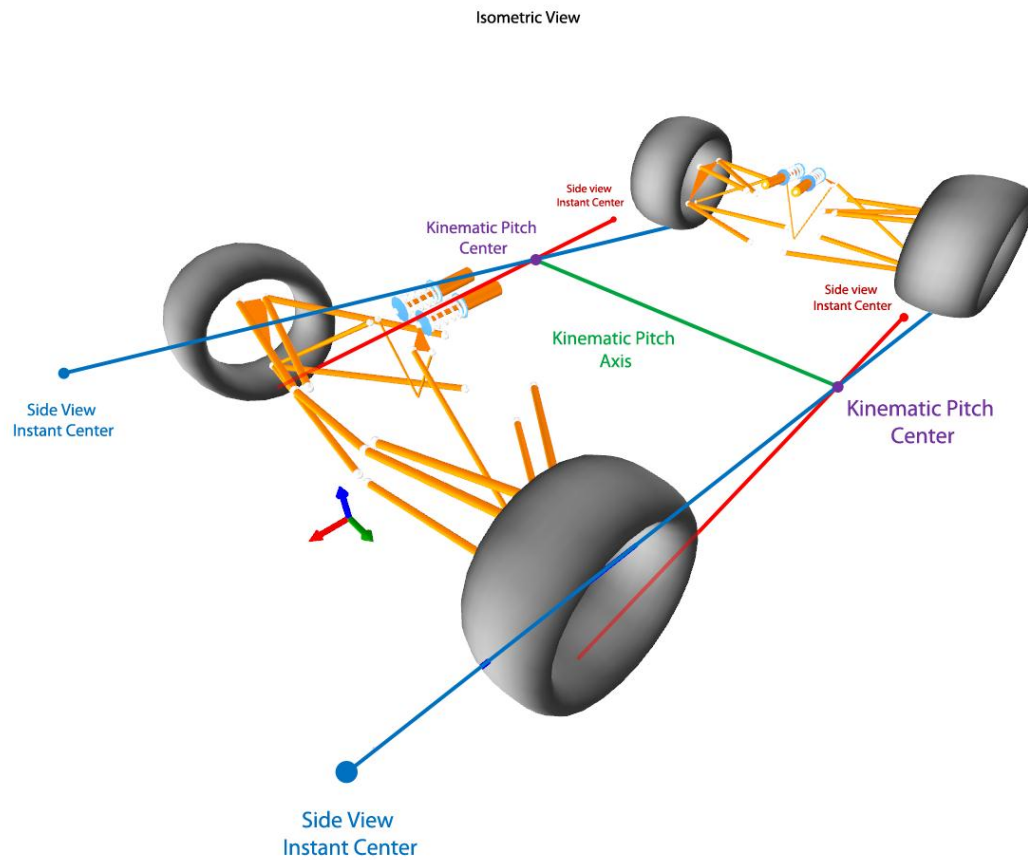


Figure 52 Kinematic Pitch Axis

8.3.3.19 Kinematic Roll Center

The Kinematic Roll Center is the intersection point between the left and right planes, which are defined by the instant axis and contact patch point, and the vertical plane between the left and right contact patches. This point is not stationary – it can move as the suspension articulates.

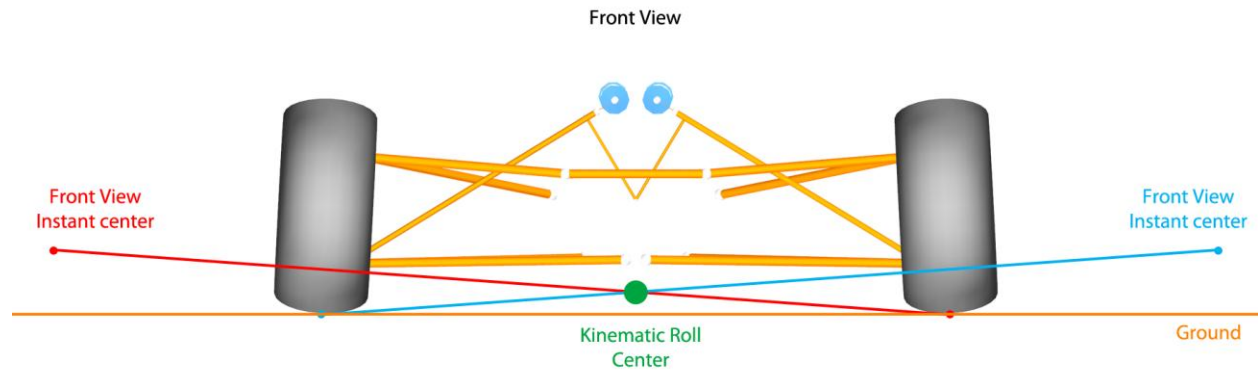


Figure 53 Kinematic Roll Center

8.3.3.20 Kinematic Roll Axis

The Roll Axis is a line drawn between the Front and Rear Roll Centers. This is the axis about which the suspended mass rotates around.

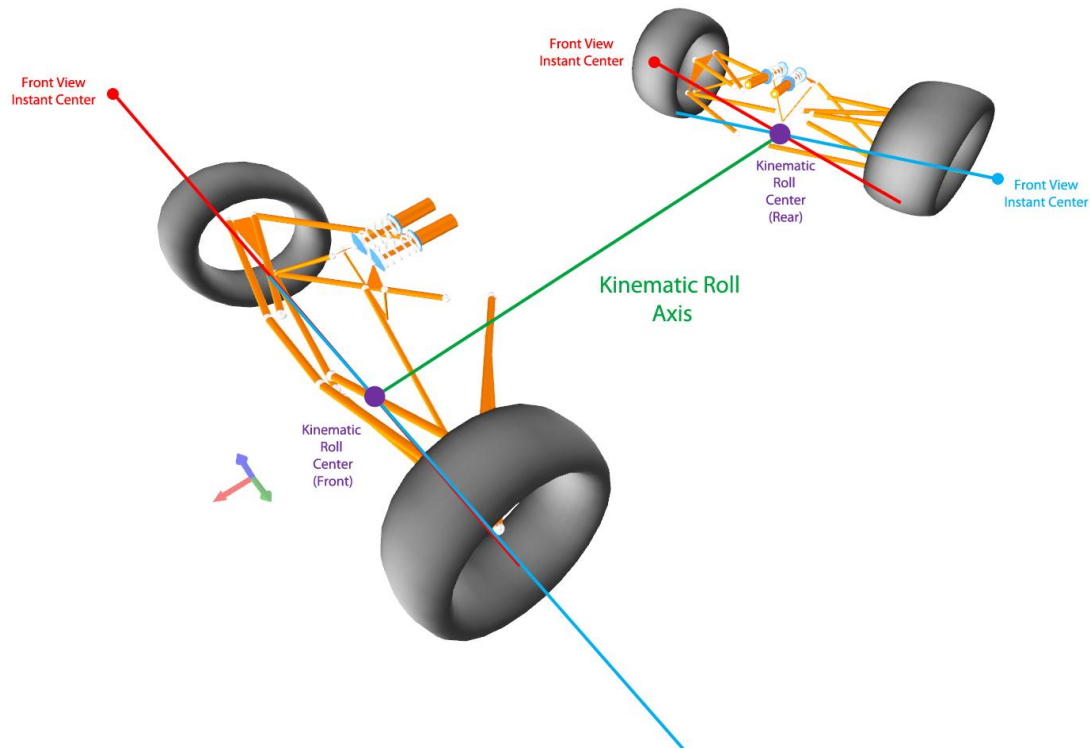


Figure 54 Kinematic Roll Axis

8.3.3.21 Roll Axis Inclination

The Roll Axis Inclination is defined as the angle between the kinematic roll axis and the ground plane, viewed perpendicular to the vertical-longitudinal plane.

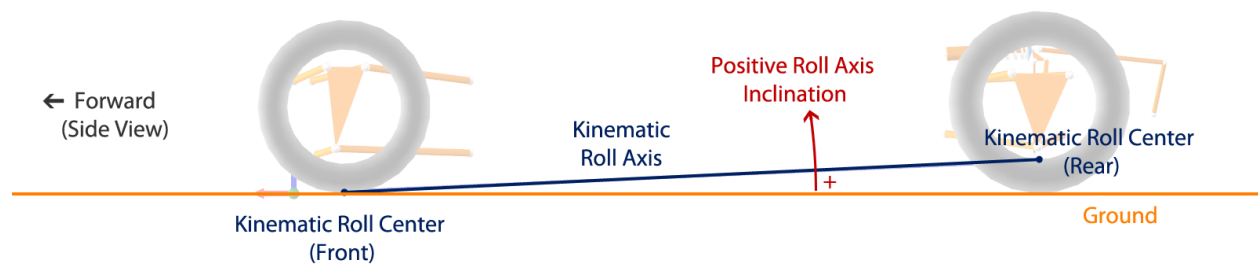


Figure 55 Roll Axis Inclination

8.3.3.22 Ackerman Percentage Definition

The Ackerman percentage is defined as a function of the inside steer angle, the outside steer angle, the front track and the vehicle wheelbase. The equations are as follows.

$$Ackerman = \tan^{-1} \left(\frac{wheelbase}{\frac{wheelbase}{\tan \delta_{outside}} - track_{front}} \right)$$

$$Ackerman_{percent} = \frac{\delta_{inside}}{Ackerman} \times 100$$

Figure 56 Ackerman Percentage

8.4 Available Results Channels

Motion [Percentage Completion]
Motion [Heave]
Motion [Roll]
Motion [Pitch]
Motion [Steering]
Kinematic Roll Center X [Front]
Kinematic Roll Center Y [Front]
Kinematic Roll Center Z [Front]
Kinematic Roll Center X [Rear]
Kinematic Roll Center Y [Rear]
Kinematic Roll Center Z [Rear]
Kinematic Pitch Center X [Left]
Kinematic Pitch Center Y [Left]
Kinematic Pitch Center Z [Left]
Kinematic Pitch Center X [Right]
Kinematic Pitch Center Y [Right]
Kinematic Pitch Center Z [Right]
Front View Instantaneous Center X [Front Left]
Front View Instantaneous Center Y [Front Left]
Front View Instantaneous Center Z [Front Left]
Front View Instantaneous Center X [Front Right]
Front View Instantaneous Center Y [Front Right]
Front View Instantaneous Center Z [Front Right]
Front View Instantaneous Center X [Rear Left]
Front View Instantaneous Center Y [Rear Left]
Front View Instantaneous Center Z [Rear Left]
Front View Instantaneous Center X [Rear Right]
Front View Instantaneous Center Y [Rear Right]
Front View Instantaneous Center Z [Rear Right]
Side View Instantaneous Center X [Front Left]
Side View Instantaneous Center Y [Front Left]
Side View Instantaneous Center Z [Front Left]
Side View Instantaneous Center X [Front Right]
Side View Instantaneous Center Y [Front Right]
Side View Instantaneous Center Z [Front Right]
Side View Instantaneous Center X [Rear Left]
Side View Instantaneous Center Y [Rear Left]
Side View Instantaneous Center Z [Rear Left]
Side View Instantaneous Center X [Rear Right]

Side View Instantaneous Center Y [Rear Right]
Side View Instantaneous Center Z [Rear Right]
Heave Motion Ratio CoilOver [Front Left]
Heave Motion Ratio CoilOver 1 [Front Left]
Heave Motion Ratio CoilOver 2 [Front Left]
Heave Motion Ratio CoilOver 3 [Front Left]
Heave Motion Ratio CoilOver [Front]
Heave Motion Ratio CoilOver 1 [Front]
Heave Motion Ratio CoilOver 2 [Front]
Heave Motion Ratio CoilOver 3 [Front]
Heave Motion Ratio CoilOver [Front Right]
Heave Motion Ratio CoilOver 1 [Front Right]
Heave Motion Ratio CoilOver 2 [Front Right]
Heave Motion Ratio CoilOver 3 [Front Right]
Heave Motion Ratio Spring [Front Left]
Heave Motion Ratio Spring 1 [Front Left]
Heave Motion Ratio Spring 2 [Front Left]
Heave Motion Ratio Spring 3 [Front Left]
Heave Motion Ratio Spring [Front Right]
Heave Motion Ratio Spring 1 [Front Right]
Heave Motion Ratio Spring 2 [Front Right]
Heave Motion Ratio Spring 3 [Front Right]
Heave Motion Ratio Damper [Front Left]
Heave Motion Ratio Damper 1 [Front Left]
Heave Motion Ratio Damper 2 [Front Left]
Heave Motion Ratio Damper 3 [Front Left]
Heave Motion Ratio Damper [Front Right]
Heave Motion Ratio Damper 1 [Front Right]
Heave Motion Ratio Damper 2 [Front Right]
Heave Motion Ratio Damper 3 [Front Right]
Heave Motion Ratio ARB Angle [Front]
Heave Motion Ratio ARB Displacement [Front]
Heave Motion Ratio CoilOver [Rear Left]
Heave Motion Ratio CoilOver 1 [Rear Left]
Heave Motion Ratio CoilOver 2 [Rear Left]
Heave Motion Ratio CoilOver 3 [Rear Left]
Heave Motion Ratio CoilOver [Rear]
Heave Motion Ratio CoilOver 1 [Rear]
Heave Motion Ratio CoilOver 2 [Rear]
Heave Motion Ratio CoilOver 3 [Rear]
Heave Motion Ratio CoilOver [Rear Right]

Heave Motion Ratio CoilOver 1 [Rear Right]
Heave Motion Ratio CoilOver 2 [Rear Right]
Heave Motion Ratio CoilOver 3 [Rear Right]
Heave Motion Ratio Spring [Rear Left]
Heave Motion Ratio Spring 1 [Rear Left]
Heave Motion Ratio Spring 2 [Rear Left]
Heave Motion Ratio Spring 3 [Rear Left]
Heave Motion Ratio Spring [Rear Right]
Heave Motion Ratio Spring 1 [Rear Right]
Heave Motion Ratio Spring 2 [Rear Right]
Heave Motion Ratio Spring 3 [Rear Right]
Heave Motion Ratio Damper [Rear Left]
Heave Motion Ratio Damper 1 [Rear Left]
Heave Motion Ratio Damper 2 [Rear Left]
Heave Motion Ratio Damper 3 [Rear Left]
Heave Motion Ratio Damper [Rear Right]
Heave Motion Ratio Damper 1 [Rear Right]
Heave Motion Ratio Damper 2 [Rear Right]
Heave Motion Ratio Damper 3 [Rear Right]
Heave Motion Ratio ARB Angle [Rear]
Heave Motion Ratio ARB Displacement [Rear]
Roll Motion Ratio CoilOver [Front Left]
Roll Motion Ratio CoilOver 1 [Front Left]
Roll Motion Ratio CoilOver 2 [Front Left]
Roll Motion Ratio CoilOver 3 [Front Left]
Roll Motion Ratio CoilOver [Front]
Roll Motion Ratio CoilOver 1 [Front]
Roll Motion Ratio CoilOver 2 [Front]
Roll Motion Ratio CoilOver 3 [Front]
Roll Motion Ratio CoilOver [Front Right]
Roll Motion Ratio CoilOver 1 [Front Right]
Roll Motion Ratio CoilOver 2 [Front Right]
Roll Motion Ratio CoilOver 3 [Front Right]
Roll Motion Ratio Spring [Front Left]
Roll Motion Ratio Spring 1 [Front Left]
Roll Motion Ratio Spring 2 [Front Left]
Roll Motion Ratio Spring 3 [Front Left]
Roll Motion Ratio Spring [Front Right]
Roll Motion Ratio Spring 1 [Front Right]
Roll Motion Ratio Spring 2 [Front Right]
Roll Motion Ratio Spring 3 [Front Right]

Roll Motion Ratio Damper [Front Left]
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Roll Motion Ratio Damper 3 [Front Left]
Roll Motion Ratio Damper [Front Right]
Roll Motion Ratio Damper 1 [Front Right]
Roll Motion Ratio Damper 2 [Front Right]
Roll Motion Ratio Damper 3 [Front Right]
Roll Motion Ratio ARB Angle [Front]
Roll Motion Ratio ARB Displacement [Front]
Roll Motion Ratio CoilOver [Rear Left]
Roll Motion Ratio CoilOver 1 [Rear Left]
Roll Motion Ratio CoilOver 2 [Rear Left]
Roll Motion Ratio CoilOver 3 [Rear Left]
Roll Motion Ratio CoilOver [Rear]
Roll Motion Ratio CoilOver 1 [Rear]
Roll Motion Ratio CoilOver 2 [Rear]
Roll Motion Ratio CoilOver 3 [Rear]
Roll Motion Ratio CoilOver [Rear Right]
Roll Motion Ratio CoilOver 1 [Rear Right]
Roll Motion Ratio CoilOver 2 [Rear Right]
Roll Motion Ratio CoilOver 3 [Rear Right]
Roll Motion Ratio Spring [Rear Left]
Roll Motion Ratio Spring 1 [Rear Left]
Roll Motion Ratio Spring 2 [Rear Left]
Roll Motion Ratio Spring 3 [Rear Left]
Roll Motion Ratio Spring [Rear Right]
Roll Motion Ratio Spring 1 [Rear Right]
Roll Motion Ratio Spring 2 [Rear Right]
Roll Motion Ratio Spring 3 [Rear Right]
Roll Motion Ratio Damper [Rear Left]
Roll Motion Ratio Damper 1 [Rear Left]
Roll Motion Ratio Damper 2 [Rear Left]
Roll Motion Ratio Damper 3 [Rear Left]
Roll Motion Ratio Damper [Rear Right]
Roll Motion Ratio Damper 1 [Rear Right]
Roll Motion Ratio Damper 2 [Rear Right]
Roll Motion Ratio Damper 3 [Rear Right]
Roll Motion Ratio ARB Angle [Rear]
Roll Motion Ratio ARB Displacement [Rear]
Pitch Motion Ratio CoilOver [Front Left]

Pitch Motion Ratio CoilOver 1 [Front Left]
Pitch Motion Ratio CoilOver 2 [Front Left]
Pitch Motion Ratio CoilOver 3 [Front Left]
Pitch Motion Ratio CoilOver [Front]
Pitch Motion Ratio CoilOver 1 [Front]
Pitch Motion Ratio CoilOver 2 [Front]
Pitch Motion Ratio CoilOver 3 [Front]
Pitch Motion Ratio CoilOver [Front Right]
Pitch Motion Ratio CoilOver 1 [Front Right]
Pitch Motion Ratio CoilOver 2 [Front Right]
Pitch Motion Ratio CoilOver 3 [Front Right]
Pitch Motion Ratio Spring [Front Left]
Pitch Motion Ratio Spring 1 [Front Left]
Pitch Motion Ratio Spring 2 [Front Left]
Pitch Motion Ratio Spring 3 [Front Left]
Pitch Motion Ratio Spring [Front Right]
Pitch Motion Ratio Spring 1 [Front Right]
Pitch Motion Ratio Spring 2 [Front Right]
Pitch Motion Ratio Spring 3 [Front Right]
Pitch Motion Ratio Damper [Front Left]
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Pitch Motion Ratio Damper 3 [Front Left]
Pitch Motion Ratio Damper [Front Right]
Pitch Motion Ratio Damper 1 [Front Right]
Pitch Motion Ratio Damper 2 [Front Right]
Pitch Motion Ratio Damper 3 [Front Right]
Pitch Motion Ratio ARB Angle [Front]
Pitch Motion Ratio ARB Displacement [Front]
Pitch Motion Ratio CoilOver [Rear Left]
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Pitch Motion Ratio CoilOver 3 [Rear Left]
Pitch Motion Ratio CoilOver [Rear]
Pitch Motion Ratio CoilOver 1 [Rear]
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Pitch Motion Ratio CoilOver 3 [Rear]
Pitch Motion Ratio CoilOver [Rear Right]
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Pitch Motion Ratio Spring [Rear Left]
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Pitch Motion Ratio Damper [Rear Right]
Pitch Motion Ratio Damper 1 [Rear Right]
Pitch Motion Ratio Damper 2 [Rear Right]
Pitch Motion Ratio Damper 3 [Rear Right]
Pitch Motion Ratio ARB Angle [Rear]
Pitch Motion Ratio ARB Displacement [Rear]
Single Left Wheel Motion Ratio CoilOver [Front Left]
Single Left Wheel Motion Ratio CoilOver 1 [Front Left]
Single Left Wheel Motion Ratio CoilOver 2 [Front Left]
Single Left Wheel Motion Ratio CoilOver 3 [Front Left]
Single Left Wheel Motion Ratio CoilOver [Front]
Single Left Wheel Motion Ratio CoilOver 1 [Front]
Single Left Wheel Motion Ratio CoilOver 2 [Front]
Single Left Wheel Motion Ratio CoilOver 3 [Front]
Single Left Wheel Motion Ratio CoilOver [Front Right]
Single Left Wheel Motion Ratio CoilOver 1 [Front Right]
Single Left Wheel Motion Ratio CoilOver 2 [Front Right]
Single Left Wheel Motion Ratio CoilOver 3 [Front Right]
Single Left Wheel Motion Ratio Spring [Front Left]
Single Left Wheel Motion Ratio Spring 1 [Front Left]
Single Left Wheel Motion Ratio Spring 2 [Front Left]
Single Left Wheel Motion Ratio Spring 3 [Front Left]
Single Left Wheel Motion Ratio Spring [Front Right]
Single Left Wheel Motion Ratio Spring 1 [Front Right]
Single Left Wheel Motion Ratio Spring 2 [Front Right]
Single Left Wheel Motion Ratio Spring 3 [Front Right]
Single Left Wheel Motion Ratio Damper [Front Left]
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Single Left Wheel Motion Ratio Damper [Front Right]
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Single Left Wheel Motion Ratio Damper 2 [Front Right]
Single Left Wheel Motion Ratio Damper 3 [Front Right]
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Single Left Wheel Motion Ratio ARB Displacement [Front]
Single Left Wheel Motion Ratio CoilOver [Rear Left]
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Single Left Wheel Motion Ratio CoilOver 2 [Rear]
Single Left Wheel Motion Ratio CoilOver 3 [Rear]
Single Left Wheel Motion Ratio CoilOver [Rear Right]
Single Left Wheel Motion Ratio CoilOver 1 [Rear Right]
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Single Left Wheel Motion Ratio Damper [Rear Right]
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Single Left Wheel Motion Ratio Damper 3 [Rear Right]
Single Left Wheel Motion Ratio ARB Angle [Rear]
Single Left Wheel Motion Ratio ARB Displacement [Rear]
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Single Right Wheel Motion Ratio CoilOver 3 [Front Left]

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Single Right Wheel Motion Ratio Spring 2 [Front Right]
Single Right Wheel Motion Ratio Spring 3 [Front Right]
Single Right Wheel Motion Ratio Damper [Front Left]
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Single Right Wheel Motion Ratio Damper 3 [Front Left]
Single Right Wheel Motion Ratio Damper [Front Right]
Single Right Wheel Motion Ratio Damper 1 [Front Right]
Single Right Wheel Motion Ratio Damper 2 [Front Right]
Single Right Wheel Motion Ratio Damper 3 [Front Right]
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Single Right Wheel Motion Ratio ARB Displacement [Front]
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Single Right Wheel Motion Ratio CoilOver 1 [Rear]
Single Right Wheel Motion Ratio CoilOver 2 [Rear]
Single Right Wheel Motion Ratio CoilOver 3 [Rear]
Single Right Wheel Motion Ratio CoilOver [Rear Right]
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Single Right Wheel Motion Ratio Damper [Rear Right]
Single Right Wheel Motion Ratio Damper 1 [Rear Right]
Single Right Wheel Motion Ratio Damper 2 [Rear Right]
Single Right Wheel Motion Ratio Damper 3 [Rear Right]
Single Right Wheel Motion Ratio ARB Angle [Rear]
Single Right Wheel Motion Ratio ARB Displacement [Rear]
Wheelbase [Left]
Wheelbase [Right]
Wheelbase Average
Percentage Ackermann [Front]
Inside vs. Outside Steer Angle [Front]
Steering Wheel Angle vs. Steering Angle [Front Left]
Steering Wheel Angle vs. Steering Angle [Front Right]
Steering Wheel Angle vs. Steering Angle Average [Front]
Roll Axis Inclination Angle
Front View Swing Arm Length [Front Left]
Front View Swing Arm Length [Front Right]
Front View Swing Arm Length [Rear Left]
Front View Swing Arm Length [Rear Right]
Side View Swing Arm Length [Front Left]
Side View Swing Arm Length [Front Right]
Side View Swing Arm Length [Rear Left]
Side View Swing Arm Length [Rear Right]
Front View Swing Arm Angle [Front Left]
Front View Swing Arm Angle [Front Right]
Front View Swing Arm Angle [Rear Left]
Front View Swing Arm Angle [Rear Right]
Side View Swing Arm Angle [Front Left]
Side View Swing Arm Angle [Front Right]
Side View Swing Arm Angle [Rear Left]
Side View Swing Arm Angle [Rear Right]

8.5 Hotkeys

Windows

Alt + Letter Activate a Main Menu item that has the letter underlined.

Alt + F File
S Save
A Save As
O Close
R Recent
N New
L License
T Options
B About
X Exit

Alt + D Design

Alt + S Simulation

Alt + A Analysis

File

Ctrl + N New Project
Ctrl + O Open Project
Ctrl + S Save Project
Ctrl + Shift + S Save As Project

Edit

Ctrl + X Cut Object
Ctrl + C Copy Object
Ctrl + V Paste Object
Delete Delete Object

General

F1 Help
F9 Options

Motion

Mouse Left Click Add Points
Mouse Right Click Remove Points
Ctrl + Mouse Left Click Modify Points

Visualization

Mouse Left Click Rotation
Mouse Right Click Pan
Mouse Wheel Zoom

Simulation

F4 Launch

Animation

F5 Play/Pause
F6 Stop
F7 Move Backward
F8 Move Forward

Bug Reporting

F2 Online Form

8.6 Contact Information



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